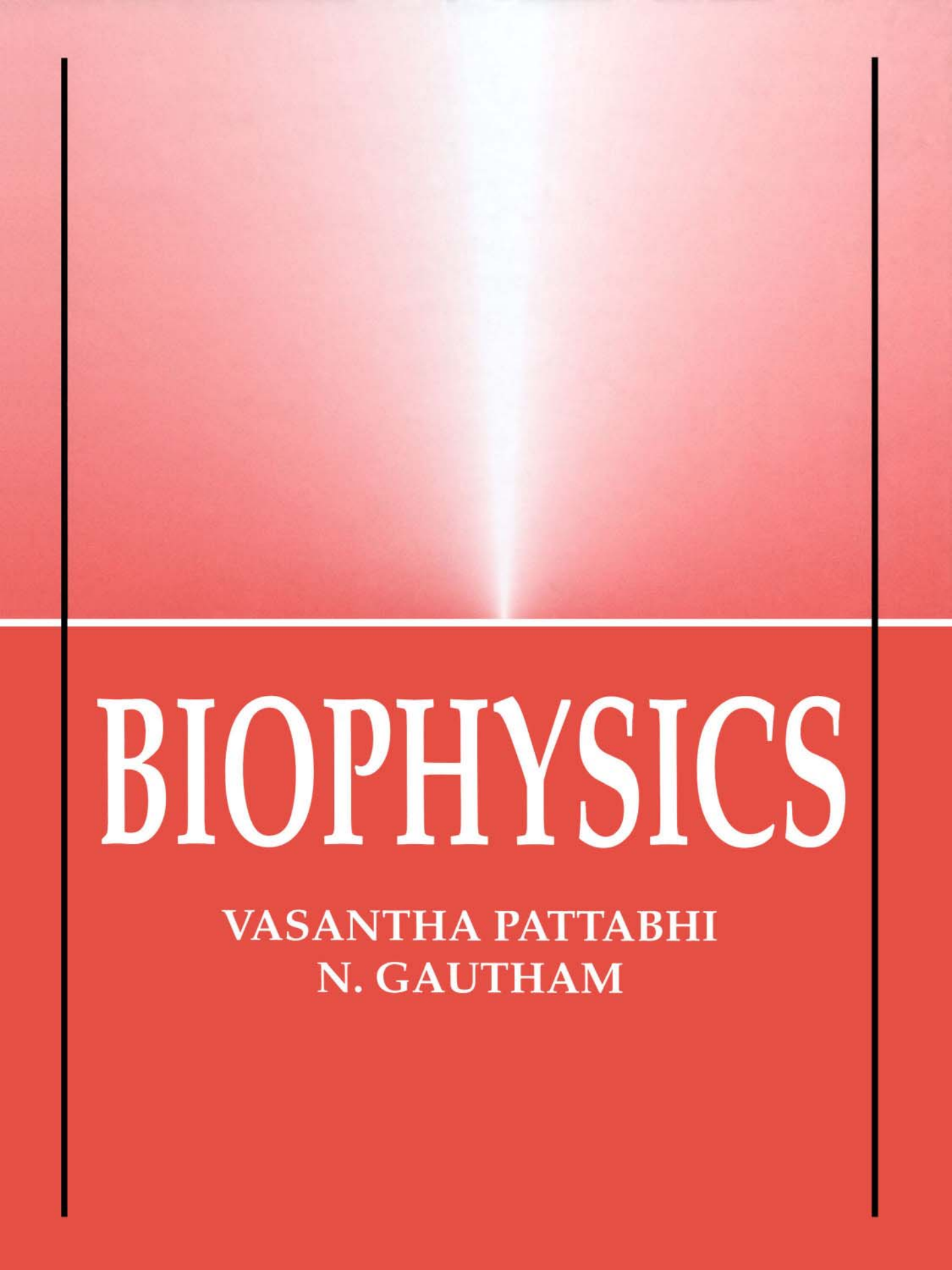


The book cover features a red gradient background. A bright, vertical beam of light emanates from the top center, creating a lens flare effect. Two thin black vertical lines run down the left and right sides of the cover, framing the text.

BIOPHYSICS

VASANTHA PATTABHI
N. GAUTHAM

This Page Intentionally Left Blank

Biophysics

Vasanthi Pattabhi
N. Gautham

Department of Crystallography & Biophysics
University of Madras, Guindy Campus
Chennai, India

KLUWER ACADEMIC PUBLISHERS

NEW YORK, BOSTON, DORDRECHT, LONDON, MOSCOW



Narosa Publishing House
DELHI CHENNAI MUMBAI KOLKATA

eBook ISBN: 0-306-47520-0
Print ISBN: 1-4020-0218-1

©2002 Kluwer Academic Publishers
New York, Boston, Dordrecht, London, Moscow

Print ©2002 Kluwer Academic Publishers
Dordrecht

All rights reserved

No part of this eBook may be reproduced or transmitted in any form or by any means, electronic, mechanical, recording, or otherwise, without written consent from the Publisher

Created in the United States of America

Visit Kluwer Online at: <http://kluweronline.com>
and Kluwer's eBookstore at: <http://ebooks.kluweronline.com>

This book is dedicated to
Professor G.N. Ramachandran, F.R.S.
who continues to inspire generations of biophysicists

*எப்பொருள் யார்யார்வாய்க் கேட்பினும் அப்பொருள்
மெய்ப்பொருள் காண்பது அறிவு*

குறள் 423

Wisdom grasps the truth of whatever and by
whomever said.

Kural 423

This Page Intentionally Left Blank

Preface

The initiation of biophysics in India is synonymous with the starting of the Department of Crystallography and Biophysics at the University of Madras, in 1953, under the leadership of Professor G.N. Ramachandran. Even today, nearly half-century later, there are only a handful of Universities in the country which have full-fledged biophysics departments. However, most life science curricula in the country now comprise a course in biophysics, owing to its increasing importance and relevance to the study of any biological system. This book has been written to fill a long-standing lacuna of textbooks in biophysics, suitable for students at the senior undergraduate and postgraduate level in various biological disciplines, such as biochemistry, molecular biology and even medicine.

Biophysics is an interdisciplinary subject, and its treatment varies with the background of both the author and the reader. For example, some aspects of theoretical biophysics are indistinguishable from pure mathematics. On the other hand, modern biophysics has made profound contributions to subjects normally considered a part of classical biology, such as evolution. In this book we have not assumed any knowledge of higher mathematics on the part of the reader as a prerequisite. We have emphasized an understanding of the underlying physical principles of the biological phenomena. Many of the explanations may therefore appear too elementary and too verbose to sophisticated readers. We believe however that the 'great divide' between students of the physical sciences who are mathematically fluent, and those of the biological sciences who are less so, is too deep to be bridged with one easy leap in a single textbook. Thus the present book does not set out to make biophysicists out of all biologists. However we believe it will cater to the needs of most biologists, who stop their acquaintance with mathematics at school. In addition, chemists and physicists may also find it useful as an introduction to the subject. The teaching notes of the authors form the skeleton of the book and this was fleshed out on the basis of several years of classroom experience.

Chapter 1 gives an introduction to laws of physics and chemistry and is designed to reinforce and enhance slightly the school-level understanding of the subject, which students with a background of purely biological science would have. This chapter covers quantum mechanics, molecular orbital theory, essentials of weak and strong molecular interactions, stereochemistry, fundamentals of thermodynamics and radiation biophysics.

Chapters 2 and 3 deal with separation techniques and physico-chemical techniques used to study

biomolecular structure. Strictly speaking this would be counted under physical biochemistry, rather than purely biophysics. The coverage is not exhaustive and is intended to give an overview of various techniques available to unravel the properties of biomolecules.

Chapters 4 to 9 describe some of the spectroscopy, microscopy, diffraction and computational techniques. These techniques are so powerful that every biophysicist must be armed with this knowledge so that it can be put to use when the need arises. These chapters aim at providing an understanding of the basic principles involved rather than thorough knowledge of the theory and practice of the techniques. The descriptions would enable most biologists to obtain an understanding of the relative merits and demerits of each technique, in order that the biochemical results, which one may come across in the general literature, become intelligible. X-ray crystallography and NMR spectroscopy have been dealt with in greater detail owing to their prominence in determining biomolecular structures.

Results obtained from the application of the techniques of structural biology are reviewed in Chapter 10. This chapter is illustrated by the largest number of diagrams. Nevertheless, in a book of this nature, in which one is always trying to keep costs low, it is not possible to include every picture the authors may feel necessary. Also we would have dearly liked to include colour illustrations.

The later chapters try and fill the gaps between biophysicists and biochemists. The treatment of Energy Pathways in Chapter 11 involves a lot of physical chemistry. Biomechanics in Chapter 12 and Neurophysics in Chapter 13 involve a lot of physiology. Though the advancements in the application of physical principles in these fields have been phenomenal they are probably too advanced and too detailed for the intended audience of this book.

We have of course referred to several books and journal articles, review papers, etc, while writing this book. At the end of the book we have included a chapter-wise list for further reading, which would amplify and extend the contents of each chapter. This list is of course neither exhaustive nor comprehensive. We have as far as possible selected books that would be available in most University libraries.

This book was written as a project sponsored and funded by the University Grants Commission, India. We take this opportunity to thank the UGC and all its officers for the invitation and support, financial and otherwise. We also thank the reviewers, whose corrections and suggestions we have tried to incorporate. Any errors that remain are, of course, ours. Our M.Sc., M.Phil and Ph.D. students have taught us (or have forced us to learn) much of the material presented in this book. We thank them all most sincerely. For actual help in preparing the book, we thank Mr A. Johnson and Mr. Prem Raj Bernard.

We would like to specially thank our respective family members for their enthusiastic support throughout. Writing this book has been a rewarding experience. But as we look back over it, we feel, with Thomas Hardy, 'the more written, the more it seems remains to be written'.

VASANTHA PATTABHI
N. GAUTHAM

Contents

Preface

vii

1. Laws of Physics and Chemistry **1**

- 1.1 Introduction 1
- 1.2 Quantum Mechanics 1
- 1.3 The Electronic Structure of Atoms 3
- 1.4 Molecular Orbitals and Covalent Bonds 5
- 1.5 Molecular Interactions 8
 - 1.5.1 Strong interactions 9
 - 1.5.2 Weak interactions 10
- 1.6 Stereochemistry and Chirality 11
 - 1.6.1 Stereochemical nomenclature 13
- 1.7 Thermodynamics 13
 - 1.7.1 Entropy 16
 - 1.7.2 Enthalpy 17
 - 1.7.3 The free energy of a system 17
 - 1.7.4 Chemical potential 18
 - 1.7.5 Oxidation-reduction potential 19
- 1.8 Radioactivity 20
 - 1.8.1 Rate of radioactive decay 21
 - 1.8.2 Measurement of radioactivity 22
 - 1.8.3 Effects of radioactivity on matter 22
 - 1.8.4 Biological effects of radiation 23
 - 1.8.5 Applications of radio isotopes 23

2. Separation Techniques **24**

- 2.1 Introduction 24
- 2.2 Chromatography 24
 - 2.2.1 Column chromatography 26

2.2.2	Thin layer chromatography	26
2.2.3	Paper chromatography	26
2.2.4	Adsorption chromatography	27
2.2.5	Partition chromatography	27
2.2.6	Gas liquid chromatography (GLC)	28
2.2.7	Ion exchange chromatography	28
2.2.8	Molecular exclusion chromatography	29
2.2.9	Affinity chromatography	30
2.3	Electrophoresis	32
2.3.1	Moving boundary electrophoresis	32
2.3.2	Zone electrophoresis	33
2.3.3	Low voltage electrophoresis	33
2.3.4	High voltage electrophoresis	33
2.3.5	Gel electrophoresis	34
2.3.6	Sodium dodecyl sulphate poly acrylamide gel electrophoresis (SDS-PAGE)	
2.3.7	Iso electric focussing	35
2.3.8	Continuous flow electrophoresis	35

3. Physico-Chemical Techniques to Study Biomolecules

37

3.1	Introduction	37
3.2	Hydration of Macromolecules	38
3.3	Role of Friction	38
3.4	Diffusion	39
3.5	Sedimentation	41
3.6	The Ultracentrifuge	43
3.7	Viscosity	46
3.8	Rotational Diffusion	48
3.8.1	Flow birefringence measurements	49
3.8.2	Electric birefringence	50
3.9	Light Scattering	51
3.10	Small Angle X-ray Scattering	54

4. Spectroscopy

58

4.1	Introduction	58
4.2	Ultraviolet/Visible Spectroscopy	59
4.3	Circular Dichroism (CD) and Optical Rotatory Dispersion (ORD)	61
4.4	Fluorescence Spectroscopy	65
4.5	Infrared Spectroscopy	67
4.6	Raman Spectroscopy	71
4.7	Electron Spin Resonance	74

5. Light Microscopy

77

5.1	Introduction	77
5.2	Elementary Geometrical Optics	77
5.3	The Limits of Resolution	79
5.4	Different Types of Microscopy	82
5.4.1	Bright field microscopy	82

- 5.4.2 Dark field microscopy 82
- 5.4.3 Phase contrast microscopy 82
- 5.4.4 Fluorescence microscopy 83
- 5.4.5 Polarising microscopy 84

6. Electron Microscopy **86**

- 6.1 Introduction 86
- 6.2 Electron Optics 86
- 6.3 The Transmission Electron Microscope (TEM) 87
- 6.4 The Scanning Electron Microscope (SEM) 88
- 6.5 Preparation of the Specimen for Electron Microscopy 90
- 6.6 Image Reconstruction 91
- 6.7 Electron Diffraction 92
- 6.8 The Tunnelling Electron Microscope 92
- 6.9 Atomic Force Microscope 93

7. X-ray Crystallography **95**

- 7.1 Introduction 95
- 7.2 Crystals and Symmetries 95
- 7.3 Crystal Systems 96
- 7.4 Point Groups and Space Groups 97
- 7.5 Growth of Crystals of Biological Molecules 99
- 7.6 X-ray Diffraction 103
- 7.7 X-ray Data Collection 105
- 7.8 Structure Solution 106
 - 7.8.1 The heavy atom or multiple isomorphous replacement (MIR) method 107
 - 7.8.2 Direct methods 108
 - 7.8.3 Molecular replacement 108
 - 7.8.4 The anomalous scattering technique 108
- 7.9 Refinement of the Structure 108
 - 7.9.1 Least squares methods 108
 - 7.9.2 Constrained-restrained refinement 109
 - 7.9.3 Computer graphics in refinement 110
- 7.10 Note on the Resolution of an X-ray Structure 111

8. NMR Spectroscopy **112**

- 8.1 Introduction 112
- 8.2 Basic Principles of NMR 112
- 8.3 NMR Theory and Experiment 114
- 8.4 Classical Description of NMR 115
- 8.5 NMR Parameters 117
 - 8.5.1 Chemical shift 117
 - 8.5.2 Intensity 118
 - 8.5.3 Line width 119
 - 8.5.4 Relaxation parameters 119
 - 8.5.5 Spin-spin coupling 120
- 8.6 The Nuclear Overhauser Effect 122

- 8.7 NMR Applications in Chemistry 122
 - 8.7.1 Chemical shift 122
 - 8.7.2 Spin-spin coupling 123
 - 8.7.3 ^{13}C NMR 124
- 8.8 NMR Applications in Biochemistry and Biophysics 124
 - 8.8.1 Concentration measurement 125
 - 8.8.2 pH titration 125
 - 8.8.3 Conformation of biomolecules 125
 - 8.8.4 Two-Dimensional NMR 126
 - 8.8.5 Determination of macromolecular structure 128
- 8.9. NMR in Medicine 128

9. Molecular Modelling 131

- 9.1 Introduction 131
- 9.2 Generating the Model 131
 - 9.2.1 Building the structure of H_2O_2 132
 - 9.2.2 Building protein structure 133
 - 9.2.3 Building nucleic acid structure 136
 - 9.2.4 Displaying and altering the generated model 137
- 9.3 Optimising the Model 138

10. Macromolecular Structure 144

- 10.1 Introduction 144
- 10.2 Nucleic Acid Structure 144
 - 10.2.1 The chemical structure of nucleic acids 145
 - 10.2.2 Conformational possibilities of monomers and polymers 147
 - 10.2.3 The double helical structure of DNA 148
 - 10.2.4 Polymorphism of DNA 151
 - 10.2.5 DNA supercoiling and unusual DNA structures 154
 - 10.2.6 The structure of transfer RNA 158
- 10.3 Protein Structure 161
 - 10.3.1 Amino acids and the primary structure of proteins 161
 - 10.3.2 The peptide bond and secondary structure of proteins 161
 - 10.3.3 Tertiary structure—supersecondary and domain structure 169
 - 10.3.4 Quaternary structure 170
 - 10.3.5 Virus structure 172

11. Energy Pathways in Biology 174

- 11.1 Introduction 174
- 11.2 Free Energy 174
- 11.3 Coupled Reactions 176
- 11.4 Group Transfer Potential 177
- 11.5 Role of Pyridine Nucleotides 178
- 11.6 Photosynthesis 179
 - 11.6.1 Photosystem I 183
 - 11.6.2 Photosystem II 185
 - 11.6.3 Photophosphorylation and carbon fixation 186
- 11.7 Energy Conversion Pathways 187

- 11.7.1 Oxidation 187
- 11.7.2 Glycolysis 187
- 11.7.3 The Krebs cycle 188
- 11.7.4 The respiratory chain 190
- 11.8 Membrane Transport 194
 - 11.8.1 Active transport 195
 - 11.8.2 Chemi-osmotic theory—passive transport 196

12. Biomechanics

199

- 12.1 Introduction 199
- 12.2 Striated Muscles 199
 - 12.2.1 Contractile proteins 201
- 12.3 Mechanical Properties of Muscles 202
 - 12.3.1 Contraction mechanism 203
 - 12.3.2 Role of Ca^{2+} ions 204
- 12.4 Biomechanics of the Cardiovascular System 205
 - 12.4.1 Blood pressure 205
 - 12.4.2 Electrical activity during the heartbeat 207
 - 12.4.3 Electrocardiography 208

13. Neurobiophysics

210

- 13.1 Introduction 210
- 13.2 The Nervous System 210
 - 13.2.1 Synapse 212
- 13.3 Physics of Membrane Potentials 213
 - 13.3.1 Membrane potential due to diffusion 213
 - 13.3.2 Voltage Clamp 215
- 13.4 Sensory Mechanisms—The Eye 217
 - 13.4.1 The visual receptor 218
 - 13.4.2 Electrical activity and visual generator potentials 223
 - 13.4.3 Optical defects of the eye 224
 - 13.4.4 Neural aspects of vision 224
 - 13.4.5 Visual communications, bioluminescence 226
- 13.5 Physical Aspects of Hearing 227
 - 13.5.1 The Ear 227
 - 13.5.2 Elementary acoustics 228
 - 13.5.3 Theories of hearing 230
- 13.6 Signal Transduction 230
 - 13.6.1 Mode of transport 231
 - 13.6.2 Signal transduction in the cell 231

14. Origin and Evolution of Life

232

- 14.1 Introduction 232
- 14.2 Prebiotic Earth 233
- 14.3 Theories of Origin and Evolution of Life 234
- 14.4 Laboratory Experiments on the Formation of Small Molecules 235
 - 14.4.1 Prebiotic synthesis of amino acids—Miller and Urey's experiment 235
 - 14.4.2 Prebiotic synthesis of purines, pyrimidines and nucleosides 237

14.4.3	Sugars	237
14.4.4	Nucleotide synthesis	238
14.4.5	Optical activity	239
14.4.6	Polymerisation	240

FURTHER READING	243
-----------------	-----

INDEX	247
-------	-----

Laws of Physics and Chemistry

1.1 Introduction

Physics is an exact science. Biology, on the other hand, can be classified as a descriptive science. However, in recent times, the latter is also being transformed into a more exact science with the advent of molecular biology and molecular biophysics. One aspect of this transformation originates from the applications of physics to physiology. Physical laws or concepts such as mechanics, hydrodynamics, optics, electrodynamics and thermodynamics are used to explain physiological observations like muscle contraction, neural communication, vision, etc. A second and more fundamental aspect of the transformation emerged from the search for universal principles governing the world around us. Almost from the time man was able to think he has been wondering as to what makes him and the living world distinct from the rest of the environment. By the beginning of twentieth century it was realized that the laws of physics and chemistry, which were applied to non-living things, could equally well explain forces controlling biology, and that no new fundamentally different principles were necessary to explain the organisms and interactions which make up the living world. In the hundred years that have passed since then, this concept has become stronger and today nobody thinks it necessary to invoke any special physical or chemical forces or laws in the study of biology. This chapter will describe, in an elementary way, some of these basic concepts and laws of modern physics and chemistry, in particular those which are directly relevant to biology.

1.2 Quantum Mechanics

By the end of the nineteenth century, physical science had arrived at a description of the external world in terms of two distinct types of entities. On the one hand were particles and particulate matter to which the laws of mechanics, as enunciated by Newton and developed by several others, were wholly applicable. Concepts such as mass, momentum, kinetic energy, etc. were descriptive of particles. On the other hand were radiation and other wave like phenomena to which the laws of wave

propagation and other optical laws were applicable. Phenomena such as diffraction and interference could be explained in terms of the wave-like picture.

It was Max Planck who first suggested that there need not be a complete separation between the concept of a particle and the concept of a wave. In order to explain the observed results when the intensity of radiation from a perfectly radiating body (i.e. a so-called 'black body') was measured as a function of the wavelength, Planck proposed that electromagnetic radiation was emitted or absorbed not continuously but in small packets, or quanta. The energy contained within each quantum was given by the expression

$$E = h\nu$$

where ν is the frequency of the radiation and h ($= 6.625 \times 10^{-34}$ joule seconds) is a universal constant called Planck's constant. The black body could only absorb or emit integral multiples of this quantity, never any fraction of a quantum. Einstein extended this idea to say that not only is radiation absorbed or emitted in packets but it also propagates only as packets, called photons, each containing an amount of energy given by the Planck expression. Thus, radiation, which was thought to be wavelike, now could be treated as a particle and the same entity could both diffract as well as possess momentum. The French scientist De Broglie suggested that the same type of wave-particle duality could be applied to entities hitherto thought to be particles. Thus, he said, electrons and other sub-atomic particles could be thought of also as wave packets, with a wavelength given by

$$\lambda = h/p$$

where p is the momentum of the particle. In fact this equation is applicable not only to sub-atomic particles but to all matter. However Planck's constant is so small that the wave nature of matter becomes observable only when the mass of the particle, and hence its momentum, is also very small. When this happens, then according to De Broglie's theory, a particle such as an electron would not only possess mass and momentum but would also show diffraction and other wave-like effects. Hundreds of thousands of experiments performed all over the world since the proposals were made have firmly established the truth of the wave-particle duality of the physical world. Quantum mechanics is now accepted as the only correct description of physics. However for practical calculations on macroscopic objects like cricket balls and space rockets it is sufficient to use Newton's laws. For calculations of atomic level phenomena we use quantum mechanics, chiefly as developed by Schrodinger and Heisenberg.

Schrodinger introduced a wave equation that replaces the classical equations of motion. The solution to this wave equation is a 'wave function' which is a function of space and time. The physical state of any system such as an electron (or an atom or a molecule or a system of molecules) can be described completely by the corresponding wave function. All physically meaningful and measurable quantities such as the energy, the momentum, the position and so on can be extracted from the wave function. The wave function is usually represented as Ψ and is obtained as a solution of the following 'eigenvalue' equation

$$H\Psi = E\Psi$$

where E is the characteristic energy of the system or the energy 'eigenvalue' and H the operator called Hamiltonian operator, which operates upon the characteristic wave function or the 'eigenfunction' Ψ to yield the energy values. A physical way of understanding Ψ is as a probability amplitude. In other words, since Ψ is a function of the coordinates x , y and z and time t , the $|\Psi|^2$ gives the probability of finding the system at the point x , y , z at time t . Written explicitly, the Schrodinger wave equation has the following form

$$\frac{d^2\psi}{dx^2} - \frac{8\pi^2m}{h} (E - V) \psi = 0$$

where m is the mass of the particle and V the expression for the potential energy. This is a one-dimensional equation, independent of time. To solve this equation one has to, first of all, define the appropriate expression for the potential energy V , which will depend on the problem studied. When this expression is inserted into the Schrodinger wave equation, the differential equation so obtained can be solved to find Ψ and E . In three dimensions the Schrodinger equation becomes

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} - \frac{8\pi^2m}{h} (E - V) \psi = 0$$

The time dependent form of the equation is as follows:

$$-\frac{h^2}{8\pi^2m} \left(\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} \right) = \frac{ih}{2\pi} \frac{d\psi}{dt}$$

where i is the imaginary root.

This scheme of calculations involving the wave equation is called *wave mechanics*. It is complemented by another, equivalent scheme developed by Heisenberg and Born, called *matrix mechanics*. The starting point for these calculations was the discovery made by Heisenberg that there exists a physical limit to the accuracy with which one can simultaneously determine both the position and the velocity of a particle. This is due to the fact that any measurement necessarily implies that the particle under observation is being subjected to a disturbance, at the very least by a photon of light. If the particle is small, then the act of observation itself leads to uncertainty in the position. (If the particle were large, then since it is extended it cannot be said to possess a very precise location). Heisenberg stated these facts in the form of the Uncertainty Principle, which declares that no two canonically conjugate observables can be measured simultaneously with arbitrary precision. Position and momentum are canonically conjugate variables. So are *energy* and *time*. Another such pair is *angular momentum* and *angular position*.

Both matrix mechanics and wave mechanics may be used to arrive at the physically meaningful and measurable variables that characterize any physical system. Which scheme of calculation is to be used depends on the actual information that is sought. The most spectacular and immediate success of quantum mechanics was in the theoretical description of the structure of atoms.

1.3 The Electronic Structure of Atoms

Rutherford's model of the atom described it as consisting of a central nucleus with a positive electric charge Ze , and Z electrons each with a negative electric charge e revolving around it. Z is the atomic number of the atom. In order to resolve certain inconsistencies in this picture, Bohr proposed an altered model, which was later shown by quantum mechanical calculations to be the correct one. Bohr postulated that the electrons, which revolve round the central nucleus, occupy only certain fixed orbits. These orbits are such that the associated angular momentum is always an integral multiple of $\hbar/2\pi$. They are called the stationary quantum states of the atom and can be calculated as the solutions of the Schrodinger wave equation (Figure 1.1). When the electrons are in one of the stationary states, no radiation occurs, in spite of the fact that according to classical electrodynamics one would expect such radiation due to the accelerated motion of the charged electrons. Bohr also postulated that radiation is emitted or absorbed only when electrons jump from one stationary state (also called an orbital) to another. The difference in energy between two states is given by

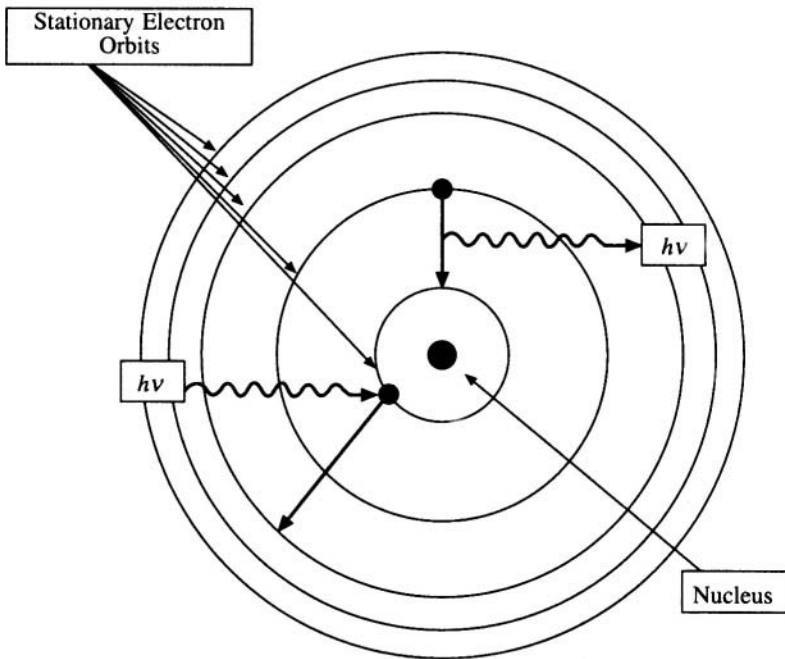


Fig. 1.1 Stationary states in the Bohr atom

$$\Delta E = h\nu = hc/\lambda$$

where λ is the wavelength of the radiation, and c the velocity of light. This equation determines the wavelength of the radiation emitted or absorbed (Figure 1.2). The motion of each of the electrons in the atom is described by a wave function ψ , known also as an atomic orbital. A combination of the wave functions of all the electrons leads to the wave function Ψ for the atom as a whole. Four quantum numbers are used to describe the orbital of an electron. These are (a) principal quantum number n , (b) angular momentum quantum number l , (c) magnetic quantum number m_l , and (d) spin quantum number m_s . The principal quantum number specifies the total energy of the electron. The angular momentum associated with the orbital motion of the electrons is described by the angular

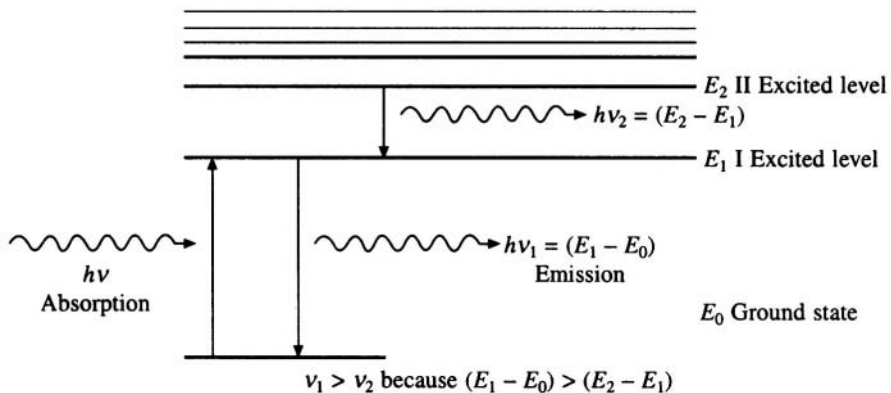


Fig. 1.2 Energy levels and the transitions between them

momentum quantum number. The principal quantum number can assume only integral values ($n = 1, 2, 3, \dots$). There can be more than one electron with the same principal quantum number, though for each electron at least one of the quantum numbers must be different. Electrons with the same value of the principal quantum number n constitute an electron shell. These shells are designated $k, l, m, \dots$ etc., corresponding to $n = 1, 2, 3, \dots$ etc. The angular momentum quantum number l can only take on the values $0, 1, 2, 3, \dots, n - 1$. In the presence of a magnetic field the orientation of the vector representing the angular momentum is described by the magnetic quantum numbers m_l , which can have values $0, \pm 1, \pm 2, \dots, \pm l$. Lastly, the spin angular momentum is due to the spin of the electron about its own axis. The spin vector can assume one of two possible orientations in the presence of a magnetic field and the corresponding values of m_s are $+1/2$ and $-1/2$. Pauli's Exclusion Principle states that no two electrons can have all the four quantum numbers the same. Two electrons may occupy the same space orbital. Then their spins must be anti-parallel to each other, i.e. if one of them has $m_s = +1/2$, the other has the value $-1/2$. The orbital with the lowest energy is known as the ground state. For an atom with many electrons the ground state is when the electrons are arranged in the lowest energy orbitals consistent with the exclusion principle. Other states are called excited states.

When an electron (or the atom as a whole) makes a transition from one quantum state to another, energy is either absorbed or emitted. The radiated energy is equal to the difference in energy levels of the initial and final states. The frequency ν of the absorbed or emitted radiation can be written as

$$\nu = \Delta E/h$$

where ΔE is the difference in the energies of the two levels. On atoms with more than one electron, the spin and orbital angular momenta of the individual electrons in an atom combine to produce a net angular momentum for the atom as a whole. Thus an atom with two electrons can have a total spin equal to zero when the electrons are paired (i.e. they have anti-parallel spins) or equal to one when they are unpaired (parallel spins). However, in the latter case the electrons must occupy different orbitals according to Pauli's Exclusion Principle. The orbitals are designated s, p, d, f etc. according to the value of the angular momentum quantum number. The $1s$ orbital is the lowest energy orbital with $n = 1, l = 0$ and $m_l = 0$. This orbital can be occupied by two electrons with spins $+1/2$ and $-1/2$ respectively. The orbital with the next higher energy value is the $2s$ orbital, followed by the $2p$ orbital, and so on (Table 1.1).

Table 1.1 Electron distribution in atomic orbitals of the first three shells

Atomic Shells	n	l	m_l	m_s	Orbital Designation	Number of Electrons Accommodated
K	1	0	0	$+1/2, -1/2$	$1s$	2
L	2	0	0	$+1/2, -1/2$	$2s$	2
		1	$+1, 0, -1$	$+1/2, -1/2$	$2p$	6
		0	0	$+1/2, -1/2$	$3s$	2
M	3	1	$+1, 0, -1$	$+1/2, -1/2$	$3p$	6
		2	$+2, +1, 0$			
			$-1, -2$	$+1/2, -1/2$	$3d$	10

1.4 Molecular Orbitals and Covalent Bonds

An exact analytical solution of the Schrodinger equation is possible only for simple systems. In order to obtain the energy eigenvalues and eigenfunctions of systems which have more than one electron,

a number of approximate methods are available. The molecular orbital method is one such approximate method wherein wave functions for the electrons in a molecule are constructed from atomic orbitals, i.e., the wave functions for single electrons in the atoms that form the molecule. When a linear combination of atomic orbitals is used to describe a molecular orbital then the method is called MO-LCAO approximation (Linear Combination of Atomic Orbitals).

Consider, for example, a hydrogen molecule where two hydrogen atoms A and B come together to form a molecule. When the atoms are far apart their electrons occupy the $1s$ orbital ($n=1, l=0$). But when they are brought together to form a molecule, the molecular orbital is given by

$$\Psi_{\text{MO}} = C_1\phi_A(1s) + C_2\phi_B(1s)$$

where ϕ_A and ϕ_B are atomic orbitals and $C_1 = \pm C_2$. Hence, we get

$$\Psi_1 = C_1\phi_A(1s) + C_1\phi_B(1s)$$

$$\Psi_2 = C_1\phi_A(1s) - C_1\phi_B(1s)$$

Ψ_1 is called an even orbital and Ψ_2 the odd orbital. In case of an even orbital the electrons are shared by the two nuclei while in an odd orbital there is no electron density between the nuclei. This orbital has higher energy (Figure 1.3). The even orbitals are bonding orbitals and the odd orbitals are anti-bonding orbitals. Note that if the hydrogen molecule is in the anti-bonding state, this does not mean that the electrons or the nuclei will fly apart. On the contrary, the bond between the two atoms remains, except that this is now unstable and of higher energy, and the molecule tends to make a transition to the lower energy bonding state. When the hydrogen molecule is formed, the bonding orbital is occupied by two electrons with anti-parallel spins. When the molecule is excited one of the electrons jumps to the anti-bonding orbital. As in the case of atoms, molecular orbitals also have associated quantum numbers. In a diatomic molecule like hydrogen, n is the principal quantum number and l is the quantum number that gives the angular momentum component along the inter-nuclear axis. $l = 0, 1, 2, \dots$ are called $\sigma, \pi, \delta, \dots$ orbitals, analogous to the quantum number l in atoms.

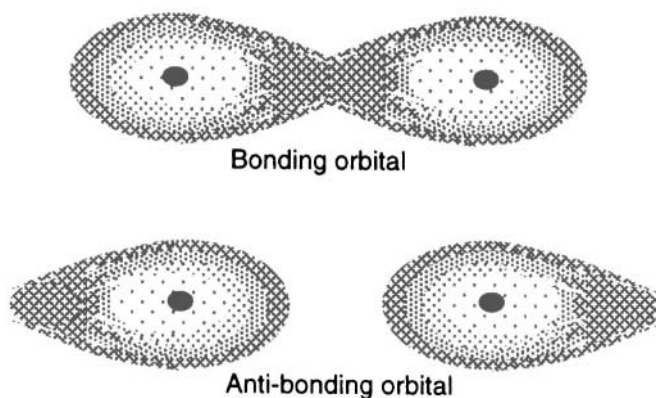


Fig. 1.3 Bonding and anti-bonding orbitals in a hydrogen molecule

In diatomic molecules, σ orbitals are symmetric with respect to the inter-nuclear axis and π orbitals are anti-symmetric. The higher orbitals are more complex (Figure 1.4). In multi-atomic molecules the atomic orbitals first combine to form a hybrid orbital before a molecular orbital is formed. Carbon, which is a pivotal atom in biology, is a classic example for the formation of hybrid orbitals. The electronic configuration of carbon is $1s^2 2s^2 2p^2$ (i.e. two electrons in the $1s$ orbital, 2 electrons in the

2s orbital and 2 electrons in the 2p orbital accounting for the six electrons in carbon). However a lower energy state is achieved if one of the electrons from the 2s level is promoted to a 2p state. Thus, the atom has four unpaired electrons—one in 2s orbital and three in 2p orbital. These four orbitals combine to form hybrid orbitals. The stable way of doing this is to combine the 2s orbital with each component p_x , p_y and p_z of the p orbital in the following way

$$\Psi_1 = 1/2 (\phi_s + \phi_{px} + \phi_{py} + \phi_{pz})$$

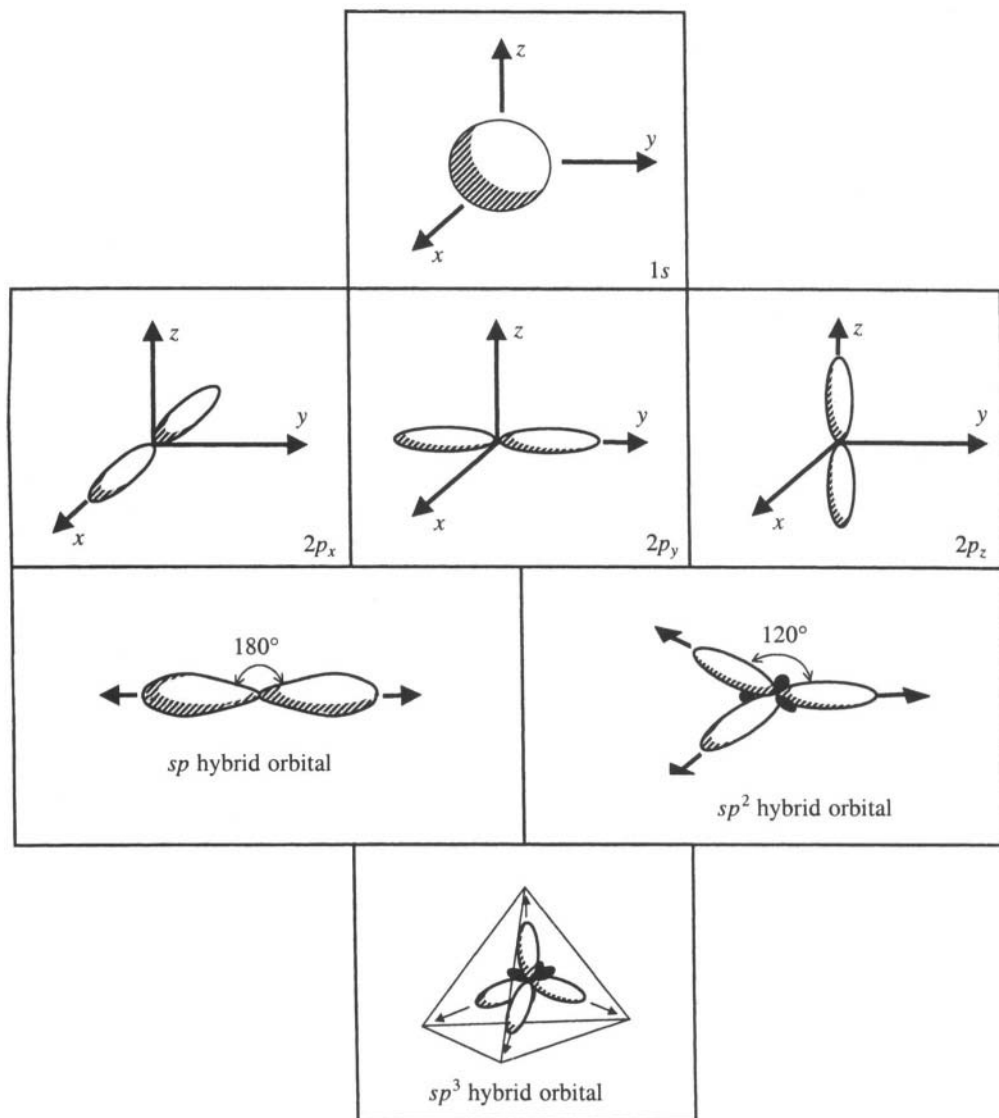


Fig. 1.4 Pure and hybrid atomic orbitals

$$\Psi_2 = 1/2 (\phi_s + \phi_{px} - \phi_{py} - \phi_{pz})$$

$$\Psi_3 = 1/2 (\phi_s - \phi_{px} + \phi_{py} - \phi_{pz})$$

$$\Psi_4 = 1/2 (\phi_s - \phi_{px} - \phi_{py} + \phi_{pz})$$

(p orbitals have a butterfly-like appearance with their nodal point at the nucleus. The p_x , p_y and p_z orbitals are at right angles to each other, see Figure 1.4). The resultant hybrid is known as the sp^3 orbital and points towards the four corners of a tetrahedron (Figure 1.4). When these hybrid orbitals combine with the $1s$ orbitals of four hydrogens, a methane molecule is formed through four σ bonds (covalent bonds). The strongest bonds are formed when the overlap of the orbitals is a maximum.

Other schemes of hybridization are also possible. For example, sp^2 hybrids, also known as trigonal hybrids, are formed when $2s$, $2p_x$, and $2p_y$ orbitals are mixed. This leads to three coplanar orbitals making an angle of 120° with each other (Figure 1.4). The p_z orbital is perpendicular to the plane. When a molecule such as ethylene (Figure 1.5) is formed, the sp^2 orbitals fuse with each other and with the $1s$ orbital of hydrogen to form localized σ orbitals. The p_z orbitals, which extend below and above the plane, also fuse together to form π orbitals. In benzene the sp^2 atomic orbitals of the six carbon atoms, fuse together to form six σ orbitals in a closed ring. The six electrons in these orbitals are not localized to a particular atom but move freely in the ring. The orbitals are therefore known as delocalized orbitals and the molecule is said to have conjugated double bond or resonance structure. If two $2p$ orbitals combine to form a π molecular orbital it could be a bonding orbital (π) or an anti-bonding orbital (π^*). Transitions from one molecular orbital to another, for example from π to π^* or from n to π^* are called $\pi\pi^*$ or $n\pi^*$ transitions respectively. As in the case of atoms, electronic transitions in molecules also lead to emission or absorption of light. Owing to the differences in the energy levels, $\pi\pi^*$ transitions are about 100 times more intense than $n\pi^*$ transitions.

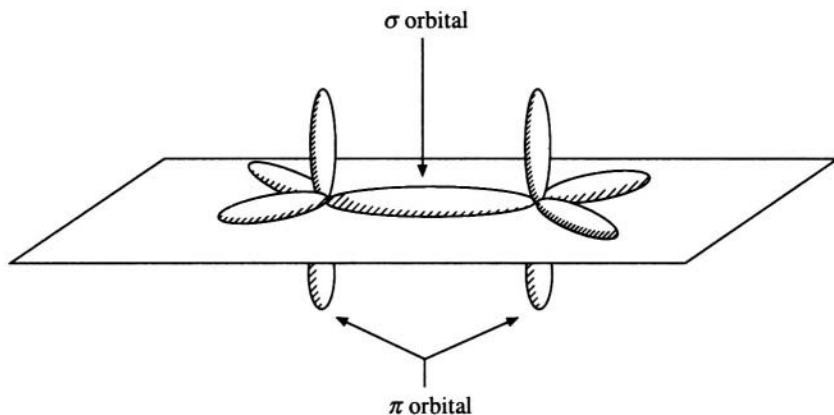


Fig. 1.5 The σ and π molecular orbitals

1.5 Molecular Interactions

The structure of a biological molecule determines its function. In turn, the forces between the atoms in a molecule determine its structure. The interactions between the atoms in the molecule are classified as strong or weak, depending on whether or not the interaction can be disrupted by weak forces like thermal motion. (It is customary to measure the strength of an interaction by the energy required to disrupt it.) Thus, for example, the primary structure of biological macromolecules is made of strong interactions such as the covalent peptide bond. Higher order structures like secondary, tertiary and

quaternary structures are governed by weak forces and can, therefore, be disturbed by a relatively small increase in temperature, or a change of pH, etc. Strong interactions are implicated mainly in the formation of the chemical structure, and, to some extent, also in the formation of the molecular structure. Weak interactions, on the other hand, not only help to determine the three-dimensional structure, but also are involved in the interactions between different molecules. Any interaction within a molecule or between molecules can be understood as a sum of the interactions between pairs of atoms. When two atoms come close to each other a potential energy is associated with the system. This potential energy is a minimum (largest negative value) when the force of attraction between the atoms equals the force of repulsion, and is zero when the two atoms are separated by an infinitely large distance (Figure 1.6). The system as whole tries to attain an equilibrium minimum energy state and the final equilibrium distance between the two interacting atoms is a resultant of all the different forces, which act on each of them. Some of these forces are described below.

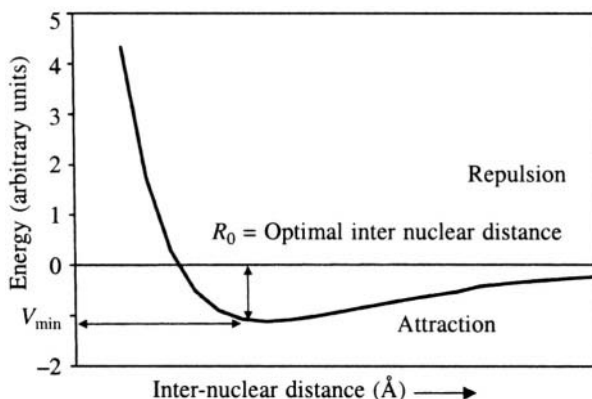


Fig. 1.6 Potential energy between two atoms plotted as a function of the inter-nuclear distance

1.5.1 Strong interactions

Covalent bonds, ionic bonds and metallic bonds are classified in this category. An ionic bond is formed due to electrostatic attraction between two oppositely charged ions. The structure of sodium chloride is an example of this type of bonding. In a crystal of common salt, every sodium ion is surrounded by six chlorine ions, and vice versa (Figure 1.7). Note that, strictly speaking, the entire crystal is one large ‘molecule’ of sodium chloride, consisting of sodium and chlorine atoms held together by ionic forces, and an isolated NaCl molecule has no independent existence¹. In such ionic compounds an electron is transferred from an electropositive to an electronegative element in order to achieve a stable electron configuration. In NaCl, sodium loses one electron and becomes an ion, **Na⁺**, with a single positive charge but a stable electronic configuration $1s^2 2s^2 2p^6$, with completely filled orbitals. In a similar fashion, chlorine acquires one electron to attain the same stable neon-like

¹When Lawrence Bragg determined the structure of NaCl by X-ray crystallography and suggested that NaCl did not exist as a single molecule by itself, he was subjected to much ridicule by chemists for daring to suggest that they give up their cherished prejudices. They said that as a physicist he should stick to subjects he understood, rather than speak what they thought was obvious nonsense. History, of course, proved Bragg right.

configuration and becomes Cl^- , again an ion, with a single negative charge. The positive ion and the negative ion come together to form a strong ionic bond and thence the crystal of sodium chloride.

Of the other strong interactions, covalent bonds have already been described. Metallic bonds are not relevant in biological systems and will not be further dealt with here.

1.5.2 Weak interactions

Hydrogen bonds and van der Waals interactions are classified in this category. Van der Waals forces (also known as residual forces) are weak forces, which hold together inert elements and molecules.

Most organic crystals, for example, are built of neutral molecules and this type of force plays an important role in their stability. Van der Waals forces act between all atoms and ions in all solids but the effect cannot be felt in the presence of strong interactions like covalent, ionic or metallic bonds. Van der Waals forces are basically electrostatic in nature in that they involve interactions between electric dipoles. When two bonded atoms have a difference in their ability to attract electrons (i.e. in their electronegativity), they form an electric dipole and hence possess a dipole moment. For example, in a hydroxyl group (OH) the oxygen atom draws the electron away from the hydrogen nucleus. There is thus a net (fractional) positive charge centered on the hydrogen nucleus, and a corresponding net (fractional) negative charge around the oxygen nucleus. In other words a dipole is formed (Figure 1.8). Note that the molecule as a whole is electrically neutral. The dipole moment may be a permanent feature arising from the molecular structure, if relatively electropositive atoms are bonded to relatively electronegative ones, and the dipoles of all the bonds do not cancel.

In such a case a residual dipole moment may exist independent of the environment of the molecule. Temporary dipoles may also arise by induction due to, for example, the presence of an external electric field. Both permanent and temporary dipoles arise in solids in which the molecules are polar. But in the case of inert elements such as helium, neon, etc., and non-polar, symmetric molecules, such as methane, hydrogen, etc., London proposed that a dispersion effect would lead to momentary polarisation of the charge, and hence a dipole. Even in the case of a highly symmetrical system with no permanent dipole moment, the electronic distribution at any instant of time may not be perfectly symmetrical, due to the constant motion of the electrons. This lack of symmetry gives rise to an instantaneous dipole moment. This instantaneous dipole moment of an atom or a molecule will polarise the neighbouring atoms or molecules, and a momentary attractive force arises between the two. This adds up to give a net attractive van der Waals force due to London dispersion. There are thus three components of the van der Waals force, viz., the permanent dipoles in the molecule, the dipoles induced by an external electric field, and the ones induced by the London dispersion effect. The

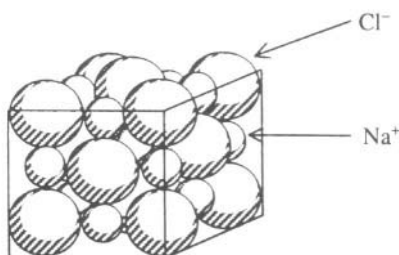


Fig. 1.7 Structure of crystalline sodium chloride salt with Na^+ and Cl^- ions placed alternately in a face-centered cubic cell.

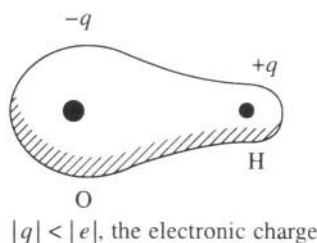


Fig. 1.8 The dipole moment in a hydroxyl group arising out of the difference in electronic density surrounding the two nuclei. The charge q is less than one full electronic charge.

relative contribution of each to the total force varies, and depends on the type of molecule. At the level of the individual atoms, others define the term ‘van der Waals radius’ as the distance of closest approach to the centre of the atom, in the presence of van der Waals forces alone. This radius can be experimentally determined by X-ray crystallography. In general, the van der Waals radius of an atom is larger than its covalent or ionic radius, showing that the effective radius of the atom within a molecule is different from (and less than) the distance of closest approach by other molecules or atoms.

Hydrogen bonds are weaker than covalent bonds but are stronger than van der Waals bonds. Hence the atoms connected by a hydrogen bond are separated by a distance (2.5 to 3.0 Å) greater than that of a covalent bond but shorter than if van der Waals forces alone were operative. Hydrogen bonds are generally found between two electronegative atoms. If X is an electronegative atom bonded to a hydrogen atom, it draws the electron of the hydrogen atom towards it, leaving the latter with a residual positive charge. The magnitude of this residual charge is less than one, since the electron is not completely stripped away from the nucleus, but only displaced more towards the nucleus of the atom X . The H ion, with a fractional positive charge, is now attracted by the electrons surrounding the second electronegative atom Y . As the hydrogen ion is almost devoid of its only electron, it can penetrate the electron cloud of atom Y till it is stopped by the repulsive forces of the nucleus of atom Y . Thus a bond is formed between atom X and atom Y , called the hydrogen bond. A hydrogen bond is generally written as $X-H \cdots Y$ since the H ion is asymmetrically disposed with respect to X and Y . It is particularly associated with X and loosely bonded to Y . Hydrogen bonds are among the most important of the interactions between biological molecules and confer specificity to the recognition of one molecule by the other. A prime example of this is formation of the highly specific A.T and G.C base pairs in DNA, which depends almost solely on the number and nature of the hydrogen bonds between the pairs.

1.6 Stereochemistry and Chirality

The three-dimensional spatial arrangement of the atoms of a molecule is termed its stereochemistry. A molecule may be identified first by its chemical formula, then by its chemical structure, and finally by its molecular structure. For example, C_2H_6O is the chemical formula of ethyl alcohol. CH_3-CH_2-OH is the chemical structure of ethyl alcohol. The molecular structure is determined by the three dimensional arrangement of the atoms in space. The word ‘configuration’ is used to describe the distinctive arrangement of various atoms or groups attached to a chiral centre (as described below). The word ‘conformation’ describes the different arrangements of atoms that are obtained when parts of the molecule are rotated about one of the bonds. A change in the configuration necessarily involves breaking at least one covalent bond and reforming it after a rearrangement of the atoms. A change in the conformation necessarily does not involve any breaking of covalent bonds. Groups that are connected by a single bond can undergo rotation leading to different relative orientations with respect to each other. For example, in ethyl alcohol, depending on the angle of rotation about the C–C bond, the relative orientations of hydrogen atoms and the hydroxyl group will vary. Energetically, the most favoured conformation about any bond is the ‘staggered’ one in which atoms attached at either end of the bond are at the largest possible distance from each other. An ‘eclipsed’ conformation increases steric contacts and is much less favourable. Arrangements of atoms or groups arrived at by rotation around a double bond are configurational isomers not conformational isomers. This is because the double bond does not allow free rotation. To produce the isomers, it becomes therefore necessary to break the double bond and then reform it after the rearrangement of the atoms. *cis-trans* isomers fall under this category and are observed in the case of the amino acid proline and in polypeptides. Linear

polypeptides consist almost exclusively of *trans* peptide units while cyclic peptides can have both *cis* and *trans* peptide units.

In some molecules the differences in the arrangement of the atoms may not affect many of the physical properties of the molecule such as melting point, density, refractive index, etc., but may affect the interaction of the different molecules with polarised light. Such molecules are said to be optically active and are called optical isomers. Optically active molecules rotate in different directions the plane of polarisation of the plane polarised light incident on them. Levo-rotatory molecules rotate the plane in the clockwise direction and dextro-rotatory molecules rotate it in the anticlockwise direction. These optically active molecules may differ in their biological behaviour. For example, only one of the optical isomers of asparagine is sweet to taste. Louis Pasteur showed that the difference in action between two optical isomers is due to the difference in molecular architecture, i.e. due to differences in the arrangement of the atoms in space. The two types of molecules are related by a mirror reflection and they cannot be superimposed. Objects of everyday life which are enantiomers (i.e. related by mirror symmetry) are the right hand and the left hand, righthanded and lefthanded screws, etc. The two alanine molecules shown in Figure 1.9 have the same number of atoms but differ in their stereochemistry and hence are known as stereoisomers. Compounds with four different substitutions at a carbon atom will exhibit optical activity. Such a carbon atom is generally known as an asymmetric carbon and a molecule containing such an atom is a chiral molecule. A mixture of two isomers, which have optical activity opposite to each other, is on the whole optically inactive since the effect of one is annulled by the other. Such a mixture is known as a racemic mixture.

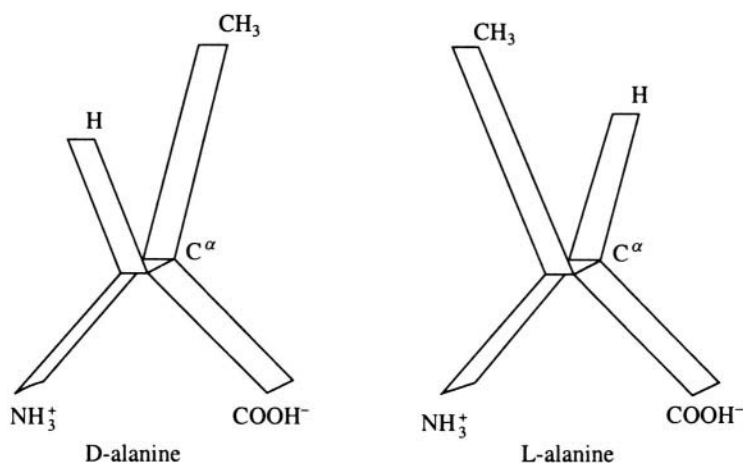


Fig. 1.9 A schematic representation of *D*-alanine and *L*-alanine

A pair of enantiomeric molecules will behave in the same way with respect to symmetrical environments or achiral chemical reagents. However, their interaction with other chiral molecules will differ. An example of this is the very specific interaction between a protein and its substrate. Stereoisomers not related by mirror symmetry are called diastereoisomers. Diastereoisomerism will occur in molecules with more than one chiral centre. Both configurational and conformational differences can lead to optical isomers— i.e. either enantiomers or diastereoisomers. Other than carbon, atoms like silicon with a valency of four can also act as chiral centres. Some molecules like hexahelicene, and polyglycine show optical activity though they have no chiral centre. This is due to the helical structures assumed by these molecules, which can be either right handed or left handed (Fig. 1.10).

All the major biological molecules, like amino acids, proteins, DNA, sugars and lipids, are optically active. Out of the two or more optical isomers available for these molecules only one kind is commonly found in biological systems. Optical activity and Life seem to be inseparable.

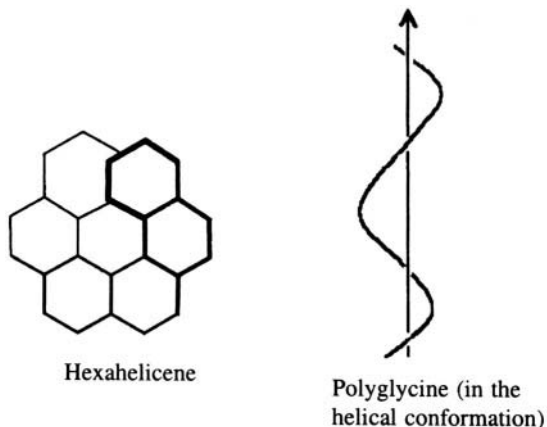


Fig. 1.10 Optically active molecules, but without a chiral centre

1.6.7 Stereochemical nomenclature

Molecules are three-dimensional entities. To represent them in two dimensions, i.e. on a piece of paper some standard methods are available. The most widely used are the Newman projection and the Fischer projection (Figure 1.11). In addition to these, there is another convention used to describe the arrangements of various groups around a chiral centre. This notation is known as the Cahn-Ingold-Prelog system and the enantiomers are called the [R] and the [S] configurations. The R-S convention is as follows: If the groups attached to the chiral centre are arranged according to the sequence rules (given below) as $a \rightarrow b \rightarrow c \rightarrow d$, then if the chiral centre is viewed from the side opposite to the lowest group (i.e. d in this case), and if $a \rightarrow b \rightarrow c$ occur in a clockwise direction, the molecule is in the [R] configuration. On the other hand if they occur in the anti-clockwise direction then it is in the [S] configuration (Figure 1.12). The sequence rules are as follows: (1) Atoms are arranged according to atomic number, e.g. $\text{Cl} > \text{O} > \text{N} > \text{C} > \text{H}$. (2) If the attached atoms are the same, then the next atom is used for ordering, e.g. $\text{CH}_3\text{---CH}_2\text{---CH}_2\text{---} > \text{CH}_3\text{---CH}_2\text{---} > \text{CH}_3\text{---}$. (3) If the substituents have atoms of the same rank then the one with more atoms of the highest rank gets preference, e.g. $(\text{CH}_3)_3\text{---C---} > (\text{CH}_3)_2\text{---CH---} > \text{CH}_3\text{---CH}_2\text{---}$ etc., and $\text{CH}_3\text{---OH---} > \text{CH}_3\text{---CH}_2\text{---}$ and so on. Since at each chiral atom there are two ways in which the attached groups can be arranged, if there are n chiral centers then there will be 2^n stereoisomers.

In the case of amino acids a different notation is used. In this molecule the C^α atom is the chiral centre. If we look down the H---C^α bond and if $\text{CO} \rightarrow \text{R} \rightarrow \text{N}$ is read in the clockwise direction, then it is the *L* configuration. If it is read in the anti-clockwise direction then it is the *D* configuration (Figure 1.9). All naturally occurring amino acids are in the *L* configuration.

1.7 Thermodynamics

Thermodynamics deals with energy, heat and work. Any system, physical or chemical or even biological, can be considered as a thermodynamical system and the laws of thermodynamics can be applied to it, if it utilizes energy in any form. To introduce some of the thermodynamic concepts, consider the

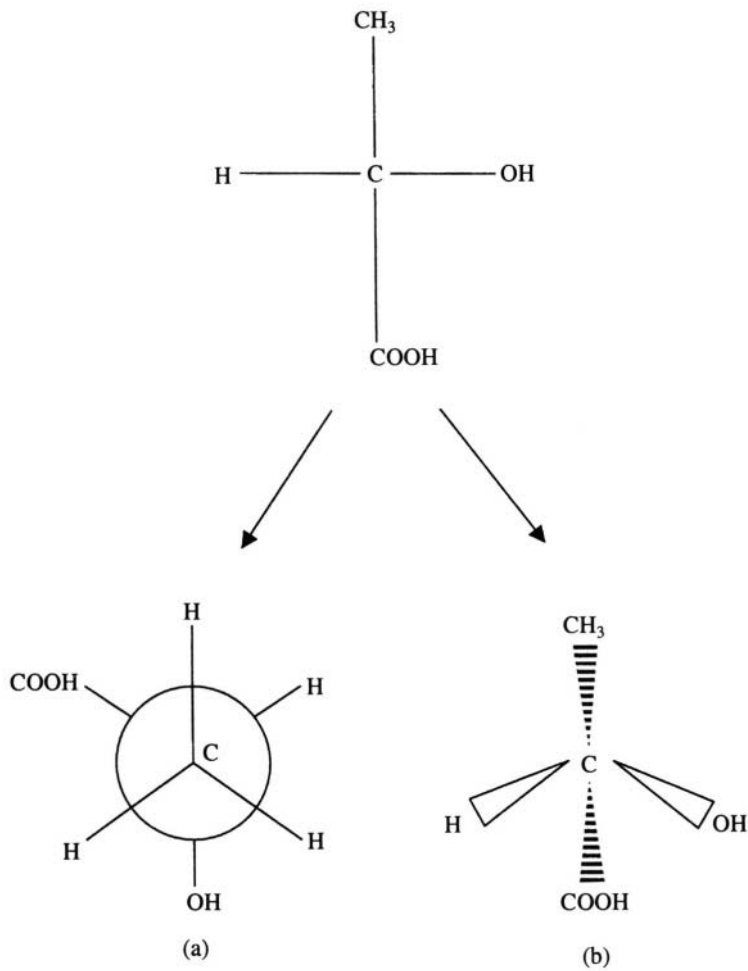


Fig. 1.11 (a) The Newman projection. (b) The Fischer convention

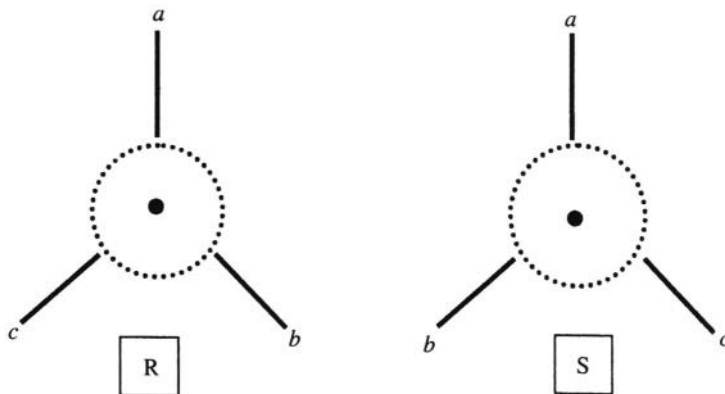


Fig. 1.12 The Cahn-Ingold-Prelog nomenclature system to distinguish R and S enantiomers

simple example of a glass of hot water on a table. If the glass is not disturbed in any way, the temperature of the water slowly decreases till it reaches the temperature of the room. This is achieved by transferring heat energy from the hot water to the environment. The glass of water is considered the system and the room is the environment. If T_s and T_e represent the temperature of the system and the environment respectively, then, if $T_s > T_e$, heat will flow from the system to the environment. If $T_s < T_e$, the flow of heat will be in the other direction. Heat energy Q is positive when heat flows from the environment into the system and is negative when the opposite happens. A system is said to be isolated when there is no exchange of energy or matter with the surroundings. When there is an exchange of energy but not of matter then the system is said to be closed. When both matter and energy can be freely exchanged with the environment, then the system is an open one.

In the previous example of the glass of water, we consider that heat, i.e. energy, but not matter, can be exchanged between the glass and the room. The system is a closed one and since initially the water is hotter than its surroundings, heat flows from the system to the environment, until $T_s = T_e$. At this point the system is said to be in equilibrium with the surroundings. The equilibrium in this case is a stable one and the temperatures remain equal, unless an external source (or sink) of heat disturbs one of them. The process by which the glass of hot water has attained this equilibrium with its environment is called an irreversible process. A process can be reversible or irreversible depending on whether or not it can trace back the same path to reach the initial state. A reversible process (considering the system and environment together) goes through infinitely small changes at an infinitely slow pace, and is always at equilibrium. Natural or biological processes almost always take place spontaneously and transfer energy from one part of the system to another in a short time, and hence are not thermodynamically reversible processes.

Thermodynamics requires a measurement of energy. The units of energy are calories or joules, depending on whether it is expressed as heat or as work. One calorie is defined as the amount of heat required to raise the temperature of 1 gm of water from 14.5 to 15.5°C. Work is generally defined as

$$W = F \times d$$

where F is the force applied on the body and d is its displacement under the application of this force. One joule is defined as the work done in moving one kilogram through one meter in the absence of any external forces. The work done by a system is taken to be positive, while work done on a system is negative. Work and heat are inter-convertible just as, in general, different forms of energy are inter-convertible. Thus one calorie equals 4.186 joules. Work and heat, unlike pressure, volume, mass, etc., are not intrinsic properties of a system.

A system can go from one state to another through an infinite number of different processes or paths. The values of Q and W for each one of these paths are different. However, the difference, $Q - W$ remains the same and is independent of the path taken. Thus

$$dU = dQ - dW = U_f - U_i \quad (1.1)$$

where dU is the change in the internal energy, an intrinsic property of the system. Equation (1.1) is the first law of thermodynamics. It can be paraphrased as 'energy can neither be created nor destroyed'.

The first law tells us about the energy balance but not about the direction in which the process proceeds. The fact that every thermodynamic process always goes in a particular direction may be illustrated as follows. Consider an iron rod lying on a table. It will never happen, spontaneously, that all the heat energy in the rod gathers together at one end making that end of the rod hotter than the other. On the other hand, if there is an imbalance in the heat at the two ends of the rod to begin with,

then it always happens that, as time proceeds, the difference in temperature between the two ends becomes smaller and smaller and the heat at both ends tends to equalize. The above process therefore has a particular direction. This intuitive concept is codified as the second law of thermodynamics, which makes it possible to predict the direction of the natural process. This law is stated in many different ways. One of the conventional statements is “It is not possible to convert heat completely to work with no other change taking place”. It is also stated in terms of heat flow, as follows: “It is impossible for heat to flow from one body to another which is at a higher temperature, with no other change taking place”.

The third law of thermodynamics states that the entropy of a pure solid or liquid is zero at the absolute zero temperature. It is sometimes stated as follows – “It is impossible to attain absolute zero temperature through a finite number of operations.”

There are several thermodynamic variables that are used to describe a thermodynamic system. Some of them are described in detail below.

1.7.1 Entropy

Entropy is a variable used in a precise statement of the second law of thermodynamics. It is defined as a thermodynamic function whose change is independent of the path of transformation of the system and is given by

$$dS = \delta Q/T \quad (1.2)$$

i.e. the change in entropy dS is measured as the ratio of the change in the heat energy δQ to the temperature at which the change takes place in a reversible process. The difference in entropy between any two states can be obtained by integrating the above equation

$$dS = S_f - S_i = \int dQ/T$$

where the integration is carried out over the reversible path of the process. S_f and S_i are the final and initial states of the system respectively. Thermodynamic equations most often involve the difference in entropy between the initial and final states of the system, rather than the absolute entropy of a system and it is this quantity which is of interest in almost all the cases. This difference is independent of the path taken during the process. Entropy (S), like temperature (T) and internal energy (U), is an intrinsic property of the system and, like T and U , is called a state function describing the state of the system. In the case of a reversible process, the change in entropy is calculated in a straightforward way, using the above equations. In the case of an irreversible process, the change in entropy between two equilibrium states is calculated by finding a reversible path between the two states and calculating the entropy change for that path. The second law of thermodynamics can be restated to say that the entropy of an irreversible process in an isolated system always increases, i.e.

$$\Delta S = S_f - S_i > 0$$

Thus naturally occurring spontaneous irreversible processes, such as biological processes, always result in an increase in entropy. Boltzmann showed that entropy could be defined in such a way that it will be a measure of order in a system

$$S = k \log (w)$$

where k is the Boltzmann's constant and w the number of configurations a system can take. A larger number of configurations imply a less ordered system. Thus to state that a spontaneous process moves toward the maximum entropy is the same as saying that the system will move toward maximum disorder or randomness.

1.7.2 Enthalpy

The enthalpy of a system H is defined as

$$H = U + PV$$

where U is the internal energy, P the pressure and V the volume. P and V are intrinsic properties and hence thermodynamic parameters. Their product is expressed in units of energy. Therefore enthalpy is also expressed in units of energy and is known as the heat content of a system. Since H is a combination of state functions and parameters that are independent of the path of transformation of the system, enthalpy too is independent of the path. The change in enthalpy is obtained by considering the differential form of the equation, which is

$$dH = dU + PdV + VdP$$

At constant pressure $VdP = 0$. Therefore,

$$dH = dU + PdV \quad (1.3)$$

But from the first law of thermodynamics we know that

$$dQ = dU + dW$$

where W , the work done, equals PdV (at constant pressure). Therefore

$$dQ = dU + PdV = dH \quad (1.4)$$

Thus enthalpy is an intrinsic property of the system. Its increase is equal to the amount of heat absorbed from the environment at constant pressure. A reaction is exothermic when heat is let out to the surroundings and the enthalpy change is then negative. On the other hand, in an endothermic reaction heat is absorbed from the surroundings, and the enthalpy change is positive.

1.7.3 The free energy of a system

A linear combination of thermodynamic state functions such as internal energy U and entropy S is also a state function. Such combinations are known as thermodynamic potentials. Enthalpy, defined above, is one such potential. The Helmholtz free energy is another, used widely in biology. It is defined as

$$F = U - TS$$

The change in free energy is thus

$$dF = dU - TdS - SdT$$

The physical meaning of this differential equation is that a change in free energy will involve a change in internal energy, in temperature and in entropy. At constant temperature, i.e. for an isothermal process, $dT = 0$. In that case

$$dF = dU - TdS \quad (1.5)$$

Now the first law of thermodynamics states that

$$dU = dQ - dW$$

But, from equation (1.2)

$$dQ = TdS$$

Therefore

$$dU = TdS - dW \quad (1.6)$$

Now combining (1.5) and (1.6) we have

$$-dF = dW \quad (1.7)$$

Therefore the work done under isothermal conditions always leads to a decrease in the Helmholtz free energy (as inferred from the negative sign of dF). It is also seen from the above relations that when the system is in an equilibrium state, the entropy is at a maximum and the free energy is at a minimum.

Another thermodynamic potential function that is used to describe biological processes is Gibb's free energy and is defined as

$$G = U - TS + PV$$

Under constant pressure and temperature (i.e., $dT = 0$, $dP = 0$) the change in this potential is given by the appropriate differential form of the equation,

$$dG = dU - TdS + PdV = dH - TdS \quad (1.8)$$

For reversible processes, using equation (1.6), we have

$$-dW_{\max} = dU - TdS$$

since the maximum work is done in such processes. Thus

$$dG = -dW_{\max} + PdV$$

Therefore

$$-dG = dW_{\max} - PdV = dW_{\text{net}}$$

At equilibrium dG , dS and dF are all zero. Once again the decrease in free energy is a measure of the useful work done under conditions of constant pressure and temperature.

1.7.4 Chemical potential

In a closed system the mass and chemical composition of the constituents remain constant. But in an open system, such as a biological process, the mass and chemical composition may change. Hence, thermodynamic quantities like U , S , F etc. will also depend on the quantities of chemicals present in the system. Let n_1 , n_2 , n_3 ... be the quantities of the 1st, 2nd, 3rd ... components. Then the change in free energy by the addition of 1 mole of component i to the system at constant pressure and temperature, when the quantities of the other components remain unaltered, is given by

$$\mu_i = \left(-\frac{\partial G}{\partial n_i} \right)_{T, P, n_j, j \neq i}$$

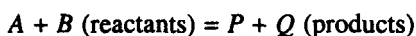
μ_i is known as the chemical potential of the component i . Generally n_1 , n_2 , n_3 ... are expressed in gram-moles. For a one component system, the chemical potential is nothing but the Gibb's free energy measured per mole of the component. For a multi-component system the total Gibb's free energy

$$G = \sum_i \mu_i n_i$$

The free energy of a system cannot be defined as an absolute number, since what is measured is only the change in the free energy during a process. Therefore a relative scale is defined for every element in its stable state, such that the free energy is zero at a temperature of 25°C and a pressure of 1 atmosphere. The free energy of a compound is therefore nothing but the change in free energy due to its formation from its elements. The change in the standard free energy in the course of a reaction (denoted ΔG°) can be computed from the standard free energy of formation of reactants and products.

$$\Delta G^\circ = -RT \log K$$

where K is an equilibrium constant, R the gas constant and T the temperature. For the reaction



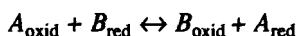
if C_A, C_B, C_P, C_Q are the concentrations of the components, then

$$K = (C_A C_B) / (C_P C_Q)$$

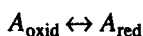
When ΔG° is negative, i.e. if the concentration of the products is higher than that of the reactants at equilibrium, then the reaction will proceed spontaneously in the direction of the products and is exothermic. If it is positive, it requires energy for the reaction to go in the direction of the products i.e. the reaction is endothermic. Thus standard free energy change is an indicator of the direction of the reaction. However, it does not quantify the rate of reaction.

1.7.5 Oxidation- reduction potential

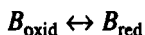
Oxidation-reduction reactions, which are very common in biochemical processes, involve the transport of electrons. A substance is said to be 'oxidized' when it loses electrons and said to be 'reduced' when it gains electrons. A biological reaction takes place in an aqueous medium and hence the reactant can pick up a proton from the surrounding medium and become reduced. Or it can bind to an oxygen atom and become oxidized. Consider the following reaction.



Since in an oxidation-reduction reaction there is an exchange of electrons, the free energy of the reaction is expressed in units of electric potential. The above equation can be split as follows:



where A_{oxid} and A_{red} are called a redox couple. Similarly,



The change in free energy for one-half of the reaction is

$$\Delta G^\circ = \Delta G^\circ + RT \log (A_{\text{oxid}}/A_{\text{red}}) = \Delta G^\circ + RT \log (\alpha/(1-\alpha))$$

where α is the fraction of A that is oxidised. Converting this to electrochemical potential, we obtain

$$E_A = E_A^\circ - (RT/ZF) \log (\alpha/(1 - \alpha)) \quad (1.9)$$

where E is the oxidation-reduction (redox) potential, F is Faraday constant, and Z the number of transported electrons in the reaction. The redox potential is measured against a standard, which is generally a hydrogen electrode. The potential E measured while inserting the hydrogen electrode in a 1 N acid solution (i.e., pH = 0) is taken to be zero. Against this standard, the redox potential of other substances can be measured. As biological processes take place at pH = 7 (i.e. neutral pH) it is more

convenient to use the redox potential measured at this value as the standard. For such a measurement equation (1.9) is generally written as

$$E_R = E_m - (RT/ZF) \log (\alpha/(1 - \alpha))$$

where E_R is the redox potential measured against a standardized electrode and E_m the midpoint potential, measured when half of the substance is in the oxidized state and the other half is in the reduced state. A strong reducing agent (such as NADH) has a negative redox potential while a strong oxidizing agent (such as O_2) has a positive redox potential.

1.8 Radioactivity

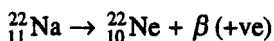
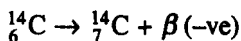
An atom consists of a nucleus surrounded by electrons. The nucleus in turn consists of protons and neutrons. The atomic number Z of the atom is the number of electrons present, which is also equal to the number of protons as the atom is electrically neutral. The mass number A of the atom is given by the sum of neutrons and protons, i.e. $A = Z + N$, where Z = atomic number and N = number of neutrons. Atoms with the same atomic number but different mass numbers are known as isotopes. The number of isotopes for a given element may vary from two to as many as 20. For example, hydrogen has three isotopes, ^1_1H (hydrogen), ^2_1H (deuterium) and ^3_1H (tritium). The superscript stands for the mass number. In general an element is represented ^A_ZZ , where A is the mass number and Z the atomic number. In an atomic nucleus when the protons and neutrons are equal in number, the isotope is generally a stable one. However, elements with higher atomic numbers frequently have a neutron to proton ratio of more than one. These isotopes are unstable and are known as radioisotopes. Radioisotopes are found in nature (e.g. uranium, plutonium, thorium and radium) though they are more often made artificially. Radioactive isotopes emit particles and/or electromagnetic radiation to become a stable isotope. This process is known as radioactive decay. A neutron can emit an electron (also known as a negatron to indicate its nuclear origin) to become a proton



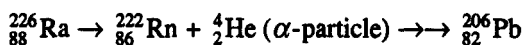
Similarly, a proton can become a neutron by emitting a positron. A positron has the same mass as an electron but is positively charged.



Both the above processes alter the N/Z ratio. For example



Sometimes the term 'beta particle' is used to denote either a positron or a negatron. Radioactive decay can also result in the emission of α particles and γ -rays. An α particle is a helium nucleus (two protons and two neutrons). The emission of an α particle leads to a reduction in mass number by 4 and atomic number by 2, e.g.



As the product of the first decay, radon, is also radioactive, it initiates a further decay series which finally leads to the formation of lead. γ -ray emission does not lead to any change in atomic number or mass. γ -rays are electromagnetic waves at the extreme short wavelength, high frequency end of the

spectrum. **γ -ray** emission is frequently accompanied by **α** and **β** particle emission. **γ -ray** emission arises when the excited nucleus returns to its ground state.

1.8.1 Rate of radioactive decay

Radioactive decay always occurs at a specific rate which is characteristic of the source and follows an exponential law. If we consider a sample containing N nuclei then the decay in a given time will be proportional to the number of nuclei available for decay. The rate of change in the number N of radioactive atoms is given by the differential equation

$$-\delta N / \delta t = \lambda N$$

where λ is the proportionality constant or the decay constant.

$$\lambda = -(\delta N / N) / \delta t$$

λ may also be considered the fractional rate of decay during the short period of time in which N may be treated as a constant. Integrating this equation over time t , we get

$$\log (N_t / N_0) = -\lambda t$$

or

$$N_t = N_0 \exp (-\lambda t)$$

where N_t is the number of radioactive atoms present at time t and N_0 = the number present at time $t = 0$ (i.e. the initial number). From the last equation we see that there will always be some parent nuclei present after any finite time, however long. It is therefore usual to specify the half-life of a radioactive material rather than its lifetime. The half-life of a radioactive nuclide is defined as the time taken for one half quantity of the sample to decay from the parent nuclide to the daughter nuclide (Figure 1.13). Substituting this definition into the above equation we have $N_t = 0.5N_0$. Thus half-life is given by

$$T^{1/2} = \log 2 / \lambda$$

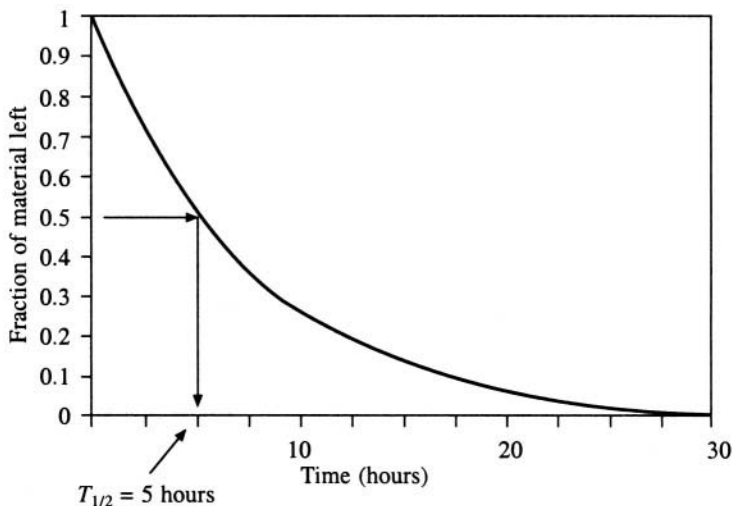


Fig. 1.13 Radioactive decay curve with decay constant $\lambda = 0.1386$, corresponding to a half-life of 5 hours.

The half-lives of radioactive materials vary from 10^{19} years to 10^{-7} secs. The half-lives of some of the isotopes are:

^3H — 12.26 years	^{32}P — 14.20 days
^{14}C — 5760 years	^{42}K — 12.40 hours

1.8.2 *Measurement of radioactivity*

Apart from the half-life, radioactive decay can also be expressed as disintegrations per minute (dpm) or per second (dps). The unit of radioactivity used is curie Ci and is defined as that quantity of radioactive substance in which the number of nuclear disintegrations is equal to that in 1 gram of radium. Thus $1 \text{ curie} = 3.7 \times 10^{10} \text{ dps}$. Millicurie ($1 \text{ mCi} = 1/1000$ of a curie) and microcurie ($1 \mu\text{Ci} = 1/1000000$ of a curie) are the units generally used in biological studies. Specific activity is the amount of radioactivity per unit mass of the sample. Hence, the unit of specific activity is mCi/mg or $\mu\text{Ci/mg}$. The amount of radiation that falls on a sample can be measured in roentgens or in rads. Roentgen is generally used in cases where electromagnetic radiation is involved, while rad can be used both for radiation and particles. One rad is defined as the dose required to produce 100 ergs of absorbed energy per gram of target. A dose of one rad can increase the temperature of one gram of material by $2 \times 10^{-6} \text{ K}$.

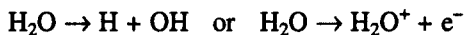
Radioactivity can be measured by counting devices or by photographic methods. In both cases advantage is taken of the ability of the radioactive emission to interact with matter. Geiger-Muller counters are based upon the ability of the radioactive emission to ionize matter. Scintillation counters are based on its ability to excite an atom leading to emission of visible light as the atom returns to the ground state. The photographic method is advantageous when we require a permanent record of the experiment conducted. In this method the ionizing radiation from a radioactive source interacts with a photographic emulsion (silver halides) and leaves an image. Autoradiography, which uses photographic method, is a sensitive technique by which the distribution of radioactivity in a biological specimen can be determined, for example, the sites of localization of a radiolabeled drug in the body.

1.8.3 *Effects of radioactivity on matter*

α particles ($^4\text{He}^{2+}$), due to their size, are slow and interact with atoms causing ionization and excitation. Ionization takes place when an outer electron of an atom is completely removed. Excitation takes place when the electrons are raised to a higher excited orbital. Though the initial energy of alpha particles is reasonably high (3 to 8 Mev) their energy is dissipated very quickly and hence they are not very penetrating. β particles with a negative charge (negatrons) are small compared to α -particles and hence their interaction with matter is very much less. They are thus more penetrating. They are rapidly moving particles causing ionization and excitation, though less than α -particles. γ -rays are electromagnetic radiation and carry no charge or mass. They travel very large distances without appreciable loss of energy, and are thus highly penetrating. γ -rays also interact with matter and produce secondary electrons. A low energy γ -ray can transfer all its energy to an orbital electron, which is then ejected out of the atom as a photoelectron. This phenomenon is known as photoelectric absorption of the radiation. A medium energy γ -ray transfers only a part of its energy to an orbital electron (which is then ejected) and proceeds further with reduced energy. The change in energy between the impinging photon and the emitted photon can be measured as a change in its wavelength and is known as the Compton effect. A high energy γ -ray, when it interacts with a nucleus, transfers all its energy to it and excites it. The excited nucleus in turn emits another photon of the same or lesser energy. The photon may then spontaneously decay to produce a positron-negatron pair in a process known as 'pair-production'.

1.8.4 Biological effects of radiation

Ionizing radiation can affect both somatic and sex cells. The somatic effects depend on the amount of radiation and it can cause reddening of the skin, loss of hair, etc. It is also known that radiation can cause carcinogenesis. Further, biological specimens consist of a large amount of water, and irradiation of these can lead to hydrolysis



Unlike genetic damage, which is caused by even very small amounts of radioactivity, serious somatic effects occur only with exposure to higher doses. Even so, a dose of about 500 rads can immediately kill a human being. Further, irradiation can release free electrons which in turn can produce secondary electrons by collision. These free electrons can cause a great deal of damage in the cell. Genetic effects occur when the reproductive cells are irradiated, leading to mutations in DNA. The damage to DNA due to irradiation can be in the form of breaks in the strands or alterations in the base sequence. For a given dose, mRNA is the most vulnerable, while DNA synthesis itself is the least affected. Thus the biological effects of radioactivity could be disastrous to life.

1.8.5 Applications of radio isotopes

³H, ¹⁴C and ³²P isotopes are used in autoradiography of biological samples. Radioactive isotopes used in biological experiments are known as radiotracers. The greatest advantage of the radiotracer method is its extreme sensitivity. Metabolites that occur in very low concentrations in the system can be identified and therefore, radiotracer experiments are frequently performed *in vivo*. Apart from their use in tracing metabolic pathways, in pharmacology radioisotopes are used to study drug accumulation, the rate of metabolism of the drug, and also the metabolic products of the drug under consideration. On the analytical side, enzyme substrate binding studies can be carried out using radio labeled substrates. Radioimmunoassay is yet another area where radioisotopes are used. Radioisotopes have other biological applications like sterilization of food, clinical diagnosis, ecological studies etc. Isotopes like ⁶⁰Co, ²²⁶Ra, ²²²Rn are used in radiation therapy of cancer.

Separation Techniques

2.1 Introduction

The identification, purification and separation of the components of a mixture of molecules is of considerable importance in molecular biology and biophysics. There are several techniques available for this purpose and the two most frequently used described in this chapter are: Chromatography and Electrophoresis.

2.2 Chromatography

Chromatographic techniques can be classified into four main categories based on the type of molecular interactions used for separation, viz. ion-exchange chromatography, adsorption chromatography, partition chromatography and molecular exclusion chromatography. The techniques can also be classified in another way, on the basis of the mode of separation and the support system used, as column chromatography, thin layer chromatography and paper chromatography. The particular technique that is used for a given sample depends on its characteristics. Sometimes, several techniques may be used one after the other to obtain extra pure sample.

The basic principle behind all the different types of chromatography is however the same. Chromatographic separation takes advantage of the fact that the sample distributes or partitions itself to different extents in two different, immiscible phases. This property is described by the partition or distribution coefficient K_d . If we consider two immiscible phases A and B,

$$K_d = (\text{Concentration of the sample in phase A})/(\text{Concentration in phase B}).$$

The effective distribution is then defined as the total amount of the substance present in phase A divided by the total amount of substance present in phase B. The two immiscible phases could be a solid and a liquid, or a gas and a liquid, or a liquid and another liquid. Generally, one of the two phases is a stationary phase and does not move. The other is a mobile phase and moves with respect

to the first. To understand how chromatography is used to separate and purify molecules, consider the column mode, with a stationary solid phase and a mobile liquid phase. To separate a mixture of molecules, a column is packed with the granular stationary phase to a height of a few centimetres and is then surrounded by the liquid phase. Assume that the column diameter is such that 1 cm^3 of liquid will surround 1 cm length of the column material. First consider the behaviour of a solvent containing a single solute. Let $32 \mu\text{g}$ of the sample in 1 cm^3 solvent be added to the column. It would occupy position A in Figure 2.1. If the effective distribution coefficient as defined above is 1.0, the solute would distribute itself equally between the liquid phase (the solvent) and the solid phase (the support material in the column). Now, if another 1 cm^3 of the solvent (this is also sometimes more properly called the 'eluent', especially when it is particularly used to elute or bring out the sample) is added to the column, half the solute will move from position A to position B, since the distribution coefficient is 1.0. There will now be $16 \mu\text{g}$ of sample each at positions A and B. If yet another 1 cm^3 of the solvent is added to the column, the situation will be as shown in the third panel of Figure 2.1, with $8 \mu\text{g}$ in position A, $16 \mu\text{g}$ in position B and $8 \mu\text{g}$ in position C. When this process is repeated, the pattern shown in the other panels in Figure 2.1 will be observed. At every step of the equilibration, 50% of the sample will be left bound to the solid phase at each position. After five equilibration steps the sample is present all through the column but has a maximum concentration at the centre of the column. If the solvent contains two solutes with different partition coefficients, at each step the amount in the solid phase will be different for the two. After a number of such steps the peaks of concentration for the two components will be at different positions along the column, thus achieving a separation of the mixture. If the process is allowed to continue sufficiently long, the different components of the mixture will flow out of the column at different times and can be collected separately. This is the basic principle of chromatography. We now consider different applications of this principle.

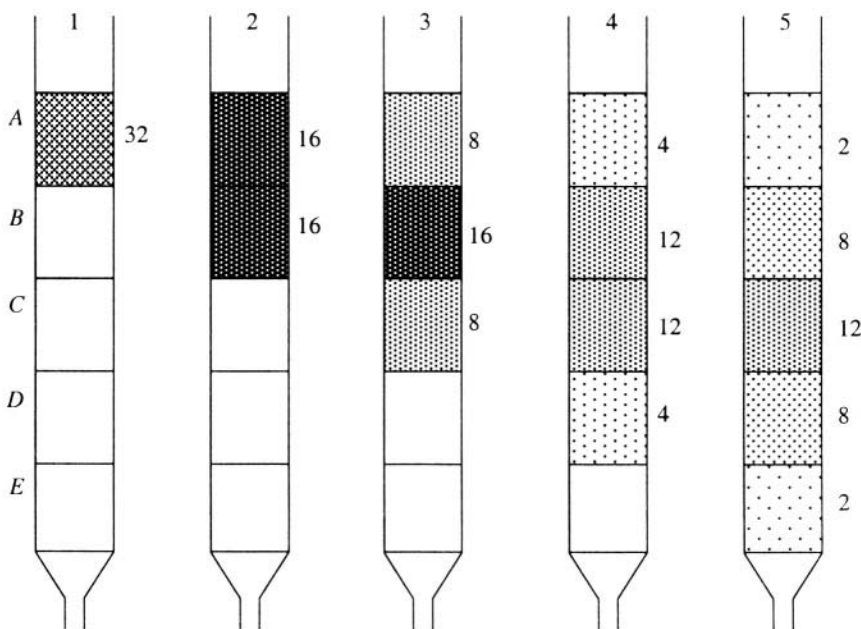


Fig. 2.1 The column chromatographic separation

2.2.1 Column chromatography

Adsorption chromatography, partition chromatography, ion exchange chromatography, exclusion chromatography and affinity chromatography all use the column mode (Figure 2.2), where the stationary phase is packed into glass or metal columns. The nature of the stationary phase depends upon the particular form of chromatography. It is generally an adsorbent, in other words, a resin or a gel.

2.2.2 Thin layer chromatography

In this case, the stationary phase is applied to a glass, plastic or metal foil plate as a uniform, thin layer and the sample is applied at the top edge of the layer using a micro-pipette or syringe. This technique can be used for partition chromatography, adsorption chromatography and exclusion chromatography. The advantage of this method is that a large number of samples can be studied simultaneously. Also, it is quick and simple.

2.2.3 Paper chromatography

In this technique, the cellulose fibres of a sheet of special chromatographic paper act as a support or the stationary phase (Figure 2.3). There are generally two methods that are employed in paper chromatography – the ascending method, and the descending method. In both of them the solvent is placed at the bottom of a jar or tank so that the chamber is saturated with solvent vapour. The chromatographic paper is held vertically inside the tank. In the ascending method, the lower end of the paper dips into the solvent and the sample spot is applied just above the surface of the solvent.

As the solvent moves up the sheet of paper by capillary action, the sample constituents move along with it and get separated. In the descending method, the upper end of the paper, where sample is

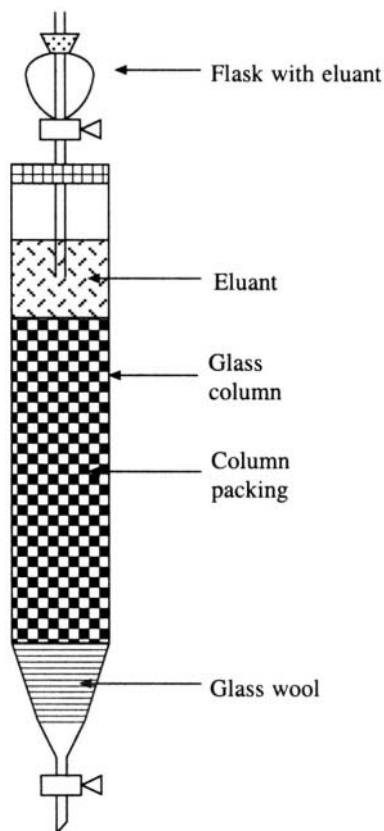


Fig. 2.2 Column mode of chromatography

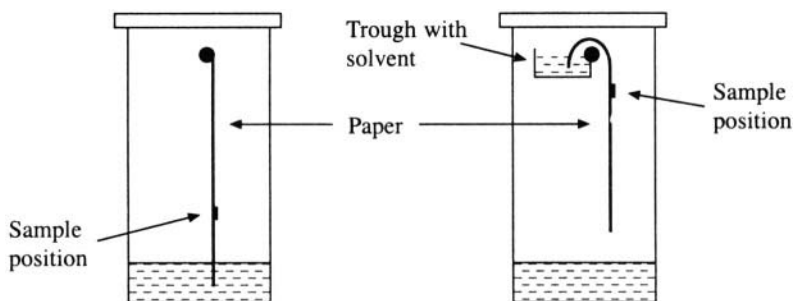


Fig. 2.3 Two types of paper chromatography

applied, is held in a trough of solvent at the top of the tank, and the rest of the paper is allowed to hang down, without touching the solvent at the bottom of the tank. Again by capillary action (and gravity), the solvent moves down the sheet of paper, taking the sample along with it and separating the components. The components are detected by spraying the paper with specific colouring reagents, for example, ninhydrin, which binds only to amino acids and proteins. Sometimes fluorescent dyes are used. For example, ethidium bromide is used with DNA samples. When the paper is subsequently examined under ultraviolet light, the position of the fluorescent or ultraviolet absorbing spots can be seen. The corresponding compounds are identified on the basis of their R_f values where

$$R_f = (\text{Distance moved by solute from the origin} / \text{Distance moved by solvent}).$$

The R_f value is the relative displacement of solute with respect to the solvent front and is a constant for a given compound under standard conditions. Thus by looking up a table of R_f values, it is possible to identify the compound after separation.

As mentioned earlier, chromatographic techniques are also classified on the basis of the molecular property of the support system or sample that is used in the separation. Some of these are considered in greater detail below.

2.2.4 Adsorption chromatography

This technique separates the sample into its components based on the degree of their adsorption² by the adsorbent and on their solubility in the solvent used. Silicic acid (silica gel), aluminum oxide, calcium carbonate and cellulose may be used as the stationary phase in this method. The optimum combination of adsorbent and eluting solvent will depend upon the type of sample separation under consideration. For example, hydroxyapatite (calcium phosphate) is used for DNA separation as this adsorbent can bind double stranded DNA but not single stranded DNA or RNA. Normally any organic solvent can be chosen as the mobile phase, e.g. hexane, heptane, propanol, butanol etc.

2.2.5 Partition chromatography

There are two types of partition chromatography. (a) Liquid-liquid chromatography: In this type of partition chromatography, the material separation is achieved based on its differing solubilities in two (immiscible) liquid phases. The sample is dissolved in a relatively non-polar solvent and is passed through a column that contains immobilised water or some other polar solvent in a supporting matrix such as silica gel. When the sample moves through the column it gets partitioned between the mobile phase and the immobile, polar phase. In the reverse phase liquid-liquid chromatographic method, the stationary phase is a non-polar substance like liquid paraffin supported by the matrix. High Performance Liquid Chromatography (HPLC) is a liquid-liquid partition chromatographic technique. It is widely used for the separation of biological compounds either as a normal phase method, or as a reverse phase method. In the latter, it is useful for separating polar compounds. (b) Counter current chromatography: This is also partition chromatography with this major difference, that there is no supporting matrix. The sample is equilibrated between two immiscible solvents (e.g. ethylacetate and water) in a test tube. The partition coefficient is defined as

$$K = (\text{Concentration } C_1 \text{ in the upper phase} / \text{Concentration } C_2 \text{ in the lower phase})$$

After equilibration the upper phase is withdrawn and transferred to a fresh volume of lower phase

²Adsorption is the process by which a substance is absorbed on to the surface of another. There is almost no penetration beyond a few molecular layers.

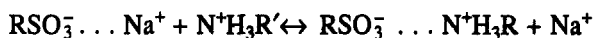
solvent in another tube. The original lower phase is added to a fresh volume of upper phase solvent, and so on. The process is repeated several times, yielding several test tubes, half of them finally containing the upper phase solvent, and half containing the lower phase solvent. A solute which is more soluble in upper phase ($K > 1$) will concentrate in tubes containing the upper phase solvent. If the solubility is greater for lower phase ($K < 1$) then it will concentrate in tubes of the lower phase solvent. Counter current chromatography is successfully used for cell organelle fractionation.

2.2.6 Gas liquid chromatography (GLC)

The two phases used are a liquid and a gas. The method is very sensitive and reproducible. The partition coefficient of a volatile sample between the liquid and the gas phases, as it is carried through the column by the carrier gas, may be largely different from unity. This feature is exploited to separate or purify the compound. The stationary or immobile phase is generally nonvolatile and thermally stable at the temperature of the experiment. Usually organic compounds with high boiling point are coated on a base to form the stationary phase. The base is generally made up of celite (diatomaceous silica). The sample to be separated is packed in a narrow, coiled, steel or glass tube and placed in an oven at an elevated temperature. An inert gas like argon, nitrogen or helium is passed through the tube. The sample is volatilised and is carried away by the mobile gas phase. It is passed over the solid phase packed into another column, and as it does so, the sample distributes itself between the gas and solid phases.

2.2.7 Ion exchange chromatography

Many biological samples like amino acids and proteins have ionisable groups. The samples can therefore be separated based on the net charge acquired by them at a certain pH. The ion exchanger is a resin and can be either a cationic exchanger or an anionic exchanger. The cationic exchanger has negatively charged groups and attracts positively charged molecules. The anionic exchanger has positively charged groups and attracts negatively charged molecules. Commercially available ion exchangers are made by co-polymerizing styrene with divinyl benzene. The two materials become cross-linked, with the permeability of the resin varying with the degree of cross-linking. When this insoluble, cross-linked resin is placed in water, a network of charged groups is formed. When the sample, suspended in the liquid phase, comes in contact with the resin, the charged groups in the sample replace the charged ions on the surface of the resin and the sample becomes strongly bound to the resin. As an example, suppose that the sodium salt of a strongly acidic resin, which contains SO_3^- groups, is placed in a suspension. Normally the Na^+ ions would be bound to the negative sulphate groups in the resin. However, on exposure to the sample, the positively charged groups present in the sample may displace some of the Na^+ ions from the surface. This exchange of ions at the exchange site leads to the distribution of the sample between the solid and liquid phases. The higher the charge on the molecule to be exchanged, the tighter it binds to the resin. At low pH and salt concentration most amino acids will firmly bind to the resin by the following reaction



The last step in the separation procedure is to elute the bound molecule from the resin by reducing the attractive force between the resin and the charged molecule. This is achieved by changing the pH or the ionic concentration, or by affinity elution. In affinity elution, an ion with greater affinity than the bound molecule is introduced into the system. If more than one type of molecule is bound to the column, a stepwise elution can also be done by using a second solvent, a third solvent and so on. A widely used matrix medium for ion exchange chromatography is DEAE-cellulose column in which

the Diethyl aminoethyl group acts as an exchanger. Cellulose fibres, polyacrylamide gel, crosslinked dextran (also known as sephadex) are other common materials used as the matrix.

2.2.8 Molecular exclusion chromatography

The separation of samples based on their molecular weights is known as molecular exclusion chromatography (MEC), gel filtration or molecular sieve chromatography. The matrix for MEC is made up of polyacrylamide, sephadex or agarose gel. Polyacrylamide and sephadex are supplied in the dehydrated form. When soaked in the solvent, they take it up rapidly and swell up forming a slurry. When this slurry is used for packing a column, pores of uniform size are formed by the gel and are used for separating the sample according to their molecular size and shape (Figure 2.4). A column of the gel (the porous material) is kept in equilibrium with a suitable solvent in which the molecule to be separated is dissolved. Large molecules in this solution, are excluded from the pores, and pass through the interstitial space reaching the bottom of the column quickly. Smaller particles penetrate through the pores of the gel and are distributed in the solvent inside as well as outside the molecular sieve. Smaller molecules are thus retarded and reach the bottom at slower rate. For a given sample the distribution coefficient K_d is dependent upon its size. If the molecule is completely excluded by the sieve then $K_d = 0$, whereas if it penetrates the gel and has accessibility to the inner solvent then $K_d = 1$. For all other intermediate sizes the K_d value will lie in the range 0 to 1. The chromatography column consists of loosely packed resin particles. V_T is the total volume of the column. It is made up of three terms: (1) The void volume V_0 or the volume of the exterior solvent. (2) V_g or the solid volume of the gel particles. (3) V_i or the internal volume of the pores that are accessible to solvent. Therefore

$$V_T = V_0 + V_g + V_i$$

The eluting volume is the volume of solvent that must flow through the column before a particular solute emerges. This is given by the equation

$$V_e = V_0 + K_d V_i \quad (2.1)$$

A solute that is completely excluded will have $K_d = 0$. Therefore $V_e = V_0$ for such a sample, i.e. the eluting volume is equivalent to the void volume. But a solute that can enter the pores will have to displace an additional volume given by $K_d V_i$. Thus the eluting volume in this case is given by equation (2.1). The partition coefficient K_d can be determined from this equation if the elution volume V_e is measured. V_0 and V_i are determined prior to the start of the experiment and are constant for a given column. V_0 is determined by measuring the elution volume of a particle larger than gel pores. V_i is

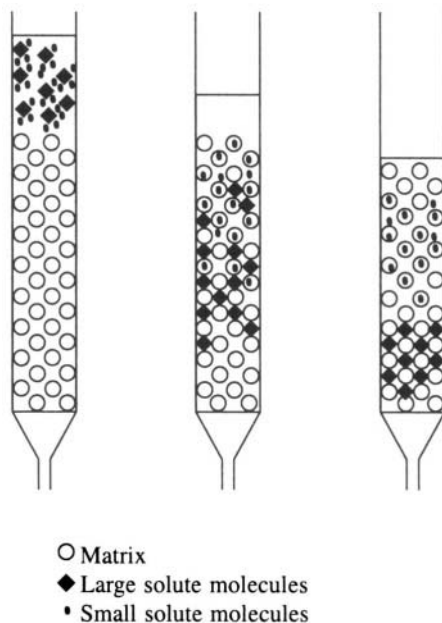


Fig. 2.4 Molecular exclusion chromatography. The column is filled with solvent containing large and small solute particles.

determined by the weight of water intake by dry resin. Two substances with different K_d values (K_d and K'_d) and size can be separated if the sample volume is less than or equal to V_s , where

$$V_s = (K_d - K'_d) V_i$$

Figure 2.5 shows the relationship between the elution volume and the molecular weight of various proteins when they pass through a sephadex G-200 column. Since the elution volume is related to the partition function by equation (2.1) above, it is possible to correlate K_d to the molecular weight. This relationship is given by the function

$$K_d = -A \log (M) + B$$

where A and B are constants and M the molecular weight. Thus by measuring the elution volume, it is possible to determine the partition coefficient and thence the molecular weight, to finally achieve a separation of the molecules based on the molecular weight.

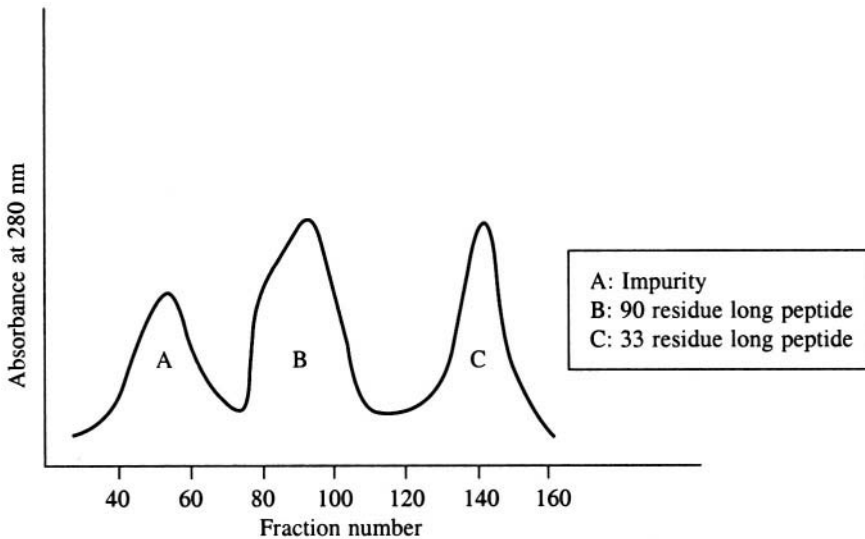
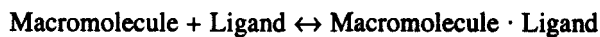


Fig. 2.5 Separation of a mixture of peptides by a sephadex column. Schematic sketch of the absorbance v/s fraction number indicating the difference in movement of the peptides through the column corresponding to the different molecular weights.

2.2.9 Affinity chromatography

Separation by affinity chromatography is based on a biological property of macromolecules rather than on a physical property. It is a highly sensitive technique and, in principle, can give absolutely pure sample in a single step. The technique requires that the macromolecule be able to bind to a specific ligand attached to an insoluble matrix under certain conditions, and then detach itself under certain other conditions. In other words the binding should be specific and reversible.



When a complex mixture of proteins is passed through a chromatography column to which a ligand is attached, then the protein with specificity for that ligand alone will bind to the column and the rest of the material will simply flow through. The bound molecule can then be recovered by a change in

the conditions such as pH of the solution, salts or temperature, or by adding a compound which interacts even more strongly to the ligand on the matrix, thus displacing the molecule of interest. The method therefore requires a preliminary knowledge of the biological specificity of the compound to be separated. In practice, agarose, cellulose, and polyacrylamide gels are used as the insoluble matrix. It is generally possible to select a ligand that has absolute specificity for the compound of interest or to select a ligand that has group selectivity. Group selectivity means that the ligand will selectively bind to a closely related group of compounds. For example, 5'AMP can reversibly bind to many NAD dependent dehydrogenases. The other end of the ligand, which is bound to the matrix, should be such that it will not interfere with the reversible binding of the macromolecule. $-\text{NH}$, $-\text{COOH}$ and $-\text{SH}$ groups are the most common groups used for binding the ligand to the matrix. In addition, there is generally a spacer molecule between the ligand and the matrix, to minimize the interference of the groups in the matrix support material (Figure 2.6). The spacer arm usually has about six to ten carbon atoms and can be hydrophobic (e.g. 1,6-diamino hexane) or hydrophilic (e.g. 6-amino hexanoic acid). Some of the ligands commonly used in affinity chromatography are: 5'AMP (with a specificity for NAD dependent dehydrogenases), lectins (specificity for cells and macromolecules containing N-acetyl- α -galactosamine residues), lysine (specific to RNA) and concanavalin A (specific to glycoproteins and glycopeptides). A highly specific ligand for any macromolecule may be produced by raising antibodies against it and then binding the antibodies to the matrix.

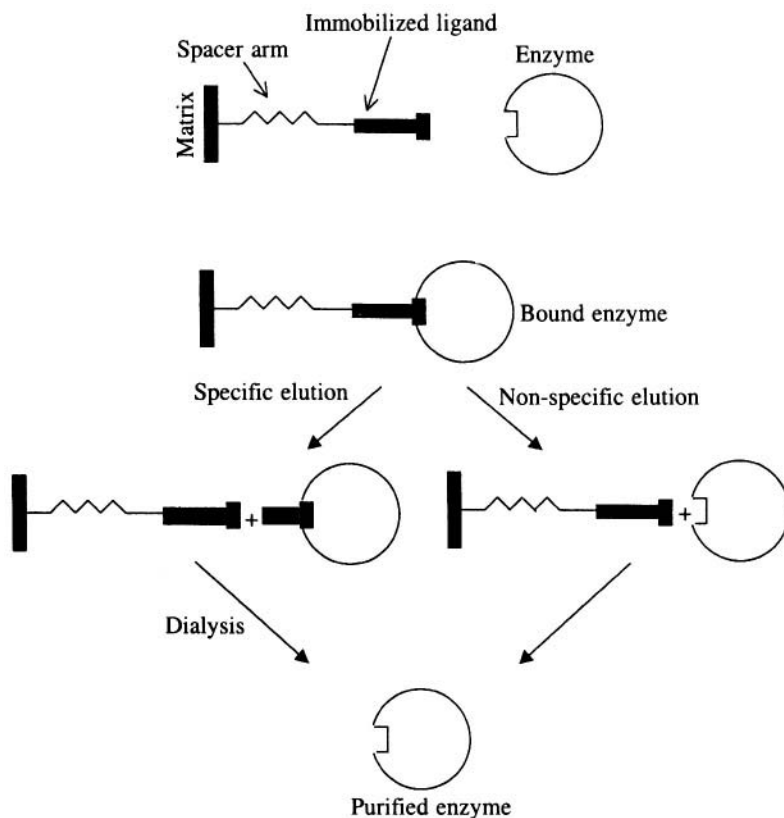


Fig. 2.6 Affinity chromatography. The enzyme attaches itself to a specific ligand that is bound to the matrix. It is then eluted out either specifically or non-specifically.

2.3 Electrophoresis

Important biological molecules like peptides, proteins, carbohydrates and nucleic acids have ionisable chemical groups and hence, under suitable conditions, exist in solution as charged species. Even non-polar molecules can be made to exist as weakly charged species by preparing phosphate, borate or similar derivatives. Molecules with similar charge may have different charge/mass ratios since their molecular weights could be different. Molecules with differing charges or differing charge/mass ratios would migrate at different rates in an electric field. The separation of molecules by utilizing these differences in rates of migration is known as electrophoresis.

If a molecule has a net charge q , the application of an electric field E will result in a force

$$F = qE$$

This force will accelerate the particle in a fluid till a steady state is reached when the frictional force is equal and opposite to the applied force. If v is the steady state velocity then

$$fv = qE$$

where f is the frictional force. Therefore

$$v = qE/f$$

For a spherical molecule with radius a and charge Ze (where e is the charge on electron and Z the atomic number),

$$v = ZeE/(6\pi\eta a)$$

since $q = Ze$ and $f = 6\pi\eta a$, where η is the viscosity of the solution. The electrophoretic mobility u is defined as the velocity per unit electric field.

$$u = v/E$$

However, due to the presence of counter ions in any aqueous solution, this equation does not completely account for the electrostatic mobility. These ions can neutralize some of the charges on the macromolecule if they bind tightly to it. Alternatively unbound or loosely bound ions can modify the ionic strength of the solution and thus alter the electric field that is felt by the polymer. In such a situation the electrophoretic mobility for a spherical particle is modified by a factor K , where K is a function of the screening parameter which has dimensions of L^{-1} (inverse of length). Therefore

$$u = (ZeK)/(6\pi\eta a)$$

Thus mobility will increase with increasing charge on the macromolecule.

Electrophoresis procedures can be generally divided into three categories: (1) Moving boundary electrophoresis. (2) Zone electrophoresis. (3) Continuous flow electrophoresis.

2.3.1 Moving boundary electrophoresis

In this method, the sample is dissolved in a buffer and placed in a cell. An additional quantity of the buffer is layered on top of it. When the electric field is applied, closely related molecules will tend to move as a band since their charges are similar, and boundaries are formed between groups of molecules with different electrophoretic mobility. The separations can be seen by direct optical observations. This method suffers from many experimental difficulties such as errors due to convection. Hence it is not very useful as a preparative technique.

2.3.2 Zone electrophoresis

In this method the mixture to be analysed is applied as a small spot or very thin band to the supporting medium. When electric field is applied the molecules move as distinct bands or zones. The molecules can be identified by staining, *UV* absorption etc. There are several different types of zone electrophoresis. We treat some of them below.

2.3.3 Low voltage electrophoresis

The electrophoresis unit (Figure 2.7) consists of a power pack, which supplies DC current across two electrodes, buffer reservoirs, a support, and a transparent insulating cover. The low voltage power pack can either give a constant voltage or a constant current and has an output of 0 to 500 V and 0 to 150 mA. Either stainless steel or platinum electrodes are used. The supporting medium, which can be paper or cellulose acetate, must first be saturated with buffer and held on a sheet of insulating material like Perspex. The sample is then applied as a small spot using a micropipette. The location of the spot depends on the nature of the molecular mixture present in the sample. For example, if there are molecules with opposite charges, then they will separate and move towards opposite electrodes and hence the sample must be applied at the centre. The two reservoirs on either side of the medium are isolated from the electrodes so that any pH change there does not affect the buffer. After application of the sample, power is switched on for the required amount of time. At the end of the experiment, the migration of the different constituents of the mixture may be analysed. The experiment may be performed in different ways. For example, instead of paper or cellulose acetate, thin layers of silica, alumina or cellulose may be applied on to a glass plate and used as the supporting medium. This technique is similar to TLC (thin layer chromatography) and is known as thin layer electrophoresis (TLE).

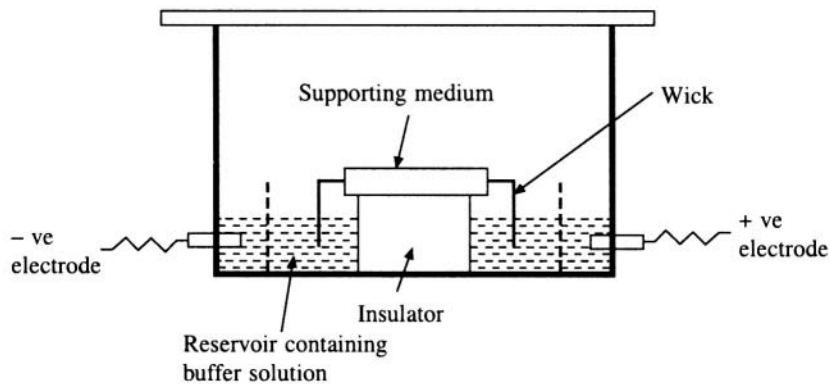


Figure 2.7 Low voltage gel electrophoresis

2.3.4 High voltage electrophoresis

A disadvantage with low voltage paper electrophoresis is that diffusion processes may influence the migration of the molecules so that it is not strictly according to the charges on them. One way to overcome this is to resort to high voltages, of the order of 10,000 volts. Such a voltage can produce potential gradients up to 200 volts/cm. If the separation between the electrodes is d metres and if V is the potential difference in volts then the potential gradient is V/d volts/metre. The force on the charged molecule is then Vq/d newtons, where q is the charge in coulombs. The migration of the molecules is caused by this force and is proportional to it, and an increase in the potential gradient leads to an increased rate of migration. The distance migrated by the ions will depend on the voltage

as well as the amount of time for which it is applied. This technique results in higher resolution and faster separation. But because of the high voltage, it generates considerable heat and the apparatus requires cooling.

2.3.5 *Gel electrophoresis*

When applied to the separation of a mixture of proteins, simple zone electrophoresis has limited resolving power, due to the similarity in mobility of the components. When zone electrophoresis is combined with molecular exclusion effects using gels, much higher resolutions can be achieved. The molecular sieving property of the gel helps in separating molecules that have similar charge properties but differ in size and shape. Starch, agar, and polyacrylamide gels are used for this purpose. The gels can be run as horizontal slabs or as vertical slabs. Vertical slabs of the type shown in Figure 2.8 are most frequently used. Gels are prepared in glass or Perspex containers. In the case of slab gels, the gel is cast between two clean glass plates that are clamped together but held apart by spacers. The samples to be separated are applied using a micro syringe into the wells set in the gel. In the vertical slab method the two ends of the plate containing the gel are kept in contact with the lower and upper reservoirs of buffer respectively, thus completing the electrical circuit between lower and upper electrodes. The components of the mixture migrate with different mobility on application of an electric field, and can be identified by staining techniques, at the end of the experiment. Normally, a known sample is run along with unknown samples in order to compare the migration characteristics. A major advantage of this method is that even very small quantities of proteins can be separated and identified.

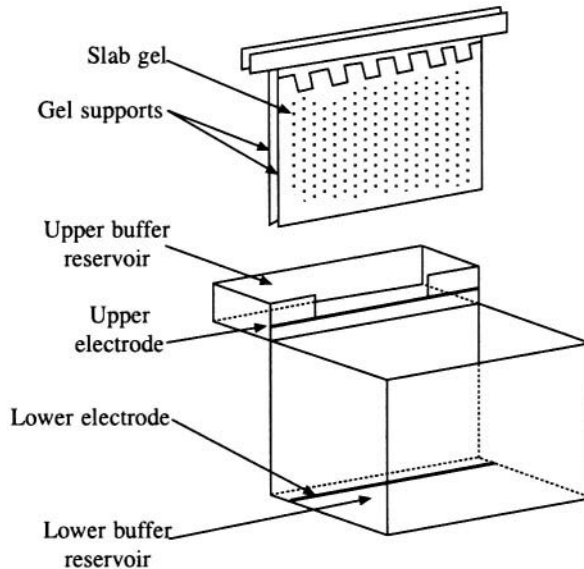


Figure 2.8 Vertical slab gel electrophoresis apparatus. Most of the equipment (except obvious parts such as electrodes and clamps) is made of perspex.

2.3.6 *Sodium dodecyl sulphate poly acrylamide gel electrophoresis (SDS-PAGE)*

In this technique, all the proteins in a mixture are converted into structures having similar charge characteristics, though they may differ in molecular weight. Sodium dodecyl sulphate is an anionic detergent that binds tightly to proteins and denatures them. When an excess of SDS is added to the

solution (about 1.4 gm per gm of amino acid), then the protein molecules in the mixture are completely surrounded by SDS. The total negative charge acquired due to SDS binding overwhelms the variations in charge in the different protein molecules. Hence, when a drop of the solution is placed on the polyacrylamide electrophoresis gel, all the protein molecules move towards the anode, with their mobility varying according to their size and shape. Reducing agents like β -mercaptoethanol may also be added to break any intrachain or interchain disulphide bonds. The apparent mobility $u(c)$ of the molecule at the gel concentration c is given by

$$\log (u(c)) = -k_x c + \log (u(0))$$

where $u(0)$ is a constant for a set of similar proteins, and k_x is a constant depending on the extent of crosslinking of the gel. It has been shown that the mobility u and molecular weight M are related by the following equation.

$$u = b - a \log (M)$$

where b is a constant which contains the term $u(0)$, and a is a function of gel concentration c . This equation fits the experimental observations extremely well and is used to determine the molecular weight of an unknown sample by running it on a gel side-by-side with another mixture containing standard proteins of known molecular weights.

2.3.7 Isoelectric focussing

Isoelectric focussing is an electrophoretic method where a pH gradient is used along with the voltage gradient. It is extremely useful in separating proteins according to their isoelectric points. The isoelectric point of a protein is defined as the pH at which the average net charge on the molecule is zero. In the electrophoretic gel the region around the anode is kept at a lower pH as compared to the region near the cathode. A pH gradient is established such that the samples to be separated have their isoelectric points within this pH range. At a pH below their isoelectric point, the particles will move towards the cathode, and at a pH above the isoelectric point they move towards the anode. At the isoelectric pH, the particles will remain immobile. Thus, when placed in the combined pH and voltage gradient on the gel, the particles move towards their isoelectric points no matter where they are initially applied. This method gives very high resolution and is especially suitable for separating isoenzymes. Even differences in pH of the order of 0.01 are enough to separate the molecules. Substances known as ampholytes generate the pH gradient between the electrodes. Ampholytes are molecules with both positive and negative charges, for e.g. polymers containing several amino and carboxyl groups. Ampholine, Biolyte and Pharmalyte are some of the commercially available ampholytes. When an electric field is applied to a mixture of ampholytes with a wide range of isoelectric points in a column or gel, they distribute themselves according to their charge and thus establish a pH gradient. A small amount of sample may then be added to the medium. The different proteins in it migrate along the column or gel till they reach the pH corresponding to their respective isoelectric points. Each protein then remains at that position as a sharp band and may be viewed and analysed by staining, absorbance etc.

2.3.8 Continuous flow electrophoresis

In continuous flow electrophoresis the experiment is conducted in free solution without a supporting medium. This reduces frictional effects between the sample ions and the solution and causes rapid migration of the components of the mixture. This is a form of the moving boundary method. The sample, in the carrier buffer, is applied through the central tube as a continuous flow. The electric

field is applied between the outer and inner concentric cylinders and the outer cylinder is rotated to maintain a stable flow of the buffer solution. Electrophoresis takes place continuously as the sample is carried upwards by the flow of carrier buffer from the bottom of the inner tube. As it moves upwards the components are separated radially. At the top a series of radial slits separates the buffer stream into thirty or more fractions, each containing a separated portion of the original mixture. Large-scale separations can be achieved by this method. This technique is therefore used for enzyme purification on an industrial scale, blood plasma fractionation. In the laboratory it is used for collection of large amounts of proteins on a preparative scale.

Physico-Chemical Techniques to Study Biomolecules

3.1 Introduction

Biological macromolecules have molecular weights of the order of 10,000 daltons. Their functions depend on their three dimensional structures. It is therefore useful to know the structure, size and shape of these molecules in order to understand the mechanism of function. There are several techniques that can give information on these aspects. Table 3.1 lists some of the physical techniques that are used in studying large biomolecules.

Table 3.1 Physical techniques used to study biological molecules and the information they yield

Technique	Information obtained
1. Osmotic pressure	Molecular weight
2. Diffusion	Molecular volume
3. Sedimentation	Molecular weight and volume
4. Viscosity	Molecular shape
5. Birefringence	Molecular shape
6. Optical Rotatory Dispersion	Helical content
7. Light scattering	Molecular size and weight
8. X-ray scattering	Molecular size and shape
9. X-ray crystallography	Three dimensional structure (atomic positions)
10. Electron Microscopy	Size and shape
11. Electromagnetic absorption, nuclear magnetic resonance, electron spin resonance	Intramolecular forces, Orientation of groups, Three dimensional structure

In this chapter we will focus on some of the hydrodynamical methods listed above and discuss in detail the principles underlying them.

3.2 Hydration of Macromolecules

The molecular volume V of any isolated molecule is given by

$$V = (M \langle v \rangle_0) / N_0,$$

where M is the molecular weight, and $\langle v \rangle_0$ is the specific volume, equal to $1/\rho$, where ρ is the density. N_0 is Avagadro's number. However, in the case of proteins and nucleic acids, there may be counter ions and water molecules tightly bound to the molecule and therefore it cannot be treated in isolation. It is normal laboratory practice to store and use proteins and nucleic acids in ~ 0.1 M buffered salt solutions, thereby simulating approximately the physiological environment. Hence, when we study the thermodynamic properties of a macromolecule solution we must treat it as a three component system consisting of salt, water and the macromolecule. Such a multi-component system is very difficult to deal with. However, since we are mainly interested in the interaction of the macromolecules with water, we may approximate the system as consisting of two components: water and macromolecule. The total volume of such an ideal solution is

$$V_{\text{tot}} = g_1 \langle v \rangle_1 + g_2 \langle v \rangle_2$$

where g_1 and g_2 are the weights of water and macromolecule respectively in grams, and $\langle v \rangle_1$ and $\langle v \rangle_2$ are the specific volumes. In addition to the bound water molecules, we may have bulk water, which may occupy holes or cavities in the macromolecule, thus changing its shape and size. Taking all this into consideration, the volume occupied by the hydrated macromolecule is given by

$$V_h = (M/N_0)(\langle v \rangle_2 + \delta_1 \langle v \rangle_1)$$

where (M/N_0) gives the weight of a single macromolecule. $\langle v \rangle_2$ is now the partial specific volume of the solute and is defined as the change in volume of the solution when a small amount of solute is added at infinite dilution, $\langle v \rangle_1$ is the partial specific volume of **water** = $1/\rho$, and δ_1 is the hydration in grams of bound water per gram of macromolecule. The hydration value of a protein gives an estimate of the amount of water that is bound to a protein. A value of the order of 0.3 to 0.4 g of water per gram of protein is usually assumed, when discussing the hydrodynamic properties of most macromolecules

3.3 Role of Friction

Hydrodynamic properties of macromolecules like diffusion, viscosity and sedimentation are affected by the frictional forces between molecules of the protein and those of the solvent. Since this frictional force is in opposition to motion we can include this in the equation of motion as

$$F - fv = m(dv/dt)$$

where f is the frictional coefficient, (dv/dt) is the acceleration and m is the mass of the molecule³. In the case of spherical particles, the translational frictional force f is proportional to the fluid viscosity η and radius r of the particle. Thus the coefficient of friction for spherical particles, known as Stokes law, is

³This is the general equation of motion $F = m(dv/dt)$ or (mass $\times$ acceleration) modified to take into account the opposing frictional force which is taken to be proportional to the velocity v .

$$f_{\text{sph}} = 6\pi\eta r$$

This relation is true when the interaction of the particle with the fluid is strong. In the other case, when there is virtually no interaction at all between the fluid and the particles, the friction coefficient is

$$f_{\text{sph}} = 4\pi\eta r$$

The frictional force can also act as a damping force for the rotational motion of the particle and in that case the coefficient is given by

$$f_{\text{rot}} = 6\eta V$$

where V is the volume. Since most biological molecules are not spheres, the previous equations will have to be modified for the shape factor. As a significant fraction of macromolecules are globular, compact irregular bodies or an ellipsoid may be a better approximation than a sphere. Ellipsoids are classified into prolate and oblate ellipsoids. An oblate ellipsoid is disc shaped and is obtained by rotating an ellipse about its minor axis (Figure 3.1). A prolate ellipsoid can be taken as an approximation for a rod and is obtained by rotating an ellipse about its long or major axis. For equal volumes, ellipsoids have greater surface area compared to spheres and hence their frictional coefficients will be larger.

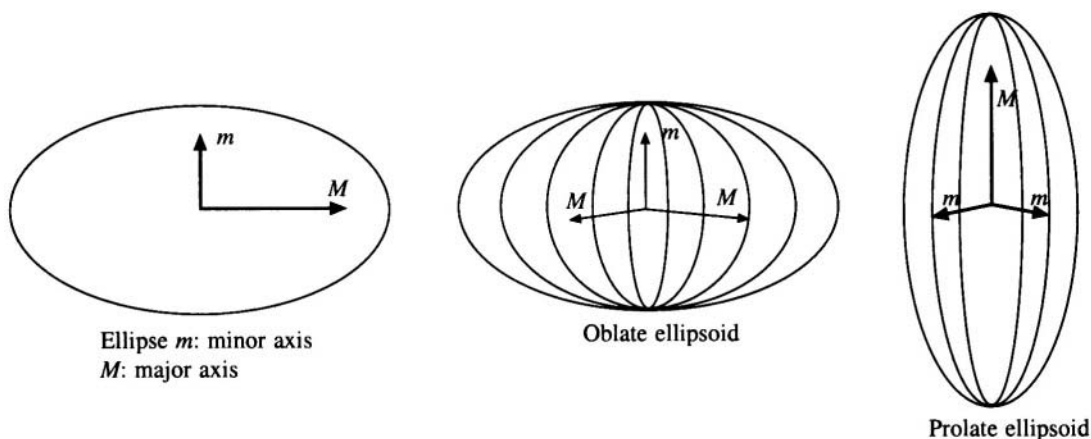


Fig. 3.1 Oblate and prolate ellipsoids

3.4 Diffusion

The frictional coefficient f comes into effect when a molecule moves through a medium. The movement of the molecule could be either diffusion or sedimentation and the driving force can be the concentration gradient, the force of gravity or the centrifugal force. Diffusion can be considered as movement of molecules from a higher concentration to a lower concentration (Figure 3.2). According to Fick's law, the rate of diffusion across a boundary (dn/dt : the number of molecules which pass through a cross section A in unit time) for a single solute component diffusing in a system at constant temperature and pressure is given by

$$dn/dt = -DA(\delta C/\delta r)$$

where $\delta C/\delta r$ is the concentration gradient and D is the diffusion coefficient. A concentration gradient

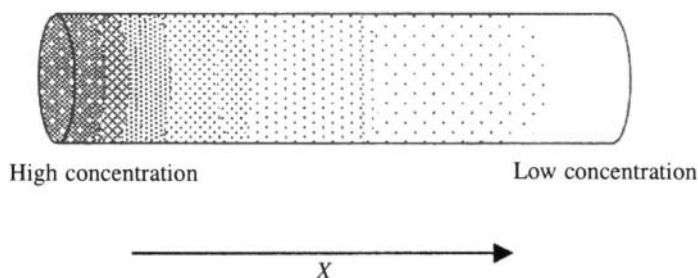


Fig. 3.2 Diffusion of molecules from higher concentration to lower concentration.

implies that the concentration of the molecules (i.e. the solute) varies with distance r in one dimension. The diffusion coefficient D is given by

$$D = kT/f \quad (3.1)$$

where k is the Boltzmann's constant, T the absolute temperature and f the frictional coefficient. Thus D is dependent on f . The concentration of the solute or the concentration gradient can be measured as a function of position, by the absorbance at various points. Alternatively it can be measured as the refractive index gradient using a light scattering technique known as Schlieren optics. Fick's second law of diffusion gives the concentration gradient across the boundary as

$$(\delta C/\delta r)_t = C_0/(4\pi Dt)^{1/2} \exp(-r^2/4Dt) \quad (3.2)$$

where C_0 is the total solute concentration difference across the boundary. In the graph of concentration gradient versus distance, plotted at different times (Figure 3.3), the downward change in the concentration

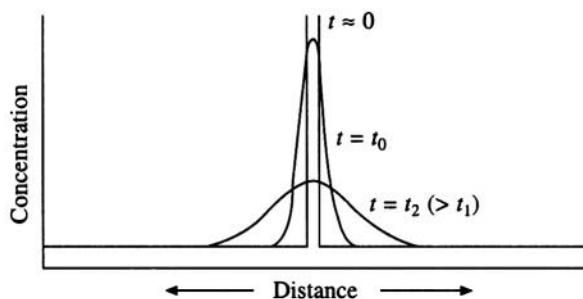


Fig. 3.3 Plot of the concentration as a function of distance at three different times. At $t = 0$, the sample is applied at the centre and is concentrated only there. It starts diffusing until at $t = t_2$ it is spread out over a wide area.

gradient as time passes is clearly observed. The x coordinate of this graph is defined in such a way that the maximum value H_m of the gradient is at $x = 0$. The rate of change of concentration gradient with time ($d/dt (\delta C/\delta r)$) is zero at the maximum value of the gradient. Differentiating equation 3.2 with respect to time and setting it equal to zero, we obtain

$$H_m = C_0/(4\pi Dt)^{1/2}$$

When H_m^2 is plotted against $1/t$, the slope of the straight line so obtained will give the value of D , which in turn can be used to determine the molecular weight using equation (3.3), provided the partial

specific volume $\langle v \rangle$ is known. $\langle v \rangle$ is defined as the volume occupied by 1 g of the solute in the solution. For non-hydrated spheres the volume V is given by

$$V = (M \langle v \rangle) / N_0$$

where M is the molecular weight and N_0 is Avagadro's number. But $V = (4/3)\pi r^3$. Therefore

$$\begin{aligned} M &= (4\pi r^3 N_0) / (3\langle v \rangle) \\ &= (4\pi N_0 / 3\langle v \rangle) \times (f / 6\pi\eta)^3 \\ &= (4\pi N_0 / 3\langle v \rangle) \times (KT / 6D\pi\eta)^3 \end{aligned} \quad (3.3)$$

The last step follows from Stoke's law, where η is the viscosity. Thus an experimental measurement of the diffusion constant of a molecule enables the calculation of its molecular weight. Diffusion experiments are not easy to perform, due to experimental difficulties such as the slow rate of diffusion of macromolecules (diffusion constants are of the order of only 10^{-7} to 10^{-6} cm²/sec), gravitational instabilities, convective mixing, etc.

3.5 Sedimentation

The sedimentation of a suspension of particles in a liquid can be achieved either by the force of gravity or by an applied centrifugal force. For example, if a muddy solution is left to stand undisturbed, it will separate out after some time as a top layer of water and a bottom layer of mud. This sedimentation of mud particles is due to the force of gravity. However, when we perform a sedimentation experiment with macromolecules, the force of gravity is insufficient to make the molecules sediment. Under these circumstances the solution containing the macromolecules is subjected to centrifugation in an ultracentrifuge. An ultracentrifuge can rotate at speeds of several thousand revolutions per minute leading to a centrifugal force of the order of 3,00,000 times the acceleration due to earth's gravity or 3,00,000 g. The force on each particle is given by Newton's second law

$$F = ma$$

where m is the mass of the particle and a is its linear acceleration. As the particles are spun in an ultracentrifuge, the radial or centrifugal force F_C is given by

$$F_C = m\omega^2 r$$

where ω is the angular velocity, and r is the radial distance of the particle from the axis of rotation. As the particles sediment, diffusion effects also take place simultaneously and hence the motion of macromolecules in an ultracentrifuge is the resultant of both the effects. Therefore the mass term in the above equation is replaced by the effective mass which is given by $(m - V\rho)$ i.e. the difference between the weight of the particle and the buoyant force due to displacement of the medium. V is the volume and ρ is the density of the solvent. $V\rho$ represents the mass of solvent displaced by the particle of mass m . Making these substitutions, the centrifugal or radial force is given by

$$F_C = (m - V\rho)\omega^2 r$$

This force is opposed by the frictional force F_f which is proportional to the velocity of the particle

$$F_f = f \times (\text{radial velocity}) = f (dr/dt)$$

where f is the frictional coefficient. When the centrifugal force is balanced by the frictional force then

$F_C = F_r$ and the particles will sediment with uniform velocity, since there is no net force on them and thus no acceleration. Writing this condition in another way

$$(m - V\rho)\omega^2 r = f (dr/dt) \quad (3.4)$$

in which ω is expressed in radians per second. Multiplying both sides by N_0 (Avagadro's number) we get

$$N_0(m - V\rho) = (N_0 f (dr/dt))/(\omega^2 r)$$

The velocity of sedimentation depends on the size and shape of the sedimenting particle and is proportional to the applied force. Therefore

$$v = s\omega^2 r$$

where s is a proportionality constant depending on shape and size of the molecule and the nature of the liquid. It is known as the sedimentation coefficient. It is defined as the velocity per unit field

$$s = \text{velocity}/\omega^2 r = (dr/dt)/(\omega^2 r)$$

Therefore

$$N_0(m - V\rho) = N_0 f s$$

or

$$N_0 m (1 - V\rho/m) = N_0 f s$$

But $N_0 m$ is the molecular weight M . Therefore

$$s = [M(1 - V\rho/m)]/N_0 f$$

in which $V/m = \langle v \rangle$, the partial specific volume. Substituting,

$$s = [M(1 - \langle v \rangle \rho)]/N_0 f$$

or

$$M = N_0 f s / (1 - \langle v \rangle \rho)$$

From equation (3.1)

$$f = KT/D$$

where f is the frictional coefficient, K the Boltzmann's constant, T the absolute temperature and D is the diffusion coefficient. Substituting,

$$M = N_0 K T s / D (1 - \langle v \rangle \rho) = R T s / D (1 - \langle v \rangle \rho) \quad (3.5)$$

where R is the gas constant and is equal to $N_0 K$. This equation is called Svedberg's equation and is arrived at by combining the sedimentation and diffusion equations. If M and D are known, either s , the sedimentation coefficient, or f , the coefficient of friction, can be calculated. Molecular weight M can also be determined using equation (3.5) if D and s are known. The sedimentation coefficient for any molecule at 20°C and in pure water is taken as the standard for comparison. The expression for this value in terms of the coefficient s at any temperature and in any liquid is given by

$$s_{20,w} = [s\eta(1 - \langle v \rangle \rho)_{20,w}] / \eta_{20,w}(1 - \langle v \rangle \rho)$$

The sedimentation coefficient is measured in Svedberg units ($1S = 10^{-13}$ sec). For example the sedimentation coefficient of one of the subunits of ribosome is 30S and the other is 50S.

3.6 The Ultracentrifuge

The ultracentrifuge is an instrument used to measure sedimentation coefficients. Figure 3.4 shows an ultracentrifuge with its component parts. The sample is held in a centrifugal cell in a titanium or aluminium rotor, which is rotated by an electric motor at speeds up to 70,000 revolutions per minute (rpm). The rotor assembly, including the sample cell, is kept inside an armored chamber to protect the users against any explosive accident. In order to avoid excessive heating that may be caused by

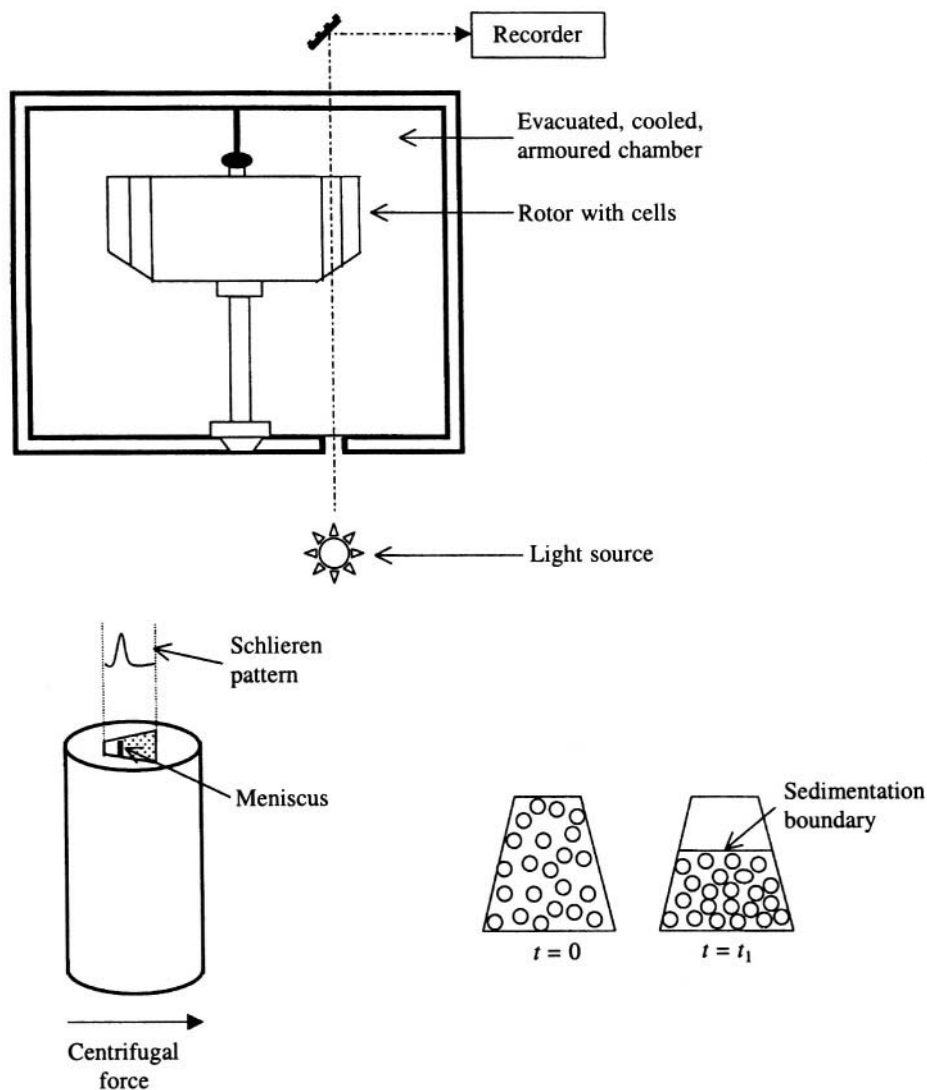


Fig. 3.4 Schematic cutaway sketch of an ultracentrifuge and its parts.

friction with air, the rotor chamber is evacuated and refrigerated. In order to avoid collision of the particles in the sample with the cell wall, the sample cell is shaped like the sectors of a circle drawn around the axis of rotation. Due to the high speeds, the balance of weight on the rotor is critical. A flexible shaft is used which can tolerate a mismatch of up to 0.5 g. A centrifuge can be either a preparative instrument or an analytical one. In a preparative centrifuge the sample is spun for a fixed length of time to separate mixtures or purify samples. In an analytical centrifuge, the movement of the boundary during sedimentation can be monitored using optical devices like Schlieren optics, Rayleigh interference or absorbance, and the period and rate of centrifugation can be accordingly adjusted. The Schlieren optical system is based on the fact that light does not deviate as long as it moves through a solution of uniform concentration but undergoes refraction in a solution with varying densities. The change in refractive index with change in concentration is recorded and is used in the determination of the sedimentation coefficient.

Molecular weight determination can be carried out by the sedimentation velocity technique or the sedimentation equilibrium method. In the sedimentation velocity method the ultracentrifuge is operated at high speeds and the movement of the sedimentation boundary is recorded as a function of time. This is a measure of rate of sedimentation. Rayleigh's interference method is used to make this record, since the Schlieren optical system is not very sensitive to small changes in concentration. The Rayleigh interference system uses a double sector cell, one for the solvent and the other for the solution. By measuring the displacement of interference fringes, the system measures the difference in refractive index between the reference solvent and the solution.

In the equilibrium method the sample is centrifuged till an equilibrium is reached between sedimentation and the diffusive movement of the particles, i.e. there is no net movement of solute particles in the cell and the concentration gradient remains stable. In other words, the net flux J is zero, i.e.

$$J = s\omega^2 rC - D(\delta C/\delta r) = 0$$

or

$$s\omega^2 rC = D(\delta C/\delta r)$$

$$s/D = (\delta C/\delta r)/\omega^2 rC$$

Substituting in the Svedberg equation, we obtain the expression for the molecular weight

$$M = [RT/(1 - \langle v \rangle \rho)\omega^2 rC](\delta C/\delta r)$$

Thus the molecular weight can be determined from a measurement of the concentration gradient. Expressing the above equation in terms of concentration at two different distances from the rotor axis, r_a and r_b we get

$$M = 2RT(C_b - C_a)/[(1 - \langle v \rangle \rho)\omega^2(r_b^2 - r_a^2)C_0]$$

where C_a and C_b are concentrations corresponding to distances r_a and r_b , and C_0 is the initial concentration. The advantage in using this equation instead of the previous one is that r_a and r_b can be measured at the meniscus and at the bottom of the cell respectively. In addition the equation is independent of the size and shape of the molecule. Moreover at the meniscus and at the bottom the equilibrium condition is surely satisfied, i.e. the flux is zero. This equation is, however, valid only for homogeneous samples and therefore the molecule to be analysed will have to be pure. The rotor should be spun at low speeds as otherwise there will be no solute molecules at the meniscus and a

pellet will form at the bottom. Molecular weights ranging from a few hundred daltons to several thousand daltons can be determined using the equilibrium method and the molecular weight decides the rotor speed. For example, a molecular weight of 50,000 daltons will require a speed of 10,000 rpm. The main disadvantage of this method is that long spinning times may be required before equilibrium is reached. Nevertheless, this technique is desirable when the diffusion coefficient D is not known. Another method suggested by Yphantis uses a high speed equilibrium. The centrifuge is operated at such a high speed that the concentration of the solute at the meniscus is zero. Now the difference in refractive index between any point in the cell and meniscus is proportional to the concentration of the solute at that point.

In another technique known as density gradient ultracentrifugation, a solution of low molecular weight (e.g. CsCl) is centrifuged in a cell, so that a density gradient is formed at equilibrium. If we add a substance of high molecular weight to this solution then it will be suspended at a position where its buoyant density is equal to the density of the low molecular weight solution. The position where it is suspended is measured and is used to calculate its molecular weight. This method is widely used in biochemical and biophysical research. If we have a multicomponent system with varying densities then the components will be suspended at different positions in the cell. In practice, a thin layer of the macromolecular sample is layered on a larger volume of the solvent and centrifuged. The components of the system will sediment according to their sedimentation coefficients and separate out as bands (Figure 3.5). This is called zonal sedimentation. The advantage of this technique is that very little ($\sim\mu\text{g}$) sample is used.

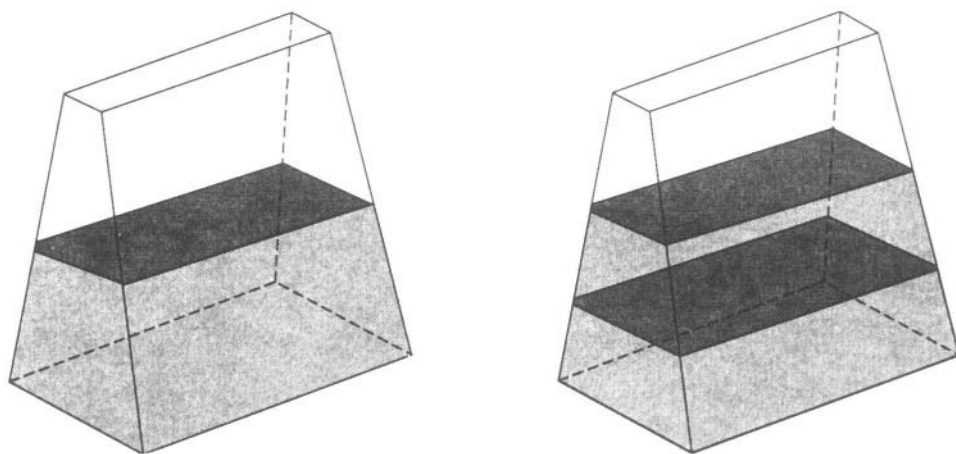


Fig. 3.5 Zone sedimentation

Apart from molecular weights, sedimentation methods can also yield useful information regarding the shape and size of the particles. An equation relating the shape and size to the sedimentation velocity is obtained by manipulating equation (3.4)

$$v_0 = \text{velocity} = [2r_p^2 (\rho_p - \rho_m) \omega^2 r] / 9\eta$$

where ρ_p and ρ_m are the densities of the particle and the medium respectively. The latter is a correction for the buoyant force. r is the distance of the particle from the axis of rotation, η is the viscosity, ω is the angular velocity. Thus the sedimentation velocity rate is proportional to the radius r_p and to the density of the particle. The equation is true for perfectly spherical particles. In the case of particles

that are not spherical, a correction factor is applied. This is called the shape factor F and is equal to (f_1/f_2) , where f_1 is the frictional coefficient of the molecule, and f_2 is the frictional coefficient of an equivalent sphere. Thus r_p , which indicates the size of the particle, and the shape factor, which gives information about the deviation of the particle from sphericity, both may be obtained from sedimentation experiments.

3.7 Viscosity

Viscosity is resistance to liquid flow. It is increased by the presence of a solute. The increase depends on the concentration and other structural properties of the solute and information regarding these properties may be obtained from measurements of viscosity. Viscosity measurement is simple and inexpensive and may be made using a capillary viscometer. A typical example is the Ostwald Viscometer shown in Figure 3.6. The solution whose viscosity is to be determined is poured into the lower bulb through arm A. It is then sucked up through arm B to fill the upper bulb and rise past the mark **a**. Now it is allowed to fall back through the capillary under gravity. The time taken for the solution to fall from point **a** to point **b** is noted. Then η , the viscosity, can be calculated using the following expression, derived from Poiseuille's equation for the viscosity of a liquid flowing through a capillary

$$\eta = (\rho g r^4 t \pi) / (8 l V')$$

where ρ is the density of the liquid, g the acceleration due to the earth's gravity, l and r are the length and radius, respectively, of the capillary, t is the time taken by the liquid to fall back from **a** to **b** and V' is a measure of the total hydrostatic pressure to which the liquid flow through the capillary is subjected and is calculated as the integral of dV/h between **a** and **b**, where h measures the height of the liquid column and V is the volume element at the point h . It is difficult to measure the absolute viscosity of a liquid by this method especially since V' is difficult to estimate. However, a comparison with a reference standard can be made. Thus if η_0 is the viscosity of the reference, and ρ_0 its density,

$$\eta/\eta_0 = (\rho/\rho_0)(t/t_0)$$

where t_0 is the outflow time for the standard. The reference liquid is usually chosen to be the solvent, which is then used to dissolve the particles of interest in order to measure the viscosity of the solution.

A theoretical treatment of the viscosity of a solution of particles is required before we can obtain shape and size information from viscosity measurements. The viscosity η of a solution, relative to the viscosity η_0 of the pure solvent, can be expressed as a power series

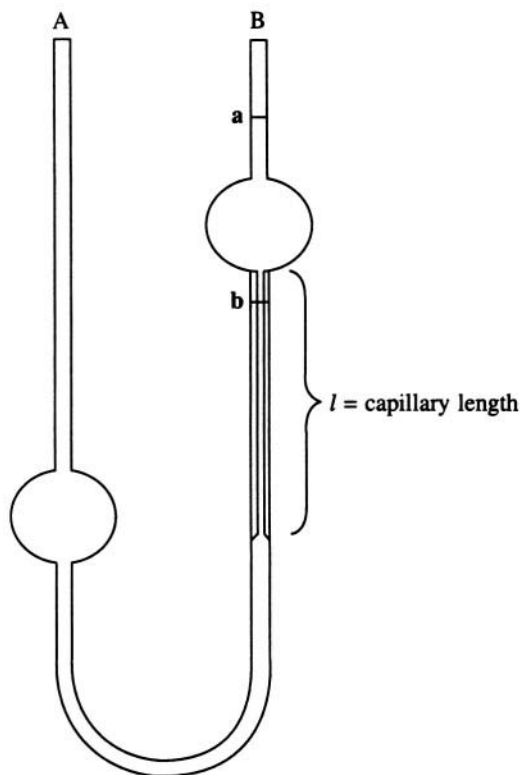


Fig. 3.6 The Ostwald viscometer

$$\eta/\eta_0 = (1 + k_1 C_2 + k_2 C_2^2 + \dots)$$

where C_2 is the solute concentration in g/cm^3 . Einstein showed that, ignoring the higher order terms, the above equation may be simplified to obtain the relative viscosity for a dilute solution of large rigid spheres as

$$\eta/\eta_0 = 1 + 2.5\phi$$

where

$$\phi = (V_h N_0 C_2)/M = C_2 \langle v \rangle$$

in which V_h is the hydrated volume, (M/N_0) is the mass of the molecule in grams, C_2 the solute concentration in g/cm^3 and $\langle v \rangle$ is the partial specific volume. Thus the change in viscosity on addition of the solute particles does not depend on their size but depends on the volume they occupy in solution.

For particles which are shaped as rigid ellipsoids, an intrinsic viscosity $[\eta]$ may be obtained from the expression

$$[\eta] = (v V_h N_0 / M)$$

v is called the Simha factor and contains information on the shape of the molecule. It varies as the ratio of the major and minor axes of the ellipsoids. The above equation can be more generally written as

$$[\eta] = KM^\alpha$$

where K and α are constants. Measurements of the intrinsic viscosity together with sedimentation or diffusion measurements can lead to a good estimate of molecular weight by eliminating shape effects. Table 3.2 lists the Simha factors for prolate and oblate ellipsoids with different axial ratios. Thus, for example, rod shaped particles always have a Simha factor greater than 15. Table 3.3 provides some

Table 3.2. Simha and Scheraga-Mandelkern factors for ellipsoids

Axial ratio	Prolate ellipsoid		Oblate ellipsoid	
	v	$\beta \times 10^{-6}$	v	$\beta \times 10^{-6}$
1	2.5	2.12	2.5	2.12
10	13.63	2.41	8.04	2.14
30	74.51	2.78	21.58	2.15
100	593.70	3.22	69.10	2.0

Table 3.3. Experimentally determined intrinsic viscosities and molecular weights

Sample	$[\eta]$	$M (\text{cm}^3/\text{g})$
<i>Near spherical particles</i>		
Ribonuclease	3.3	13,700
Hemoglobin	3.6	68,000
<i>Rod like particles</i>		
Myosin	217	4,93,000
DNA	5,000	60,00,000

experimentally determined molecular weights using the Simha shape factor. Another factor, which is related to shape of the particles is given by the Scheraga-Mandelkern equation

$$\beta' = \{N_0^{1/3} v^{1/3}\} / \{(162\pi^2)^{1/3} F\} = \{N_0 s [\eta]^{1/3} \eta\} / \{M^{2/3} (1 - \langle v \rangle \rho)\}$$

where $F = f/f_{\text{sph}}$ is the shape factor, f is the frictional coefficient of the particle, f_{sph} the frictional coefficient of an equivalent sphere, N_0 is Avogadro's number, s is the sedimentation coefficient of the solute particles, $[\eta]$ and η are the intrinsic and relative viscosity of the solution respectively, M is the molecular weight of the solute particles, $\langle v \rangle$ is the partial specific volume of the solute and ρ is the density. The values of the Sheraga-Mandelkern factor are tabulated in Table 3.2 as $\beta = 100^{1/3} \beta'$. β is less sensitive to the axial ratio as compared both to the Simha factor v , as well as to another factor called the Perrin factor. For most globular proteins the axial ratio is less than 10 and the β value is close to 2.20×10^6 . Using these values, and the Sheraga-Mandelkern equation, molecular weights can be determined to an accuracy better than 10%. Table 3.4 shows some experimentally determined molecular weights using the Scheraga-Mandelkern equation.

Table 3.4. Experimentally determined molecular weights compared with theoretical values from chemical structure

Molecule	From Chemical Structure	From Sheraga-Mandelkern equation	From sedimentation equilibrium
Ribonuclease A (bovine)	13,683	15,200	13,700
Lysozyme	14,211	12,400	14,500
Serum albumin (bovine)	66,296	59,000	68,000

3.8 Rotational Diffusion

Molecules in solution undergo both translational and rotational displacements due to the Brownian motion of the solvent. Each solute molecule undergoes rotation about its centre of mass. This movement is characterized by a coefficient called the rotational diffusion coefficient $[\theta]$ which represents the average angle of rotation per unit time and is dependent on the rotational friction experienced by the particle. For non-spherical particles, a single frictional coefficient is insufficient to describe the rotational friction. In such cases, the frictional forces acting on the particle are described by a tensor⁴.

Therefore $[\theta]$ is also a tensor and is given by the equation

$$[\theta] = kT/f_{\text{rot}}$$

where f_{rot} , the coefficient of rotational friction, is a tensor. The determination of $[\theta]$ is especially useful in the case of molecules with high axial ratios (length/width) since it gives a measure of the length of the molecule in solution. $[\theta]$ may be measured by a variety of techniques like flow birefringence, electric birefringence and the polarisation of fluorescence. In order to observe rotational diffusion, the equilibrium orientations of the molecule in normal solution must first be perturbed, and then some property of the molecule, which is dependent on the orientation of the molecule, must be

⁴ A tensor is matrix of numbers that describes a given property in different directions. In the present instance, the frictional tensor will consist of a 3×3 matrix with the diagonal elements giving the frictional coefficients along the three principal axes of rotation, and off-diagonal elements giving the coefficients for rotation about the other axes.

measured.

3.8.1 Flow birefringence measurements

Flow and electric birefringence methods are based on the differences between the refractive index measured in different directions that are produced by oriented molecules in solution. The solution to be studied is placed between two concentric cylinders (Figure 3.7), of which one (generally the outer one) is rotated with respect to the other. This establishes a velocity gradient, which produces a torque on the molecule, and the molecules align their long axis along the flow lines, perpendicular to the radius (Figure 3.8). The higher the axial ratio of the molecules, the better the alignment. Now a parallel beam of light is passed through a polariser⁵, and then through the solution, and then through an analyser. To start with, as the molecules are randomly oriented, the solution is optically isotropic and all the incident light is blocked by the analyser. When the outer cylinder is rotated, the molecules become oriented as described above and the solution becomes birefringent, i.e. the refractive index of the solution is not the same in all directions. The light transmitted through the solute now becomes elliptically polarised and hence some of it will pass through the analyser. An image of the analyser appears bright everywhere excepting in four perpendicular regions where the optic axis of the molecule is parallel to the plane of polarisation of the incident light (Figure 3.7). These four regions form a dark cross, corresponding to regions of zero birefringence. The cross is seen initially at 45° to the polariser

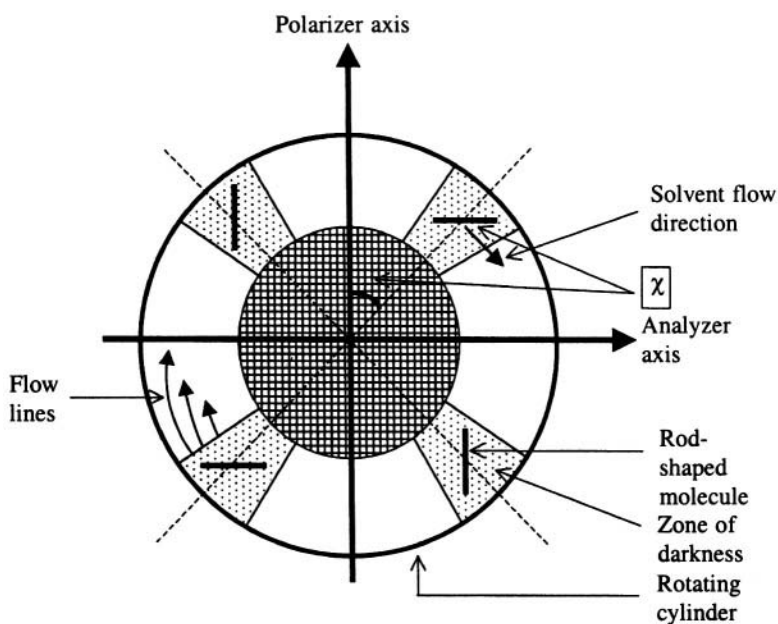
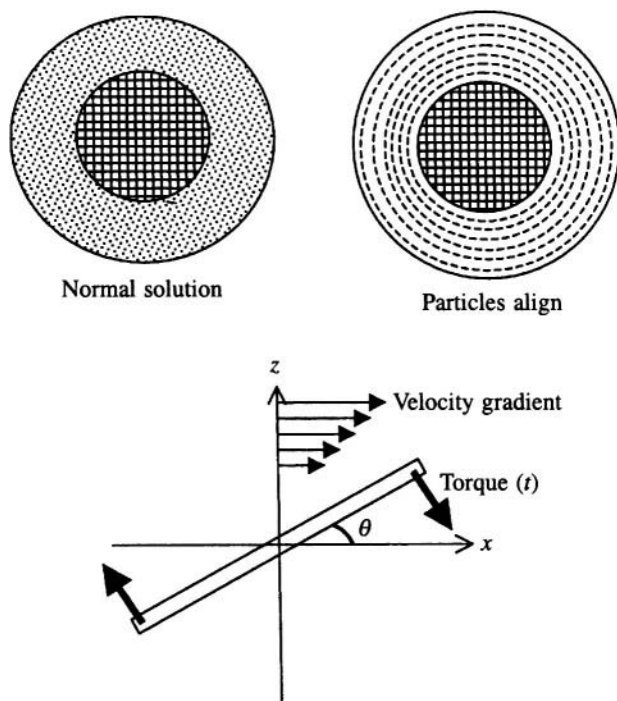


Fig. 3.7 Apparatus for the measurement of flow birefringence.

axis when there is a only slight degree of orientation of the molecules. As the degree of orientation increases, the dark cross rotates to align itself with the cross formed by the polariser and analyser.

The position of the cross is measured by the extinction angle χ (Figure 3.7). A plot of χ versus

⁵See Chapter 5 for an elementary discussion of the polarisation of light.



At $\theta = 0^\circ$, t is minimum; at $\theta = 90^\circ$, t is maximum, where t is the torque causing the molecules to rotate in the clockwise direction.

Fig. 3.8 Alignment of particles in flow birefringence

shear gradient G is made to determine $[\theta]$. The shear gradient depends on the speed of the outer cylinder, and the distance between the cylinders. Therefore χ , which is the angle between the macroscopic optic axis of the solution of oriented molecules and the direction of the solvent flow, is a measure of the degree of orientation. Rod-shaped molecules tend to orient as shown in Figure 3.7, with their long axis parallel or perpendicular to the polariser axis. The measurement of $[\theta]$ then enables the calculation of the frictional tensor using the equation above.

3.8.2 *Electric birefringence*

Instead of a velocity gradient, an electric field can be used to orient the sample. Electric birefringence is then an optical measurement of the orientation caused by the electric field. An electric pulse, generally a square wave, is applied to the solution for a sufficiently short duration such that there is no electrophoretic mobility, only rotational reorientation of the molecules (Figure 3.9). The macromolecules in the solution get preferentially aligned in a particular direction in the electric field. The degree of alignment depends on the electric properties of the molecule and is measured by the difference between the refractive index of the solution parallel to the applied field and the one perpendicular to it. This is known as form birefringence. Intrinsic birefringence refers to the difference in the refractive index between two perpendicular directions within a rod-like molecule such as DNA. The experiment to measure the electric birefringence consists of passing light through crossed polarisers, with the sample in the electric field placed in between. The light is finally received by a photomultiplier

tube connected to an oscilloscope. Since the molecules are oriented by the electric field, a static measurement of the birefringence at the equilibrium state will yield little information about the particle's shape. However, the decay in the birefringence, when the electric field is removed will depend on the molecular shape. This is measured as the decay of the birefringence signal and the rotational diffusion coefficient of the molecule is obtained.

3.9 Light Scattering

Light scattering is a technique widely used to determine the molecular weight of macromolecules in solution. A beam of light of a particular wavelength illuminates the sample. The experiment is sensitive to impurities hence great care must be taken to exclude dust particles from the solution. Radiation scattered by the sample, at the same wavelength is measured as a function of the angle between the incident beam and the detector. The chief difference between light scattering and visible range spectroscopy is that in the former technique, only elastic scattering by the particles in the solution is considered. This is the reason the measurement of scattering is made at the same wavelength as the incident beam. In this respect, the technique is similar to diffraction. However, in diffraction, we measure the interference patterns formed by the radiation scattered from different particles. In light scattering the solution is dilute enough to consider that the molecules scatter independently and are randomly oriented. Moreover they do not absorb the incident radiation, and therefore the scattered radiation has the same wavelength. Hence the term elastic scattering. In general, there are two

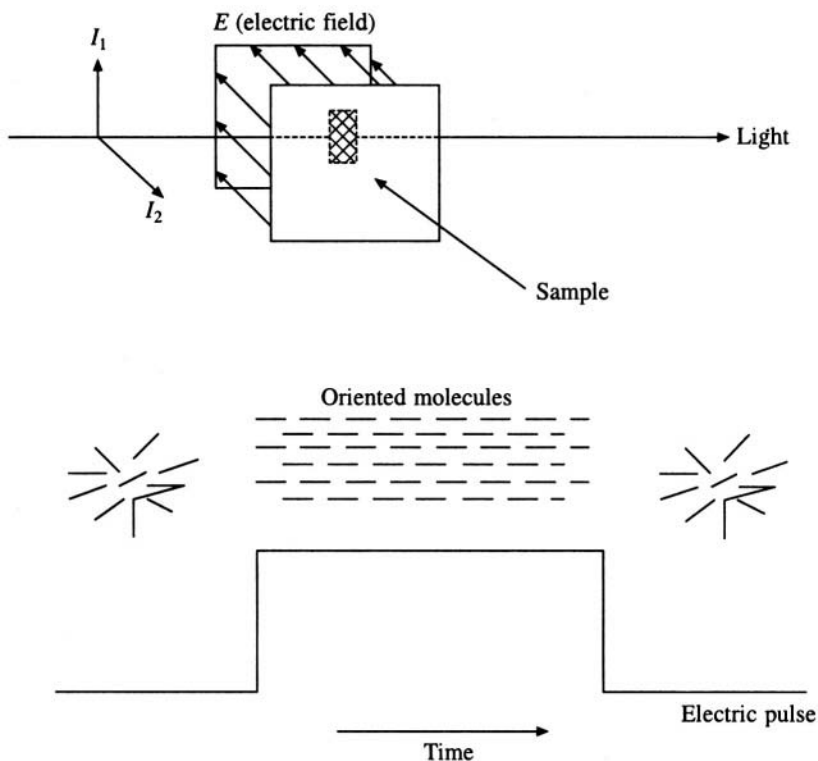


Fig. 3.9 Electric dichroism. The sample is placed in an electric field. On application of an electric pulse, the molecules in the sample become oriented along one direction. Removal of the electric field results in the molecules reverting to a disoriented state

situations: (a) when the particles are smaller than the wavelength, and (b) when they are comparable in size to wavelength.

Consider first the situation when the molecules are smaller than the wavelength of the incident and scattered radiation. Since light is electromagnetic radiation, it has electrical as well as magnetic components. But for the purposes of the present discussion we concentrate only on the electric field component E , and ignore the magnetic component. If the radiation is of frequency ν , and the maximum amplitude is E_0 then

$$E(t) = E_0 \cos(2\pi\nu t)$$

where t is the time. According to classical electromagnetic theory, when light strikes matter, electrons are displaced from their equilibrium positions with respect to the nucleus and a dipole is formed. The dipole moment p so induced is proportional to the electric field and is given by

$$p = \alpha E = \alpha E_0 \cos(2\pi\nu t)$$

where α is a constant of proportionality called polarisability. The polarisability is a measure of the ease with which an electron can be displaced from its equilibrium position. Since the incident electric field is oscillating constantly, the dipole also oscillates in magnitude, going from zero to a maximum and back. Again from electromagnetic theory, an oscillating dipole gives rise to scattered radiation, whose electric field is given by

$$E_s = [4\pi^2\nu^2\alpha^2 E_0 \cos(2\pi\nu t)]/c^2 r$$

in which c is the velocity of light and r the distance of the dipole from the detector. The ratio of the scattered radiation I_s to the incident intensity I_0 is

$$I_s/I_0 = E_s^2/E_0^2 = (16\pi^4\alpha^2)/\lambda^4 r^2$$

in which we have substituted $\nu = c/\lambda$. Thus the ratio is inversely proportional to the fourth power of the wavelength λ . This is called Rayleigh scattering. If, instead of unpolarised or circularly polarised light, we use light polarised along a particular axis then the induced dipole oscillates only in that direction. As a consequence the scattered radiation will also be polarised and the above equation has to be modified for such effects. Thus

$$I_s/I_0 = [(8\pi^4\alpha^2)/\lambda^4 r^2](1 + \cos^2 \theta)$$

for a single scattering particle. If there are N particles, then

$$I_s/I_0 = N[(8\pi^4\alpha^2)/\lambda^4 r^2](1 + \cos^2 \theta)$$

The Rayleigh ratio R_θ is the relative intensity of the scattered light and is given by

$$\begin{aligned} R_\theta &= r^2/(1 + \cos^2 \theta) V [I_s/I_0] \\ &= (8\pi^4\alpha^2 N')/\lambda^4 \end{aligned}$$

where $N' = N/V$ is called the number density of particles and is equal to CN_0/M (where C is the concentration, N_0 is Avogadro's number and M the molecular weight), and V is the volume of the solution. When we substitute for the polarisability α in terms of refractive index and concentration of the solution, we obtain

$$R_\theta = KMC$$

where K is a constant for a given solution incorporating the refractive index of the pure solvent, Avagadro's number, the incident wavelength and the rate of change of refractive index with concentration. Thus if R_θ is measured, and the concentration C is known, the molecular weight can be determined. When the sample is a heterogeneous mixture of several molecular species of concentrations $C_1, C_2, \dots, C_i$, then, instead of a single molecular weight, the light scattering experiment will yield

$$\sum_i M_i C_i$$

where M_i is the molecular weight of the component i . The weighted average molecular weight is then obtained as

$$\langle M_w \rangle = (\sum_i M_i C_i) / \sum_i C_i$$

The other situation is when the molecules in the solution have a size, which is comparable to the wavelength of the incident radiation. In the previous case, when the particle size was small, we could assume that the dipoles were oscillating in phase for all the scattering elements within the particle. However, in the present case, different parts of the molecule can be considered to scatter independently and thus the radiation emanating from the different portions undergo interference (Figure 3.10). This effect is taken care of by applying a correction factor P which is a function of θ to the Rayleigh scattering ratio. $P(\theta)$ can be expressed in terms of radius of gyration R_G of the molecule. The radius

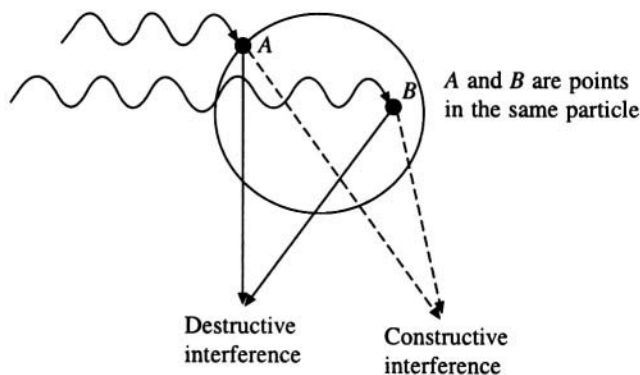


Fig. 3.10 Interference in light scattered from particles with size comparable to λ , the wavelength of the incident light

of gyration gives information about the shape of the molecule. Thus

$$R_\theta = KMCP(\theta) = KMC(1 - 16\pi^2 R_G^2 \sin^2 \theta / 3\lambda^2)$$

Or at the limit of infinite dilution (and since $1/(1-x)$ is approximately equal to $(1+x)$ for small values of x),

$$(KC/R_\theta) = (1/M)(1 + 16\pi^2 R_G^2 \sin^2 \theta / 3\lambda^2)$$

KC/R_θ is measured for various values of the scattering angle θ , and plotted against $\sin^2(\theta/2) + k'C$, where k' is an arbitrary constant used for scaling (Figure 3.11). This is known as the Zimm plot. The concentration data at each angle is extrapolated to $C = 0$. Similarly the angle-dependent scattering data is extrapolated to $\theta = 0$. Both the extrapolations intersect on the y axis giving $1/M$. The radius

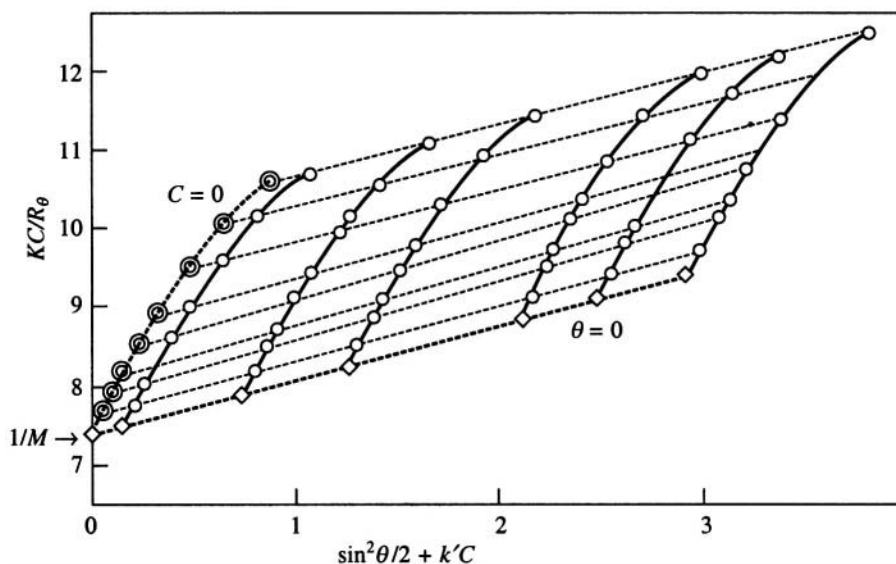


Fig. 3.11 Zimm plot for light scattering from a biomolecular solution. Experimental data ($\circ$), extrapolations to zero concentration ($\odot$) and extrapolations to $\theta = 0$ ($\diamond$), k' is a constant

of gyration can be obtained from the slope of the curve $C = 0$. The value of R_G depends on the shape of the particle. For a sphere of radius r , the radius of gyration is given by $R_G^2 = (3/5)r^2$, whereas for an ellipsoid of semi-axes a and b , $R_G^2 = (a^2 + 2b^2)/5$. For a long rod of length l , $R_G^2 = l^2/12$. In this manner one can obtain information about the molecular weight and the radius of gyration, and therefore the shape of the molecule, from the light scattering experiment.

3.10 Small Angle X-ray Scattering

This technique is analogous to that of light scattering. For an average small protein having a maximum dimension of 100 Å, light scattering can give information about the molecular weight but not about the shape of the protein, since the correction term $P(\theta)$ is close to unity. In order to obtain information on the shape of these molecules, radiation with a smaller wavelength, like X-rays, need to be used. Small angle X-ray scattering or SAXS is a diffraction technique. In order to understand the need for measurements at small angles, from the diffraction condition or Bragg's law, the angle of diffraction for an inter-planar spacing of 100 Å can be calculated to be

$$\sin \theta = \lambda / (2 \times 100) = 0.0077$$

for $\text{CuK}\alpha$ radiation of wavelength 1.5418 Å and $\theta \sim 0.45^\circ$. For a spacing of 1000 Å, θ is 0.045° . Thus, the angle (with respect to the direct beam) at which the measurements will have to be made to obtain information of this nature are very small. The experiment consists of allowing a narrow X-ray beam of known wavelength to be incident on a dilute solution of the sample, and then measuring the scattered intensity as a function of the angle. Since the scattering angle of interest is only a few degrees, at the most, from the direct beam, great care must be taken to collimate the beam to make it as thin as possible. Also in the measurement of the scattered intensity, the direct beam must be carefully and completely excluded. As in light scattering experiment, since the solution is sufficiently dilute, we assume that the molecules scatter independently. For each molecule, the molecular structure

factor is defined as the sum of individual scattered X-rays from all the atoms in the molecule. Taking into consideration the difference in scattering between the different relative orientations of neighbouring molecules and integrating the structure factor over all possible relative orientations, we obtain the expression for the scattering intensity as

$$\langle I(\theta) \rangle = n_e^2 (1 - 16\pi^2 R_G^2 \sin^2 \theta / 3\lambda^2)$$

for small values of θ , where R_G is the radius of gyration and n_e is the number of electrons in the molecule (as obtained from a knowledge of the molecular weight). This equation indicates that the zero-angle X-ray scattering from the solution is approximately proportional to the square of the molecular weight of the particle. A plot of the intensity versus $\sin^2 \theta$ yields the radius of gyration. A more common way of using this expression is to rewrite it as

$$\ln [I(\theta)/I(0)] = -(4\pi R_G \sin \theta / \lambda)^2 / 3$$

Now the small angle scattering can be plotted as $\ln [I(\theta)/I(0)]$ versus $\sin^2 \theta / \lambda^2$. This plot is called the Guinier plot and it yields the radius of gyration even in the absence of information about the molecular weight. R_G in turn gives information on the shape of the molecule.

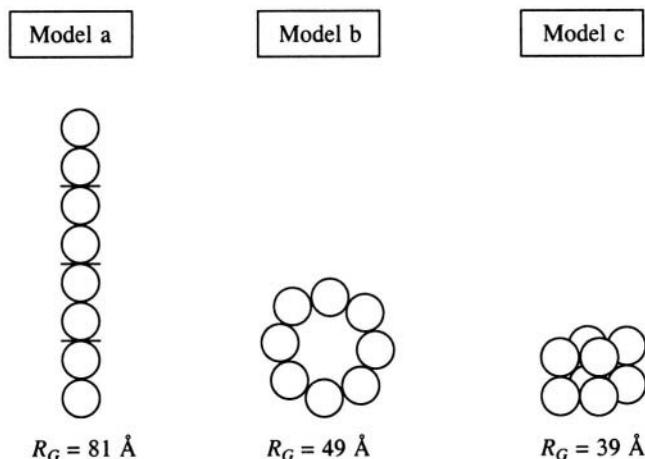


Fig. 3.12 Models for the quaternary structure of α -lactalbumin at pH 4.5

As an example of the application of SAXS consider the experiments on α -lactalbumin. At pH 4.5 these gave the radius of gyration as $34 \text{ \AA}$. From previous experiments it was known that this protein exists in solution as a tetramer of four dimeric units. Various models for the arrangement of these units (Figure 3.12) were considered and the radius of gyration calculated for each was compared with the experimental value. It is clear that model c in Figure 3.12 is likely to be the correct one.

The scattering of neutrons from solutions can also be used to analyse the shape and size of molecules in solution. The techniques and the theory are similar to SAXS. The major difference is that neutrons are scattered by the atomic nucleus, while X-rays are scattered by the electrons around the nucleus. Therefore, the two techniques may be used in a complementary fashion. In SAXS and other diffraction experiments using X-rays, the intensity of the radiation scattered is a measure of the electron density in the sample and is proportional to the atomic scattering factor. On the other hand, the intensity of neutron scattering is proportional to a factor called the scattering length of the atomic

nucleus. Each element has a different value for the scattering length (Table 3.5). This value depends on the precise nature of the interactions of the incident neutrons with the nucleus. It therefore does not show any relationship with the atomic number or atomic weight. This is unlike the case for X-ray scattering, which is related to the atomic number. The scattering length is a measure of the scattering

Table 3.5. Values of the neutron scattering lengths for some biologically relevant elements

Element	Neutron scattering length ($b \times 10^{-13}$ cm)
H	-3.74
D	6.67
C	6.65
N	9.40
Fe	9.51
Pt	9.50

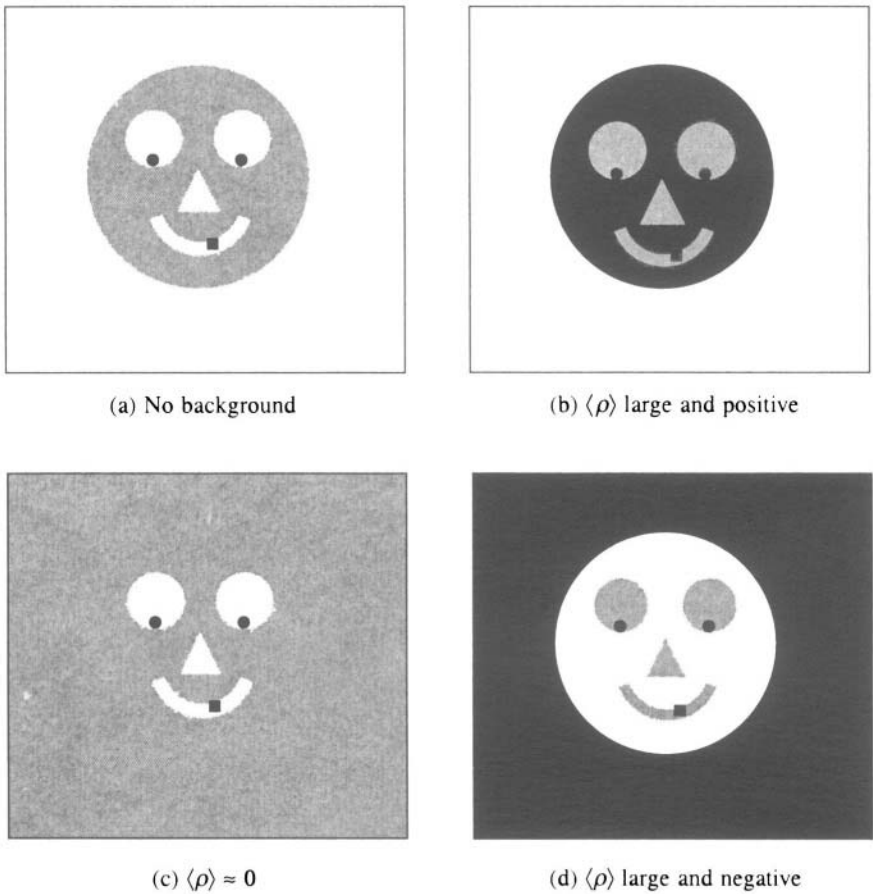


Fig. 3.13. Effect of contrast in neutron scattering

power. Neutron scattering lengths can be calculated from absolute scattering intensity measurements. Note that the scattering length for hydrogen is negative and of a magnitude comparable to heavier atoms such as carbon. Thus unlike as in the case for X-rays, the contribution of hydrogen atoms to neutron scattering is clearly distinguishable. Neutron scattering can give supplementary information on hydrogen atoms which is not available from X-ray scattering.

Another interesting feature is that the neutron scattering from deuterium is positive and of higher magnitude compared to scattering from hydrogen. By replacing different proportions of water (H_2O) with deuterated water (D_2O) in the solvent, its scattering density can be varied at will. In this way the contrast between solute and solvent can be manipulated to observe details of interest. The contrast $\langle\rho\rangle$ is the mean difference in scattering density between the solute and the solvent (Figure 3.13). Thus,

$$\langle\rho\rangle = \langle\sigma_{\text{macromolecule}}\rangle - \langle\sigma_{\text{solvent}}\rangle$$

where $\langle\sigma\rangle$ represents the average neutron scattering intensity of the appropriate component of the solution. When there is no background, the object will appear as in Figure 3.13(a). When $\langle\rho\rangle$ is positive and large, the object will appear as in Figure 3.13(b). When $\langle\rho\rangle$ is zero then the shape of the object will not be seen (Figure 3.13(c)) but internal details will be visible. When $\langle\rho\rangle$ is large and negative we see the object as in Figure 3.13(d). Thus contrast plays an important role in determining the amount of detail that can be obtained.

Deuterium can also be covalently substituted for hydrogen in cases where the sample is composed of two components. One of the components is labeled with H and the other with D . Then by adjusting the H_2O – D_2O composition of the solvent, the situation which can be arrived at is

$$\text{either} \quad \langle\sigma_1\rangle \equiv \langle\sigma_{\text{macromolecule}}\rangle$$

$$\text{or} \quad \langle\sigma_1\rangle \equiv \langle\sigma_{\text{solvent}}\rangle$$

and the scattering due to one of the components will be blacked out. For example, when dealing with icosahedral viruses, the protein coat encapsulating the nucleic acid molecule of viral genome usually makes it difficult to observe the latter in scattering experiments. If we consider that the neutron scattering length per unit volume for fully protonated protein is $3.1 \times 10^{14} \text{ cm}/\text{\AA}^3$, while that of nucleic acid is $4.4 \times 10^{14} \text{ cm}/\text{\AA}^3$, and that for fully deuterated protein it is $8.5 \times 10^{14} \text{ cm}/\text{\AA}^3$, and for fully deuterated nucleic acid it is $7.4 \times 10^{14} \text{ cm}/\text{\AA}^3$, we see that by selective labeling, the nucleic acid core and the protein coat of viruses can be observed independent of each other.

Spectroscopy

4.1 Introduction

Electromagnetic energy depends on the wavelength (or frequency) of the radiation. Different wavelengths are useful in probing different aspects of materials. Table 4.1 indicates how the various regions of the

Table 4.1 The electromagnetic spectrum and the corresponding physical experiments

Name	Wavelength (meters)	Frequency (Hz)	Use
X-rays	$10^{-12} - 10^{-8}$	$10^{20} - 10^{16}$	Diffraction, small angle scattering
Ultraviolet (UV)	$10^{-8} - 4 \times 10^{-7}$	$10^{16} - 7.5 \times 10^{14}$	Electronic structure of molecules
Visible	$4 \times 10^{-7} - 7.5 \times 10^{-7}$	$7.5 \times 10^{14} - 4 \times 10^{14}$	Electronic structure of molecules
Infrared (IR)	$7.5 \times 10^{-7} - 10^{-3}$	$4 \times 10^{14} - 10^{11}$	Vibrational and rotational spectra of molecules
Microwave	$1 \times 10^{-3} - 1$	$10^{11} - 10^8$	Rotational spectra
Radiowaves	$1 - 10^3$	$10^8 - 10^5$	NMR

electromagnetic spectrum are used.

In a spectroscopic experiment electromagnetic radiation of a particular frequency or a range of frequencies is allowed to fall on the sample under study. The radiation coming out of the sample is then analysed in terms of the intensity at different frequencies. This indicates which particular frequencies are absorbed (or emitted) by the molecule, thus giving a picture of the molecular energy levels. These, in turn, can be interpreted in terms of the chemistry or stereochemistry of the molecule. Under ordinary circumstances, all the atoms and molecules would exist in the ground state energy level. When, due to the incident radiation, electrons in the outer shell are raised by one energy level, the phenomenon is known as resonance absorption and the technique of atomic absorption spectroscopy

is based on this. The energy required lies in the ultraviolet region of the electromagnetic spectrum. Molecular spectra have additional complexities arising from vibrational and rotational motion. In this case, transitions between electronic levels are found in the ultraviolet and visible regions, those between different vibrational levels but within the same electronic level are in the infrared region, and those between rotational levels are in the infrared and microwave regions. Ultraviolet absorption spectra in molecules appear as wide bands. This is because the molecule usually contains sets of closely spaced electronic energy levels, and the transitions between these levels, as indicated by the spectra, are difficult to resolve. Fluorescence Spectroscopy of molecules also makes use of the absorption phenomenon but in a different way. A part of energy gained by a molecule by absorption of electromagnetic radiation may be dissipated away, for example, as heat. The remaining portion of the energy is radiated as the molecule returns to its ground state. Since the energy now is not as much as was originally absorbed, the frequency of emitted radiation is less than that of the incident radiation. This process is called fluorescence. For example, many organic compounds fluoresce in the visible region after being irradiated with ultraviolet radiation. Many molecules have the ability to rotate the plane of polarisation of plane-polarised light. This property is called optical activity. Its variation with wavelength of the light is called optical rotatory dispersion. A related phenomenon is circular dichroism, where the polarised light is split into left and right circular polarisation and the difference in the absorbance of the two components is measured. All these spectroscopic techniques give a greater or lesser amount of information about the stereochemistry and conformation of the molecules in the sample. They are of great utility in studies of biological molecules. We will now treat each of them in greater detail.

4.2 Ultraviolet/Visible Spectroscopy

If a beam of light of intensity I_0 falls on a cell containing the sample, the emergent radiation intensity I is less intense than the incident radiation, since a portion of it is absorbed by the sample. The absorption is different at different wavelengths and is characteristic of the sample. This characteristic quantity is called the absorbance and may be calculated from the Beer-Lambert principle. The Beer-Lambert law states that in a sample, each successive portion along the path of the incident radiation, containing an equal number of absorbing molecules absorbs an equal fraction of the radiation that traverses it. If we consider an infinitely thin slice of the sample (dl), the light traversing it may be considered a constant. Then the fraction of the light absorbed is simply proportional to the number of absorbing molecules,

$$-dI/I = C\epsilon' dl$$

where C is the concentration of the molecule in moles per litre, ϵ' is a proportionality constant called the molar extinction coefficient and the negative sign indicates a diminution of the intensity of the radiation as it traverses the sample. ϵ' is a constant for a given wavelength for a given molecule and is independent of the concentration. Integrating this equation over the entire sample thickness l , we obtain

$$A = \log(I_0/I) = C\epsilon(\lambda)l \quad (4.1)$$

where $\epsilon = \epsilon'/2.303$, i.e. molar extinction coefficient converted to log base 10. It is expressed as a function of wavelength λ . The term $\log(I_0/I)$ is called the absorbance A . In an experiment, if A is measured at each wavelength λ , $\epsilon(\lambda)$ can be calculated and tabulated. Usually, the wavelength $\lambda_{\max}$ at the maximal extinction and the extinction coefficient $\epsilon_{\max}$ at this wavelength are used to characterise the sample. Sometimes, especially for biopolymers, the molar extinction coefficient is difficult to

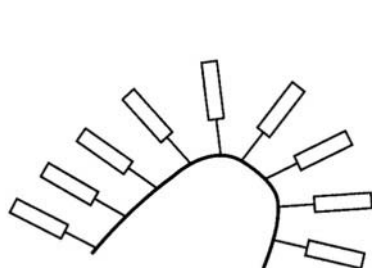
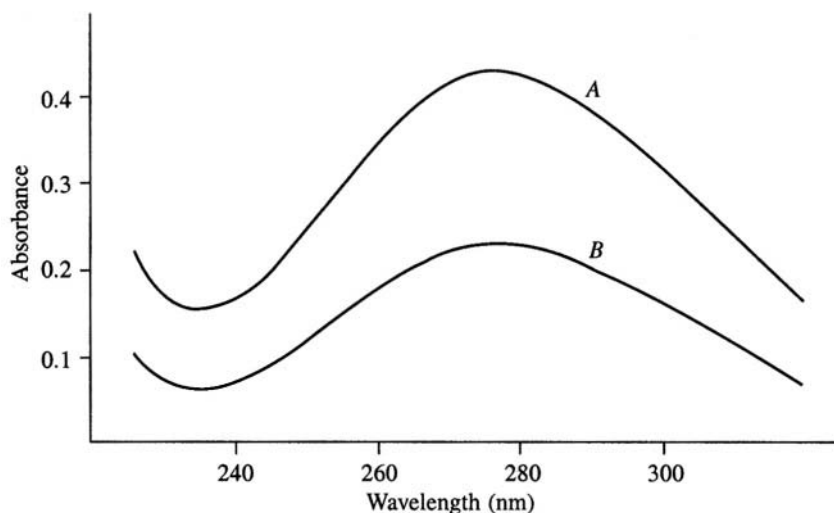
measure. Instead the average extinction per residue is calculated. In the actual experimental set up, in order to eliminate the effect of the solvent, the effect of the path length, etc., what is measured usually is the difference in absorbance between a cell containing the sample, and one containing the solvent alone. A double beam spectrophotometer has a beam splitter that splits the beam into two equidistant paths. The sample is placed in the path of one half of the beam and the reference in the other.

UV/Vis absorption spectroscopy is used frequently in biophysical research for simple tasks such as determining the concentration of a sample, as also for answering complex structural questions. Electronic transitions between molecular orbitals are responsible for the spectrum and each spectrum characterises the quantum mechanical state of the molecule. The most commonly observed transitions in biomolecules are Between the: (a) ground state π orbital and the excited π^* orbital, i.e. $\pi \rightarrow \pi^*$ transition and (b) nonbonding n orbital and the π^* orbital, i.e. $n \rightarrow \pi^*$. Though the σ orbitals do not usually participate, $n \rightarrow \sigma^*$ transitions have also been observed. At any given wavelength, one particular chemical group in the molecule will dominate the absorption spectrum. Such groups are called chromophores. In biological molecules such as nucleic acids and proteins the chromophores absorb light at wavelengths less than about **300 nm**⁶. Biologically interesting macromolecules cannot be studied in the gas phase and the presence of the solvent itself becomes a constraint since the chromophores of the solvent will dominate at certain wavelengths. Thus, the most common solvent, water, absorbs very strongly below 170 nm, restricting useful study to wavelengths above this value. Protein molecules have three types of chromophores, namely the peptide group, the amino acid side chain groups and any prosthetic groups that may be present. The peptide group has an $n \rightarrow \pi^*$ absorption band at 210–220 nm. The intensity of this band is much smaller than the intense $\pi \rightarrow \pi^*$ transition at 190 nm, which may be considered a ‘signature’ absorption of the peptide group. A third transition corresponding, tentatively, to the $n \rightarrow \sigma^*$ transition in the peptide, is also sometimes observed. As far as the amino acid side chains are concerned, many of them absorb in the same region as the peptide group. In the case of the aromatic amino acids tryptophan, tyrosine and phenylalanine, the dominant band is between 230 and 300 nm, i.e. above the peptide bands. The most intense is the 280 nm absorption of tryptophan. Tyrosine also has an absorption band at almost the same wavelength, though somewhat weaker. Phenylalanine is much weaker and at approximately 260 nm. It is usually observed only in the absence of significant amounts of tryptophan or tyrosine in the protein. The absorption bands of the side chains change with the pH of the solvent. This is mainly due to the change in the protonation of the chromophores. For example, the removal of the OH proton in tyrosine at pH > 10.9 shifts the absorbance band from 274 nm to about 295 nm. Disulphide formation changes the spectrum of cysteine, but the absorption is weak and difficult to observe. The other possible chromophores in protein arise from the prosthetic groups such as the heme group, flavin, etc. Routine measurements of the concentration of protein solution assume that 1 mg per ml of protein will have an absorbance of 1.0 at 280 nm, i.e. **$A_{280} = 1$** . This number was arrived at after a series of experiments on proteins with a known number of tyrosines and tryptophans. If the measurement is made at 230 nm, **$A_{230} = 3$** for a solution containing 1 mg of protein per ml of solution.

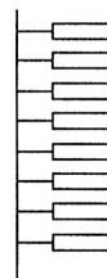
The UV absorption of nucleic acids is almost entirely due to the bases. The sugar-phosphate backbone has little or no contribution at wavelengths greater than 200 nm. The spectra of the individual bases are quite complex, but are all marked by a strong, broad band near 260 nm. Each of the bands corresponds to several $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ transitions. Unusual bases such as the Y base (**$\lambda_{\max} = 325$ nm**) and 4-thiouridine (**$\lambda_{\max} = 340$ nm**) found in tRNA have the absorption maxima far away from the

⁶nanometre = 10^{-9} metre

positions for the normal bases. This leads to their use in monitoring tRNA structure and interactions. Nucleic acid polymers show dramatic hypochromism and hyperchromism. These are effects that arise due to aggregation of chromophores. Quantum mechanical theory says that due to interactions between the electric dipoles present in the chromophores, groups of chromophores, which aggregate together, have different effects on the absorption spectrum. In a polynucleotide, if the bases are stacked one on the other, as for example in the double helical structure, this leads to a diminution of the intensities of absorption. The effect is most noticeable at the 260 nm band and is called hypochromism. If on the other hand the bases are arranged in almost the same plane, i.e. in an end-to-end arrangement, the absorption intensity is increased leading to hyperchromism (Figure 4.1). Hyperchromism is commonly used to follow the denaturation behaviour or the melting behaviour of the DNA double helix. At higher temperatures or at other disruptive conditions, when the base-base stacking is disturbed and the double helix is denatured, the intensity of absorption at 260 nm increases.



Curve A: Bases are destacked



Curve B: Bases are stacked

Fig. 4.1 The hyperchromic A and B effects in nucleic acids.

4.3 Circular Dichroism (CD) and Optical Rotatory Dispersion (ORD)

Most biological molecules or systems are optically active; i.e. they rotate the plane of polarised light. If the rotation is clockwise when seen looking towards the light source, it is said to be dextro-rotatory.

If the rotation is counter clockwise, it is levo-rotatory. For any compound, the extent of rotation depends on the number of molecules in the path of the polarised light, or in other words, for solutions, on the concentration and on the path length of the beam through it. It also depends on the wavelength of the radiation and the temperature. Optical activity is quantified by the specific rotation

$$[\alpha]_{\lambda}^t = \frac{\alpha^0}{dc}$$

i.e. the specific rotation $[\alpha]$ at a temperature t and for wavelength λ is the measured rotation α^0 divided by the concentration c in grams per cubic centimetre and the light path d in centimetres. Optical activity may also be quantified as the molar rotation

$$[M]_{\lambda}^t = \frac{\alpha M}{dc}$$

where M is the gram molecular weight of the substance. For polymers, it is common to use the mean residual rotation

$$[m]_{\lambda}^t = \frac{\alpha M_0}{dc}$$

where M_0 is the mean residue molecular weight, i.e. the average molecular weight of the monomers. Rather than the rotation at a single wavelength, measurement of the change in optical activity with change in wavelength yields more useful structural information. This is called Optical Rotatory Dispersion (ORD). A closely related phenomenon is Circular Dichroism (CD). Plane polarised light can be resolved into two beams, each circularly polarised, but in opposite senses (Figure 4.2). The

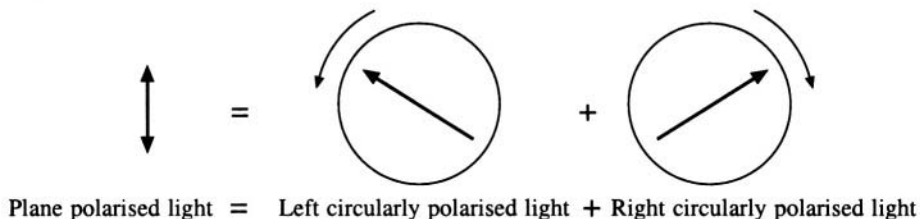


Fig. 4.2 Plane polarised light can be resolved into two components

refractive index for the right circularly polarised light, n_R may be different from that for the left circularly polarised light n_L . Similarly the absorbance A_R for the right circularly polarised light may be different from A_L . When n_R and n_L are unequal, each component will be retarded to a different extent as it passes through the sample. This leads to a change in the phases of the two components when they make up the plane-polarised light, leading to a rotation of the plane of polarisation. This is in fact the theoretical basis for the phenomenon of optical activity. Since the refractive index is always dependent on the wavelength, the angle of rotation is also wavelength-dependent. At a given wavelength λ , the angle of rotation

$$\alpha_{\lambda} = (180d/\lambda)(n_L - n_R)$$

where d is the path length. α , or more commonly, the specific rotation $[\alpha]_{\lambda}$ can be plotted as a function of wavelength (Figure 4.3). This is called the Optical Rotatory Dispersion (ORD) spectrum. The difference in the absorption between the right and the left components does not introduce a phase

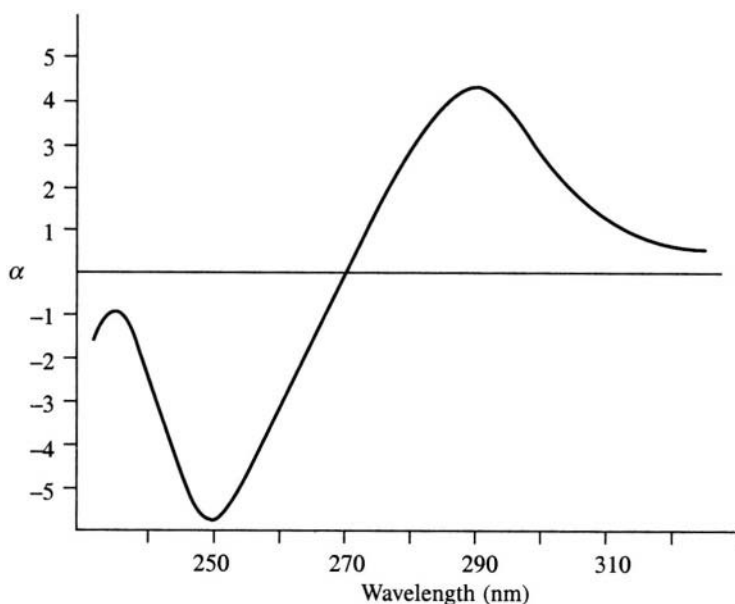


Fig. 4.3 A typical ORD spectrum.

difference between the two, but instead produces a difference in intensity, i.e. a difference in magnitude of the wave. The resultant beam is no longer plane polarised but is elliptically polarised (Figure 4.4). The difference in absorbance $A_L - A_R$ is called the circular dichroism or CD. The ellipticity of the ellipse is also a measure of the differential absorbance, known as the ellipticity θ .

$$\theta(\lambda) = 2.303(A_L - A_R)180/4\pi = 33(A_L - A_R)$$

where the absorbance A is defined as in equation (4.1) for a particular wavelength λ . The molar ellipticity is defined as

$$[\theta(\lambda)] = M\theta(\lambda)/10dc$$

where M is the molecular weight, d is the path length in centimetres and c is the concentration in

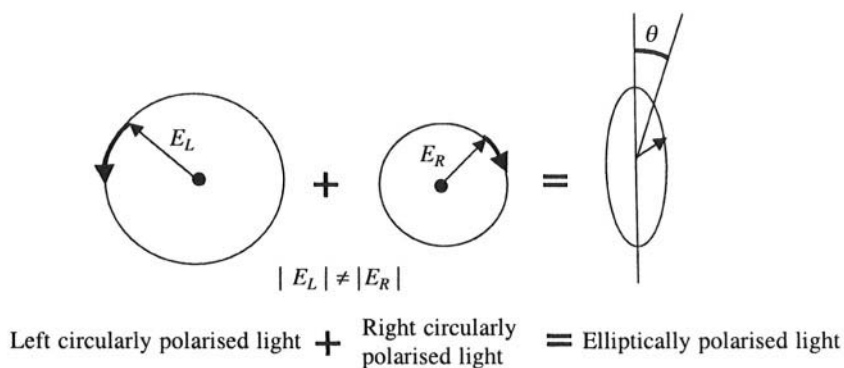


Fig. 4.4 Elliptically polarised light.

grams/ml. $[\theta(\lambda)]$ is measured in degrees. A plot of the difference in the refractive index, i.e. $(n_L - n_R)$ versus the wavelength is called the ORD curve and a plot of the difference in absorbance, i.e. $(A_L - A_R)$ versus the wavelength is called the CD curve. As seen in Figure 4.5, the two phenomena are related and together are called the Cotton effect. Both CD and ORD curves can be negative or positive and go by the names 'negative Cotton effect' and 'positive Cotton effect' respectively. Rigorously, the measurement of either CD or ORD implies the other. However, in practice, especially with the availability of appropriate instrumentation, CD is the measurement of choice.

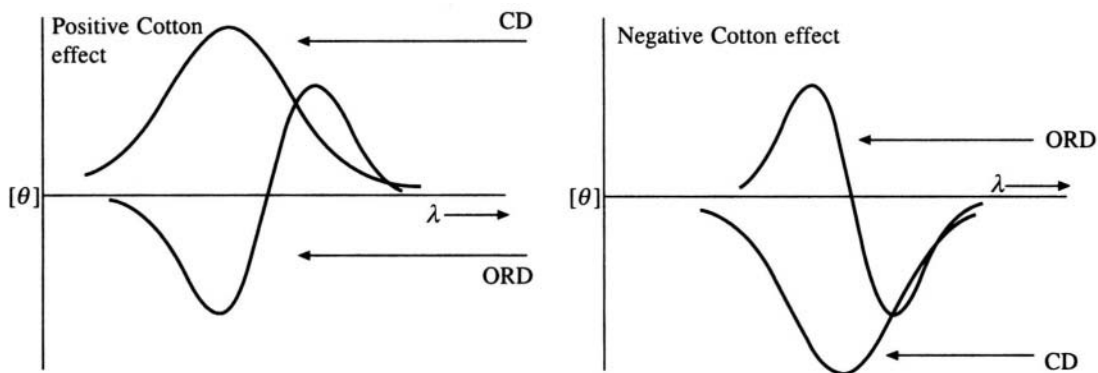


Fig. 4.5 Positive and negative Cotton effects illustrated by CD and ORD curves, respectively

The CD spectrum of a polymer is different from that of a monomer, the difference reflecting the three dimensional arrangement. CD, therefore, is immensely useful to study three-dimensional structures of biopolymers such as proteins and nucleic acids. The $\pi \rightarrow \pi^*$ transitions discussed previously dominate the spectra, and in principle, quantum mechanical calculations of the CD of the biopolymers can be performed. This would allow the interpretation of the spectrum in terms of the structure. This however is an extremely complex task and, in practice, the existing optical activity data for polypeptides, proteins and nucleic acids with known structure are used in conjunction with semi-empirical equations to analyse unknown systems.

For proteins the aim is to analyse the CD spectrum to obtain secondary structural features such as the percentage of α -helical regions or β -sheet regions or random coil regions. The protein CD spectrum is dominated by the polypeptide backbone chromophores and the effects of the side chains are usually negligible. It can be considered as a linear combination of the spectra of each of the regular secondary structural regions. One may thus write

$$[\theta(\lambda)] = k_\alpha[\theta_\alpha(\lambda)] + k_\beta[\theta_\beta(\lambda)] + k_\gamma[\theta_\gamma(\lambda)]$$

where $[\theta(\lambda)]$ is the measured ellipticity k_α , k_β and k_γ are the fractional compositions of α -helix, β -sheet and random coil regions respectively, and $[\theta_\alpha(\lambda)]$, $[\theta_\beta(\lambda)]$ and $[\theta_\gamma(\lambda)]$ are the measured CD of polypeptides in the conformation indicated. This equation can be used to estimate k_α , k_β and k_γ for an unknown protein, provided $[\theta_\alpha(\lambda)]$, $[\theta_\beta(\lambda)]$ and $[\theta_\gamma(\lambda)]$ are known. There are two methods of obtaining this information; both of them giving equivalent results when applied to unknown proteins. In the first method homopolypeptides that are known to possess one of the above secondary structures are chosen and the CD spectrum is measured for each. These spectra are used as the basis for calculating the fractional compositions. The other semi-empirical method obtains the basis values of the $[\theta_\alpha(\lambda)]$, $[\theta_\beta(\lambda)]$ and $[\theta_\gamma(\lambda)]$ from the CD spectra of proteins of known three-dimensional structure.

The second method is theoretically more attractive as the spectra will automatically include effects of length of the polypeptide on tertiary interactions.

In the case of nucleic acids, the most intense chromophores are the bases. One cannot therefore neglect the sequence and consider the backbone alone. A further contribution to the intensity arises from the stacking of the bases. Semi-empirical theoretical calculations of the CD spectrum of di- or tri-nucleotides involve treating the ellipticity due to the molecule as the sum of those due to the individual bases plus a strong contribution from the base-base stacking interactions. Thus for a trinucleotide with the bases B_1 , B_2 and B_3

$$[\theta_{B_1B_2B_3}(\lambda)] = 1/3[\theta_{B_1}(\lambda)] + 1/3[\theta_{B_2}(\lambda)] + 1/3[\theta_{B_3}(\lambda)] + I_{B_1B_2} + I_{B_2B_3} + I_{B_3B_1}$$

where $I_{B_1B_2} \dots$ etc., are due to the base-base interactions. Note that the term $I_{B_3B_1}$ represents a 'next nearest neighbour' interaction rather than the 'nearest neighbour' interaction of the other two terms. Normally one needs to consider only the 'nearest neighbour' interactions. This approach of building up the CD spectrum of a polynucleotide from the spectrum of the monomer can be extended quite successfully to longer chains, though the number of contributions then increases. For DNA the base-paired double-helical structure must also be taken into account. Accurate determinations of the average structures of several different synthetic polynucleotides with several different structures have been made. A dramatic example of the use of CD was when an inversion of the high salt spectrum of poly d(CG), led to the prediction of a novel left-handed double helical structure.

4.4 Fluorescence Spectroscopy

Fluorescence is a phenomenon by which light is absorbed by a system at one wavelength and emitted by it at a different, and longer, wavelength. Measurements of fluorescence give information about molecular conformation and dynamics and form an important part of the techniques used in biophysics. The basic principle of fluorescence can be understood in terms of the electronic energy levels in one of which, according to quantum mechanics, the molecule is usually found (Figure 4.6). Each of these electronic levels themselves has a so-called 'fine structure' due to the vibrational motion of the molecules, i.e. each electronic level is not a single energy value but consists of a band of closely spaced energies which arise due to the different vibrational modes. The molecule usually exists in the ground state. When a beam of light (visible or UV) falls on it, some of the energy is absorbed and electrons jump up to the excited state where they can go to any of the vibrational levels. So far the experiment is the same as UV/visible absorption spectroscopy. However, now the electron in the upper electronic level can lose some of its energy in many ways. One of these is by going to a lower vibrational level within the same electronic level. When now the electron jumps back to the ground state, it does so by emitting radiation, but the radiation is of less energy and therefore longer wavelength than the incident light. The molecule is said to fluoresce. A study of the intensity of the fluorescent radiation as a function of the wavelength is known as fluorescence spectroscopy. Fluorescent radiation is spontaneously emitted radiation as opposed to stimulated emission. It is governed by a rate constant k_F

$$k_F = A_{ba} = 1/\tau_F$$

where τ_F is called the radiative lifetime of the excited electronic state (b) of the molecule. A_{ba} is the Einstein coefficient of spontaneous transition from state (b) to state (a), which is the lower, ground state. A_{ba} can be evaluated in terms of the frequency of the incident light and the strength of electric moment dipole corresponding to the states (a) and (b). The molecule can also go from the excited state

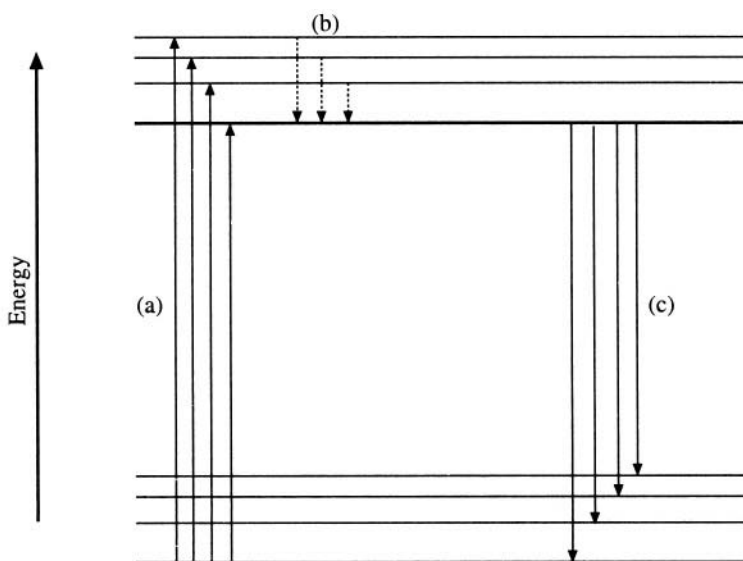


Fig. 4.6 Electronic energy levels in a molecule, indicating the fine structure, and possible transitions. (a) Transitions from ground to excited state. (b) Non-radiative loss of energy. (c) Radiative transitions to lower state

to a lower state by several other mechanisms that do not involve emission of radiation. For example, the excitation energy may be lost through collisions with the solvent molecules or by increasing the vibration of the molecules. The rate at which energy is lost through this process is called k_{ic} for 'internal conversion'. Another way in which the energy can be dissipated is by a phenomenon called quenching, resulting from the formation of complexes with solute molecules. The rate constant governing this process is called k_Q . A third process is when the excited state is converted to another, excited state of a lower energy though the transition that may be normally disallowed by quantum mechanics. The new excited state then dissipates its energy by internal conversion or by emission at a much longer wavelength than the fluorescence. The rate constant for this process is designated as k_{is} for 'intersystem crossing'. Thus the actual proportion of the absorbed incident light which is reemitted as fluorescent radiation is small and is called the quantum yield. It is given by the expression

$$\phi_F = k_F / (k_F + k_{ic} + k_Q + k_{ia})$$

Chromophores or UV/visible absorption groups in biological molecules like proteins and nucleic acids have rather low quantum yields and very short radioactive lifetimes. For example, the most intensely absorbing group among the constituents of proteins is the tryptophan side chain. Its absorption maximum is at 280 nm while its fluorescence emission maximum is at 348 nm. It has $\phi_F = 0.20$ and $\tau_F = 2.6$ nanoseconds. The tyrosine side chain absorbs at 274 nm, emits at 303 nm, $\phi_F = 0.14$, $\tau_F = 3.6$ nanoseconds. Phenylalanine absorbs at 257 nm, emits at 282 nm, has a very poor quantum yield of 0.04 and a lifetime of 6.4 nanoseconds. Other side chains have yields much lower than this. In nucleic acids, the bases have yields of the order of 10^{-4} and lifetimes less than 0.02 nanoseconds. It may be noticed that the strongest fluorescent chromophores are rigid planar moieties, since in such groups, the energy dissipation by vibrational and rotational motion is much less. The flexibility of the nucleic acid backbones and of the other side chains and backbones of proteins is responsible for the fact that virtually no fluorescence is detected from them.

Nevertheless, since fluorescence is extremely sensitive to the environment of the chromophore, it is an effective technique to study conformational changes and ligand binding. The studies are enhanced by the use of fluorescent probes (Figure 4.7). Some of these have the property that their fluorescence is strongly quenched in an aqueous solvent, but is significantly elevated in a rigid or non-polar environment. The binding of ethidium bromide to double-stranded DNA or RNA by intercalation between the base pairs can therefore be followed very easily by fluorescence spectroscopy and this ligand is used as a probe to detect the double helical structure of nucleic acids. The use of extrinsic chromophores, as these ligands are called, helps also in studying phenomena related to proteins such as the binding of the substrate, the position of various residues in the protein, etc.

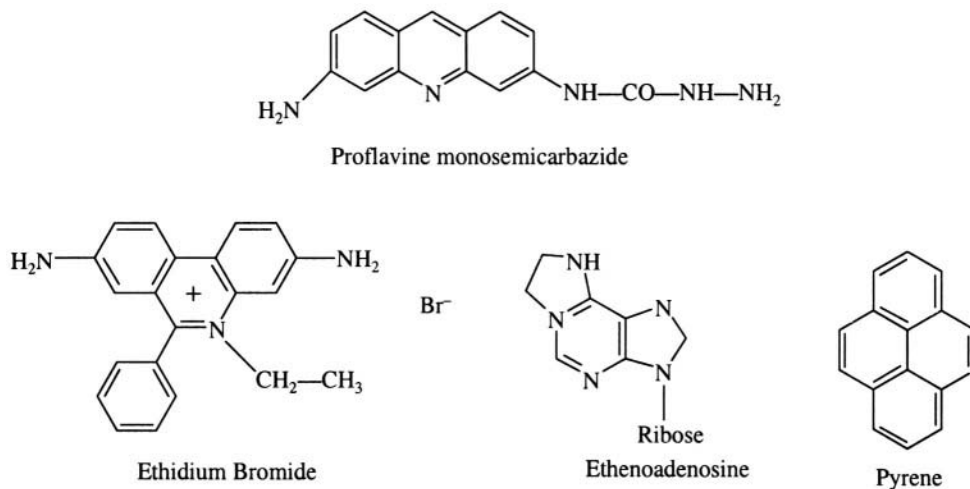


Fig. 4.7 Some common fluorescent probes

If polarised light is used as the incident beam, a measurement of the polarisation state of the emitted fluorescence yields information on the mobility of the chromophores. This is because as mobility increases, the plane-polarised light becomes more and more circularly polarised. Applications of polarisation fluorescence spectroscopy include the study of the helix-coil transition in proteins and the orientation of DNA in chromosomes.

A word about the experimental set up (Figure 4.8). Unlike as in absorption spectroscopy or in CD/ORD, the source of radiation, the sample and the detector are not placed along the same straight line. This is because fluorescence emission occurs in all directions and placing the detector at right angles increases the sensitivity by cutting out the direct beam.

4.5 Infrared Spectroscopy

Infrared radiation is heat radiation and term 'infrared' extends from 10^{14} to 10^{11} Hz on the electromagnetic spectrum. The energy of infrared radiation corresponds to energy differences between different vibrational modes in molecules. Infrared spectroscopy is therefore a probe of the vibrational motion of the molecules. Not all types of vibrations can be detected by IR spectroscopy. Since the technique involves the interaction of electromagnetic radiation with the molecules only those vibrational transitions which are accompanied by a change in the dipole moment can be detected. This means that diatomic molecules such as H_2 , O_2 , N_2 , do not exhibit infrared absorption and the vibration states of such molecules cannot be examined through IR spectroscopy. This is because the dipole moment is a

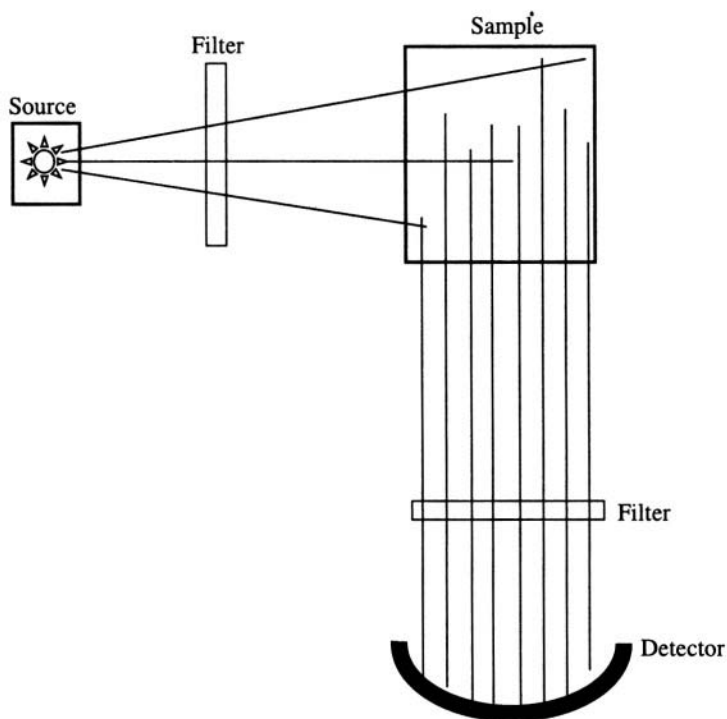


Fig. 4.8 A schematic sketch of the experimental setup used in fluorescence spectroscopy

measure of the separation of the centres of total positive charge and the total negative charge of the molecule. Diatomic molecules can only undergo symmetric vibrations, which will not change the dipole moment. For similar reasons, certain types of vibrations in particular molecules are IR-active, while others are IR-inactive. For example, in carbon dioxide, a symmetric stretching vibration of the oxygen-carbon bonds (Figure 4.9) is IR-inactive, while asymmetric stretching or bending vibrations are IR-active.

Figure 4.10 shows, as an example, the IR spectrum of 6-benzoyl adenine. It may be seen that vibrational absorption spectra consist of bands rather than lines. The spread in the energies is usually due to conformational differences in the molecules and free rotation about the bonds. Still, the spread is not as large as for UV absorption. IR spectral bands are narrower in the solid state and at low temperatures. IR spectra are plotted as a function of wave number, i.e. the reciprocal of wavelength

$$\text{Wave number} = 1/\lambda \text{ cm}^{-1}$$

Sometimes the percent transmittance rather than percent absorbance is plotted. The two quantities are complementary, i.e.

$$\text{Transmittance} = (1 - \text{Absorbance})$$

Except for the simplest molecules, infrared spectra are often characterised by many absorption bands that are broad and closely overlapping. Therefore group vibrations have been extensively used to determine whether particular chemical groups are present and to chemically characterise the samples. Groups that absorb in those regions of the spectrum, which do not have too many other absorption bands, are selected. One such region for biological molecules lies between 1800 and 2300 cm^{-1} . The

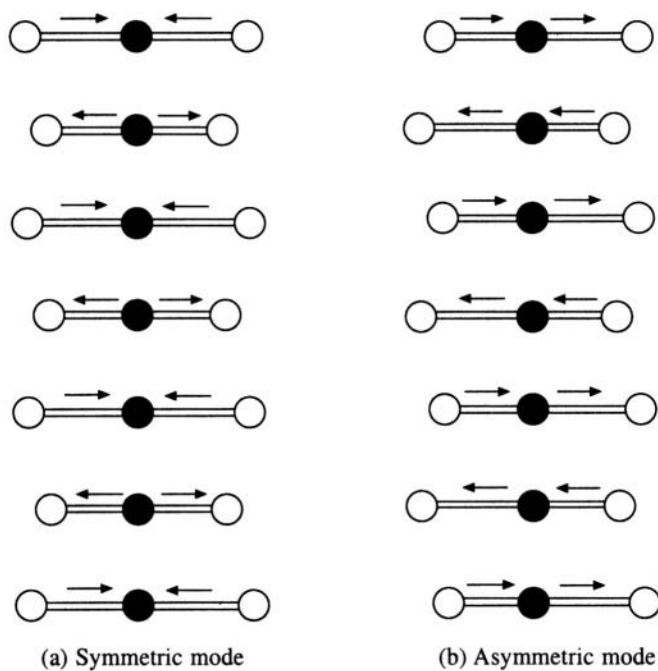


Fig. 4.9 The IR (a) inactive and (b) active stretching modes in carbon dioxide

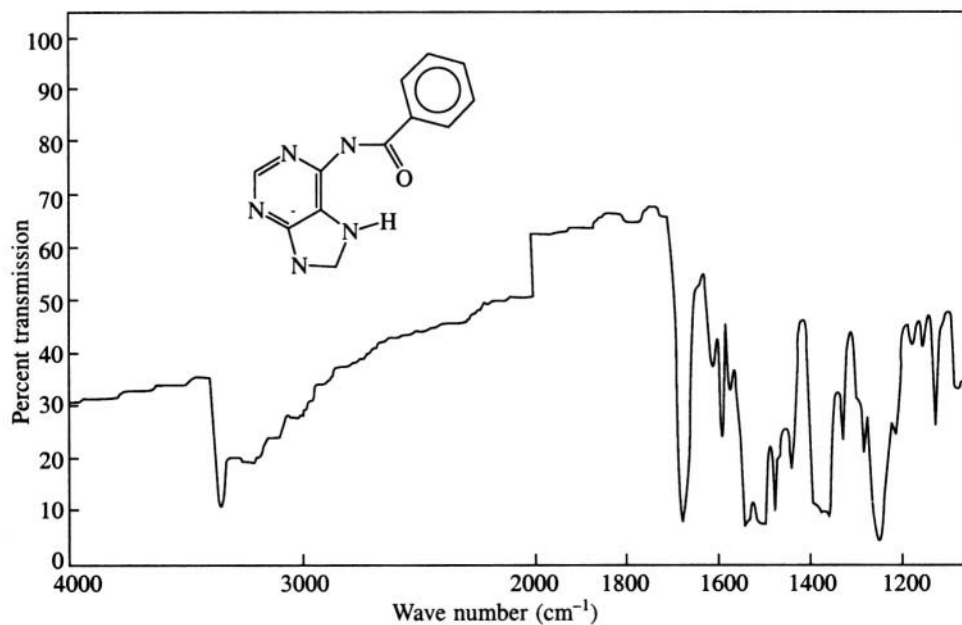


Fig. 4.10 Infrared spectrum of 6-benzoyl adenine

lower $400\text{--}1800\text{ cm}^{-1}$ region of the spectrum has many overlapping bands, and is called the 'finger print' region. For relatively simple structures, this region is used for identification. Table 4.2 gives a list of group absorption bands.

Table 4.2 Characteristic IR bands in polypeptide chains

Bond	Mode of vibration	Wave number (cm^{-1})
C-H	Stretch	2700-3300
C-H	Bend	1300 – 1800
O-H	Stretch	3030 – 3700
O-H	Bend	1200 – 1800
N-H	Stretch	3000 – 3700
N-H	Bend	1500 – 1700
C=O		
Amide I hydrogen bond	Stretch	1630 – 1660
C=O		
Amide I non-hydrogen bond	Stretch	1680 – 1700
C-N-H		
Amide II hydrogen bond	Stretch	1520 – 1550
C-N-H		
Amide II non-hydrogen bond	Stretch	< 1520

The amide I and the amide II bands are the principal infrared absorption bands of the peptide group in proteins. As may be seen from the table, both bands absorb in different regions according to whether they do or do not participate in hydrogen bonding interactions. Hydrogen bonds affect the vibration frequencies of the participating atoms. The presence of H-bonds is an indication of the polypeptide chains assuming regular secondary structures. In fact both amide I and amide II bands absorb in slightly different regions if they are in **α -helices** than if they are in **β -sheets**. This helps in the study of the conformation of biopolymers.

Carbon-monoxide stretching frequencies serve as a marker to study the conformational changes on the binding of this ligand to haemoglobin, the oxygen carrier protein. Studies on haemoglobin have also used sulphhydryl group frequencies of the cysteine group in the protein. The S-H absorption frequency centres around 2500 cm^{-1} and is slightly different for the three different cysteine groups in haemoglobin. The utility of this group as a conformational marker lies in the fact that its absorption band lies in a region away from that of water, thus making it possible to study the protein in aqueous solutions. Other absorption bands would be swamped by water frequencies. Dichroic effects on oriented polypeptides can be observed using polarised infrared radiation. For example, if the polypeptide is a **α -helix**, the N-H and C=O groups and the hydrogen bonds between them are oriented along the helix axis. The bonds that make up the amide II band are perpendicular to the helix axis (Figure 4.11). In **β sheets** the situation is just the opposite, with the N-H and amide bands being perpendicular to the chain direction, and the amide II bands being parallel to it. The absorption of polarised IR radiation consequently depends on the orientation of the sample relative to the beam. The spectrum of the actual polymer is however much more complex than the above discussion suggests. All the same, infrared dichroism of oriented polypeptide films is an established technique to study conformation.

Infrared biospectroscopy has in recent times benefited a great deal from improvements in measurement techniques. These improvements have resulted in large increases in the signal-to-noise ratio. One of the improvements was the change from using a continuous scanning mode to sweep across the

frequencies, to an interferometric, Fourier transform spectroscopic mode. The principle behind this technique is to irradiate the sample by a sharp pulse of IR radiation that contains all the relevant frequencies and then detect the response as a function of time. A Fourier transform of this detected signal yields the conventional absorption spectrum. [The Fourier Transform technique is explained in greater detail in the chapter on NMR spectroscopy.]

4.6 Raman Spectroscopy

The Raman effect was discovered by C.V. Raman and his associates in 1928. It arises when a beam of light (electromagnetic radiation in the visible region) is allowed to fall on a transparent substance such as a solution of a biological molecule. Most of the incident light will be transmitted through the sample unaltered, but a small portion is scattered at angles different from the incident direction. Again much of this scattered light is at the same frequency as the incident light. This scattering is called Rayleigh scattering. (The intensity of Rayleigh scattering is an inverse function of wavelength. Therefore a polychromatic incident beam will have a larger proportion of its smaller wavelength components scattered than its longer wavelength components. This is why the sky is blue). Apart

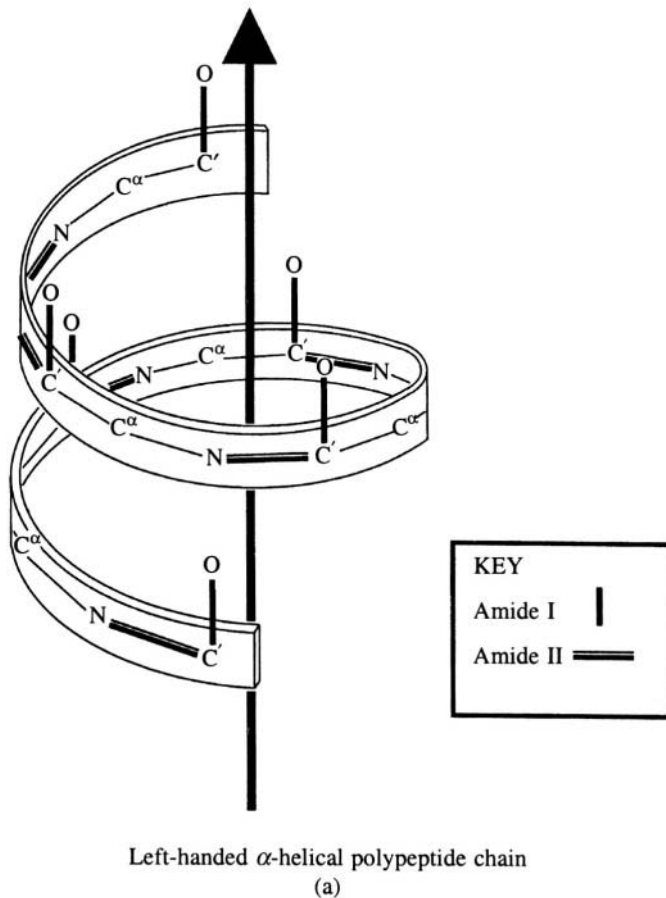


Fig. 4.11 (Contd)

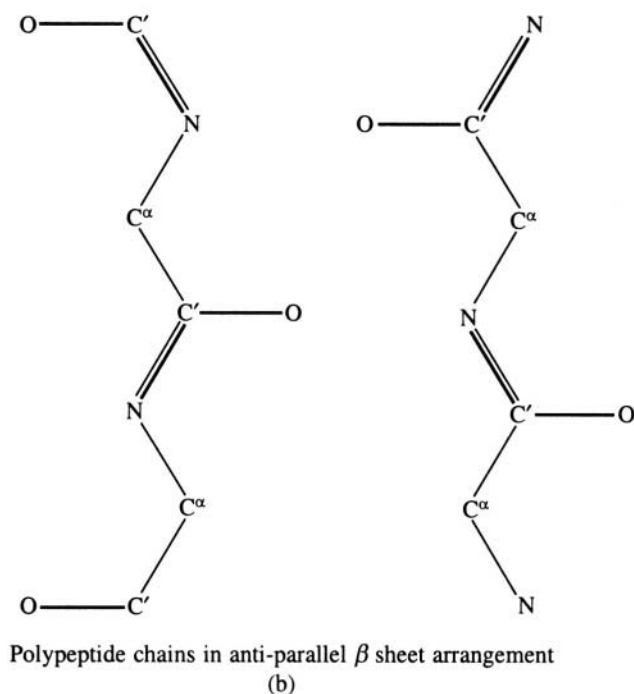


Fig. 4.11 A schematic representation of the orientation of the bonds which produce the amide I and the amide II bands in polypeptide chains. (a) The α -helical conformation is a left-handed α helix, but the orientation of the bonds in a right-handed helix is the same. (b) The β -sheet conformation is an anti-parallel sheet, but the conformation of the bonds in a parallel sheet is approximately the same

from Rayleigh scattering, a small portion of light is scattered with an altered frequency. This phenomenon is called Raman scattering. The shift in the frequency corresponds to the vibrational energy states of the molecule. Raman scattering is thus a probe of the vibrational states of the molecule, just like IR spectroscopy, except that the incident radiation is not infrared, but visible light and what is measured is not the absorbance but the frequencies of the scattered light.

The oscillating electric field of the incident light induces a dipole moment in the sample, different from any permanent dipole moment that may be present. As already seen, the permanent dipole moment gives rise to IR absorption. The induced dipole moment gives rise to Raman scattering. The strength of the induced dipole moment depends on the polarisability of the sample. Both classical mechanics and quantum mechanics show that if a particular transition from one vibration mode to another is accompanied by a change in the polarisability, the incident light will be absorbed and re-emitted with an altered frequency, leading to the Raman spectrum. The Raman effect is thus a coupling of the electronic and vibrational states of the molecule. Since, before the transition, the molecule could exist either in the higher or lower vibrational state, the scattered light could be either of lower or higher energy than the incident light. The light of lower frequency is called the Stokes spectrum or the Stokes lines. The longer frequency emissions are called the anti-Stokes lines, and are usually quite weak as compared to the Stokes lines. If it appears paradoxical that the emitted light should be of a higher frequency than the incident light, the Raman effect may be considered as a 'two-photon' phenomenon. Two photons are absorbed, the energy is exchanged and then re-emitted so that

one has lesser energy and the other higher energy than the incident photons. Usually only the Stokes spectrum is plotted, with intensity along the y axis and wave number increasing along the negative x direction (Figure 4.12).

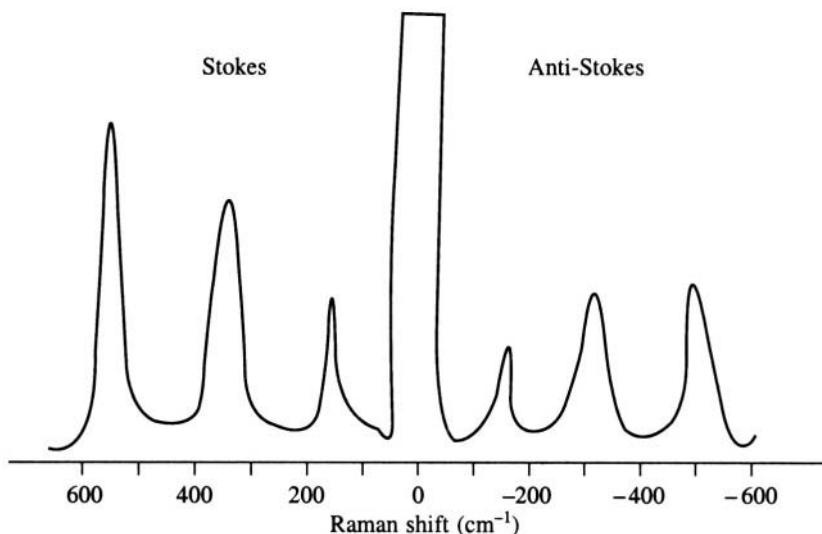


Fig. 4.12 Schematic of a Raman spectrum.

Since only a small portion of the incident energy is re-emitted as Raman scattering, the source has to be quite intense in order that the effect is observed. Also the concentration of the sample needs to be high. For these reasons, it was only after the introduction of lasers that Raman spectroscopy of biomolecules was possible. Though Raman spectroscopy, like IR, also probes vibrational modes of the molecules, there are differences and advantages. In the first place, since the mechanism of the Raman effect is different from that of IR absorption, several vibrational transitions that do not change the dipole moment but change the polarisability, can be detected only by Raman spectroscopy. Secondly, and this is of great advantage, water does not interfere very much in the Raman spectrum. This makes possible the study of aqueous solutions of biomolecules. Other advantages lie in the sensitivity of the Raman effect to conformational changes and in the possibility of using various types of samples, including powders, liquids, films, etc.

In the application of Raman spectroscopy to proteins the amide bands are again used to probe the structure. The ones that are most sensitive are the amide I band ($1645\text{--}1680\text{ cm}^{-1}$) which is predominantly a C=O stretching mode, the amide III mode ($1235\text{--}1300\text{ cm}^{-1}$) which results from a mixture of C-N stretching and N-H in plane bending, and the C-C stretching at $900\text{--}1000\text{ cm}^{-1}$. The amide II band between 1230 and 1310 cm^{-1} is also used to distinguish between α -helix, β -sheet and random coil conformations of biomolecules. Raman spectra of entire proteins have been obtained and as expected they are quite complicated. The peaks however are quite sharp and in general one may consider the spectrum of such a molecule as the sum of the spectra of the constituents, in this case, the amino acids. This, of course, is especially true in proteins such as lysozyme, where a large portion of the chain is in the random coil conformation. Raman spectroscopy has been used effectively to study thermal unfolding. In ribonuclease, a decrease in the intensity of the amide I and amide III lines corresponding to the α -helical and β -sheet frequencies was seen as the temperature was increased. This gave an indication of the unfolding of the protein.

The application of Raman spectroscopy to nucleic acids has been somewhat more extensive, since these polymers are constituted from only four monomer units, the bases, unlike proteins where there are 20 amino acids. The Raman spectra of nucleic acids has two distinct classes of lines; those arising from the bases, and those due to the sugar-phosphate backbone. The phosphate group has two types of P–O symmetric stretching modes. One arises from the phosphoester bonds and is around 810 cm^{-1} . This band is very sensitive to the conformation and on the basis of this band alone, a change of the double-helical conformation from *A* type to the *B* type may be monitored. The other symmetric P–O stretching arises from the bonds to the charged pendant oxygens. This occurs around 1100 cm^{-1} and being relatively insensitive to conformational change, serves as an internal marker. Certain Raman bands also give information about base pairing. The Raman hyperchromic and hypochromic effects, which arise from base-base interactions, are probes of the helical ordering of the polynucleotide chains.

Raman spectra of liquids and membranes are used to follow order-disorder transitions, through intermediate liquid crystalline phases. Some of the bands which are monitored are the C–H stretching region between 2700 and 3100 cm^{-1} , the C = C vibrations at $1650\text{--}1700\text{ cm}^{-1}$ and the skeletal C–C stretching at $1050\text{--}1150\text{ cm}^{-1}$. Raman spectra of intact organelles or organisms such as bacteriophages, chloroplasts, and ribosomes have also yielded significant structural information.

With the availability of tunable lasers, the technique of resonance Raman spectroscopy has become feasible. In this technique the incident laser beam is tuned to be in resonance with electronic transitions in the chromophore. Sample concentrations as low as 10^{-5} M may be used. A further advantage is that resonance Raman scattering yields information only about the immediate environment of the chromophore. This allows experiments where the chromophore is diffused into the active site of a protein, whose structure is then probed by Raman scattering. Heme groups and retinols are especially useful in this technique.

4.7 Electron Spin Resonance

Electrons possess charge and spin. Thus they have a magnetic moment. For a free electron, the magnetic moment is given in terms of the Bohr magneton, similar to the nuclear magneton. When such an electron is placed in a magnetic field H , the interaction of the field with the magnetic moment gives rise to a Larmor precession, the frequency of which is proportional to βH , i.e.

$$\nu = gBH/h$$

where g is a proportionality constant called the splitting factor or the ‘ g ’ factor and h is Planck’s constant. The value of g is 2.00232 for free electrons. The precession frequency ν is usually of the order of gigahertz (GHz, 10^9 Hz) as compared to nuclear precession frequencies of MHz. Just as in NMR, the free electrons absorb energy from an electromagnetic field oscillating exactly at their Larmor frequency. This absorption, when the resonance condition is satisfied, can be detected and is called electron spin resonance or electron paramagnetic resonance. The utility of this phenomenon in chemistry and biology comes from the variation that the splitting factor undergoes in different chemical environments.

When the electrons are not free, but bound to some nucleus, ESR can be detected only for an unpaired electron. According to Pauli’s exclusion principle, pairs of electrons will always be arranged so that the directions of the two spins are antiparallel. This implies that the interactions of the magnetic moments of the two electrons with the field will cancel each other. Only for an unpaired electron will there be a detectable interaction. The magnetic moment of an unpaired electron is about 700 times that of protons. Hence the sensitivity of an ESR experiment is higher than that of an NMR

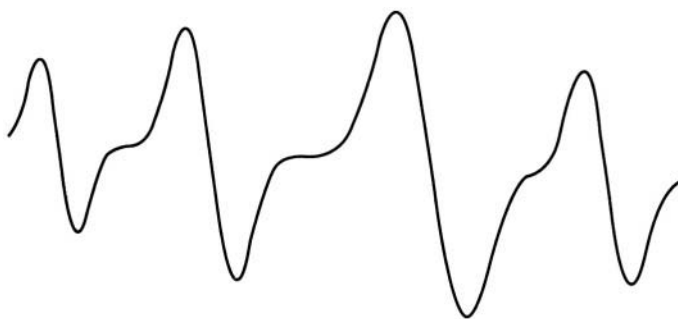


Fig. 4.13 Schematic of an ESR spectrum

experiment and the quantity of the sample required to detect the signal is comparatively small. In many ways the ESR experiment is very similar to the NMR experiment and factors such as the T_1 and T_2 relaxation times play a key role. Since ESR absorption lines are broad compared to the NMR spectrum, it is usual to plot and analyse the first derivative of the absorption spectrum rather than the spectrum itself (Figure 4.13). The role of the chemical shift in NMR is played by the g factor in ESR. The values of g for some organic radicals are shown in Figure 4.14. Note that only the oxygen and the nitrogen radicals have large differences in the value of g from that of a free electron. The magnetic moment of the electron also interacts with the nuclear magnetic moments in the neighbourhood. This interaction leads to a splitting of the absorption lines known as the hyperfine structure or the hyperfine splitting. The origin of this splitting may be understood as follows. Consider an unpaired electron in the external magnetic field H . The field experienced by this electron is a combination of local fields due to surrounding nuclei and the external field. Thus the effective field on the electron is

$$H_{\text{eff}} = H + \Delta H_{\text{loc}}$$

where ΔH is the (small) local field. In the case where the electron can be considered to be localised around just one nucleus, of spin quantum number I , ΔH_{loc} will assume $2I+1$ values along the direction of the external field. Resonance therefore will occur at the following $2I + 1$ values of H_{res} :

$$H_{\text{res}} = H_{\text{res}}^0 - a m_I, m_I = -I, -I + 1, \dots, I - 1, I$$

where a is a constant such that $a m_I$ is the value of the local field and H_{res}^0 is the field strength at which resonance would occur if $a = 0$. Figure 4.15 shows the situation for a hydrogen atom with its single electron and the single proton in the nucleus, $I = 1/2$. The dashed lines show the case when $a = 0$. The solid lines are the two transitions that actually occur, giving rise to the indicated hyperfine splitting. In general, if the spin quantum number of the nucleus is I , the ESR spectrum is split into $2I + 1$ lines.

Organic radical	'g' value
— H	2.00226
— N $\begin{array}{l} \diagup \text{O} \\ \diagdown \text{O} \end{array}$	1.999
— Cl $\begin{array}{l} \diagup \text{O} \\ \diagdown \text{O} \end{array}$	2.01

Fig. 4.14 Typical value of the splitting factor or g factor for some common organic radicals

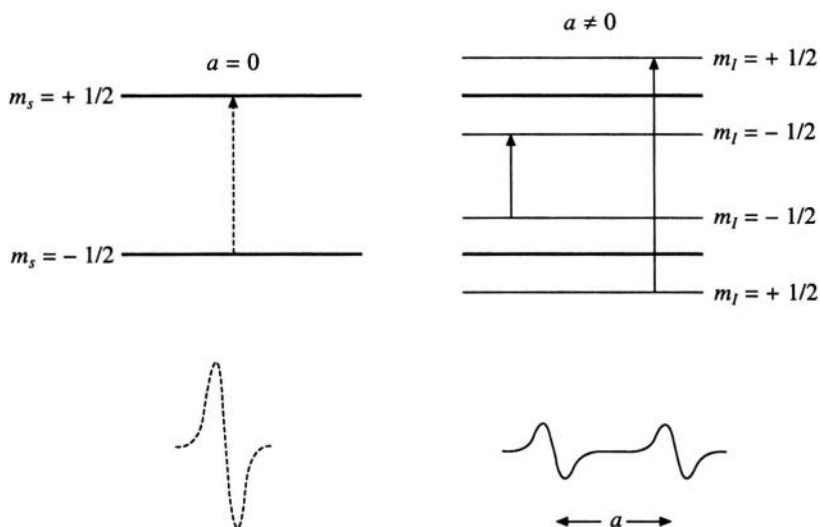


Fig. 4.15 The hyperfine splitting in hydrogen

ESR spectroscopy yields useful information in biological systems. The binding of paramagnetic metal ions such as Cu^{2+} , Mn^{2+} , etc., (i.e. those with an unpaired electron) to proteins or other molecules can be studied by the splitting that the binding induces in the absorption lines. Both the common isotopes of nitrogen, i.e. N^{14} as well as N^{15} will lead to a splitting of the absorption line of Cu^{2+} when the metal ion binds. The resonance frequency of a paramagnetic species also depends on the orientation of the spin with respect to the external magnetic field. In a liquid, the tumbling motion will average out all the orientations. But in a crystalline solid, the ESR spectrum contains information regarding the orientation. This property was used, for example, in determining the angle of the heme group, containing paramagnetic iron, with respect to the crystal axes. Also, the orientation effect leads to an inverse correlation between the tumbling motion of the molecule and the width of the absorption line – the wider the line, the slower the tumbling. This has helped in the study of fluidity of membranes and the dynamics of proteins.

In the case of molecules which do not have intrinsic paramagnetic atoms (as in most biological macromolecules), ESR studies are made by attaching one or more organic or other radicals. Such paramagnetic groups are known as spin-labels and act as a reporter or probe. For most studies spin-labels containing the nitroxide group are used. The ESR spectrum of the spin label can then be analysed in terms of the width of absorption line and the hyperfine splitting, to yield information about the conformation and dynamics of the biological molecules. This technique is especially useful in the study of membrane bilayers.

Light Microscopy

5.1 Introduction

Many of the interesting features of biological systems are too small to be seen with the naked eye. When molecular detail is not required, the light microscope is an ideal, and hence essential, instrument for a biologist. Electron microscopy can also be used to elicit highly detailed information, but suffers from several limitations, including complex operating procedures. Light microscopy is easy to use and inexpensive. An additional, crucial advantage it has over the electron microscope is its ability to focus at different levels within a three-dimensional object. In addition, the new methods and applications, which are even now being constantly developed, have helped to retain the importance of the light microscope in biological studies, in spite of the availability of advanced instruments of all kinds based on a variety of other physical principles.

Light microscopes are built up from a stack of lenses, usually made of glass and in this chapter, the optical principles which govern their arrangement are discussed first. A discussion of the resolving power of microscopes follows. Finally we discuss the newly developed methods in microscopy.

5.2 Elementary Geometrical Optics

Optical lenses can be either concave or convex. Convex lenses (Figure 5.1 (a)) are also called converging lenses. Concave lenses (Figure 5.1(b)) are known as diverging lenses. Each lens has two focal points as shown. In the figure the focal lengths, defined by OF_0 and OF_1 are equal. This is not always so and depends on the relative refractive indices of the lens and the medium on either side of it. A thin concave lens always gives a minified (i.e. made smaller) virtual image of any object viewed through it (Figure 5.2). Virtual images are images that cannot be captured on a screen or on photographic film, except with the help of additional lenses. Concave lenses are thus not normally used in microscopes except in combination with other lenses. A real image, in contrast, can be captured on a screen. Convex or converging lenses may form real or virtual images, depending on the positions of the

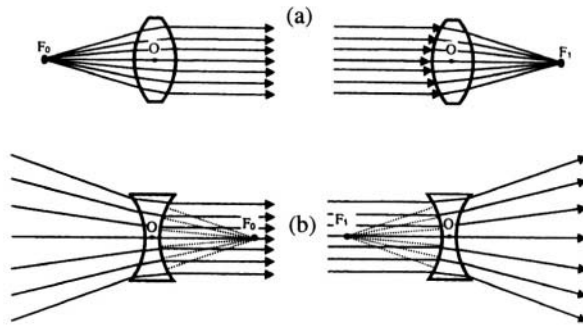


Fig. 5.1 (a) Convex and (b) Concave lenses.

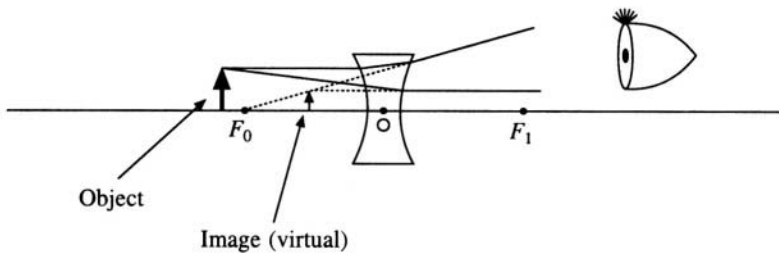


Fig. 5.2 Image (virtual) formation by a concave lens

object with respect to the focal points of the lens. Also the images may be minified, magnified, or of the same size. Figure 5.3 illustrates the various possibilities. When the object is at a distance greater than twice the focal length from the lens, a real, inverted, minified image occurs at a point between one and two times the focal length on the other side of the lens (Figure 5.3(a)). When the object is at a point exactly twice the focal length, the image is real, inverted and the same size (Figure 5.3(b)). When the object is at a distance between one and two times the focal length, the image is real, inverted and magnified, and occurs at a distance greater than twice the focal length (Figure 5.3(c)). When the object is at the focal point, the image occurs at infinity (Figure 5.3(d)). The basic compound microscope uses two convex lenses in the arrangement shown in Figure 5.4. The lens closest to the object is called the 'objective'. The lens closest to the eye is called the 'eyepiece'. The microscope is adjusted so that the object (a) is at a distance between one and two times the focal length. The image (b) formed is therefore real, magnified and inverted. This real image acts as the object for the second lens, i.e. the eyepiece. The eyepiece magnifies this intermediate image, and produces a large virtual image (c) which then becomes the image for the eye or for the camera lens. The final real image (d) is formed on the retina of the eye or the photo film. The relative positions of the eyepiece and the objective are kept constant and focusing the microscope requires movement of both as a single element. The overall magnification of the instrument is the product of the magnifications of the objective and the eyepiece. In most microscopes these are changeable and various combinations of the two lenses can be used to obtain the required magnification. In some microscopes, the magnification can be 'zoomed' up and down in a continuously variable fashion. Lenses with aspherical surfaces are also available, to reduce optical aberrations. In fact, even the simplest of compound microscopes available today will not be as simple as the one shown in the figure. The objective is a combination

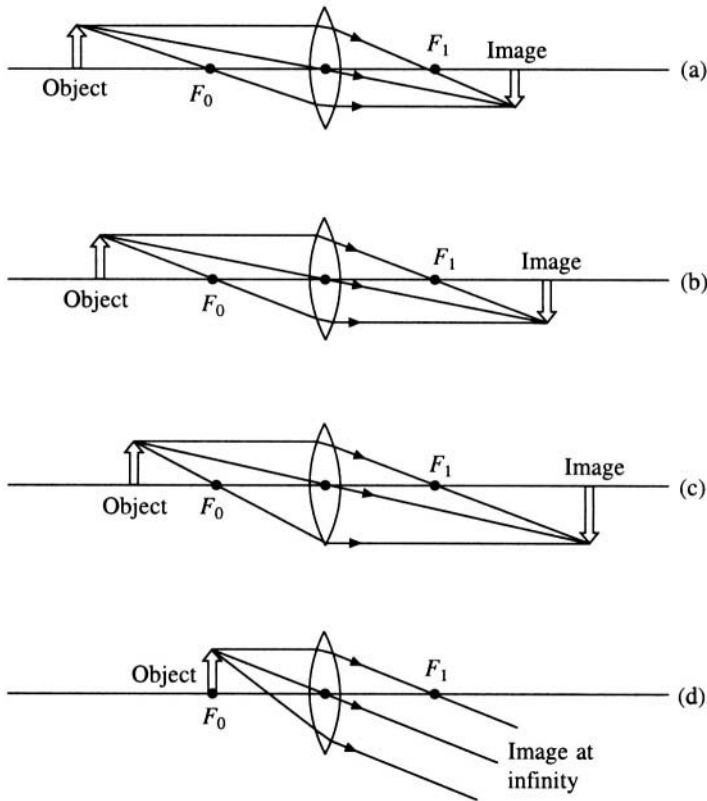


Fig. 5.3 Image formation by a convex lens

of convex and concave lenses to produce the desired magnification and eliminate distortions. Similarly the eyepiece is built up of many lenses. Nevertheless, the principles underlying the magnifying systems remain as illustrated here.

5.3 The Limits of Resolution

The resolution of a microscope is its ability to distinguish closely spaced objects. Heisenberg's Uncertainty principle (see Chapter I) states the theoretical limits to the amount of magnification that an object can be subjected to and this figure is far, far greater than the magnifying power of any light microscope. For the magnification to make sense, however, the final image should make visible the individual features. The smallest distance at which two point like objects can be placed and still appear as two distinct objects governs the actual, effective magnifying power of the microscope. The smaller this distance, the higher the resolution of the microscope and, therefore, the higher the magnification possible. The resolution limit of an instrument can be defined on the basis of Rayleigh's criterion, which, in turn, comes from an analysis of the diffraction of light. Light waves coming from different points of the object form the final image after travelling through slightly different distances. As explained in the chapter on X-ray diffraction, the waves can interfere constructively or destructively to form a diffraction pattern. Figure 5.5 shows the image, i.e. the diffraction pattern that is formed by a small pinhole. A central bright image is surrounded by concentric rings, which rapidly become fainter. The central image is called the first maximum, and the dark ring immediately surrounding it

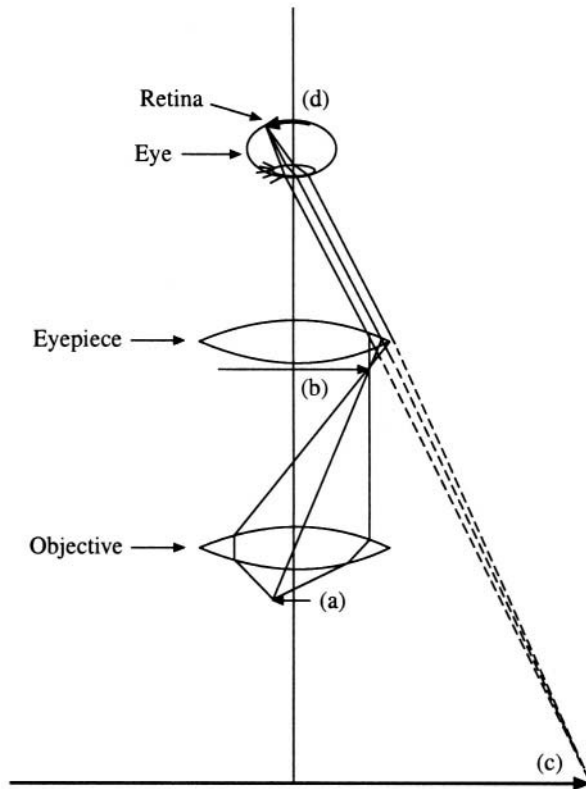


Fig. 5.4 Schematic diagram of a compound microscope

is the first minimum. The diffraction pattern due to two pinholes close to each other will overlap to different extents depending on the distance between the holes. Rayleigh suggested that two such objects could be considered as resolved, if in the diffraction pattern, the first maximum of one object overlapped with the first minimum of the other (Figure 5.6). Rayleigh's criterion is especially useful in deciding the resolution limit when dealing with self-luminous objects, such as for example, stars viewed through a telescope.

Another obvious limit to the magnifying power of a microscope is the wavelength of the light used for illumination. If the spacing between the objects is less than about $1/2$ the wavelength of the light, then the images of the two objects will almost exactly superimpose on each other and they cannot be resolved. For the usual type of light used in microscopes this limit is about 250 nm. Considering the fact that the eye (or the photographic film) imposes its own limits, the total magnification that is usually possible with light

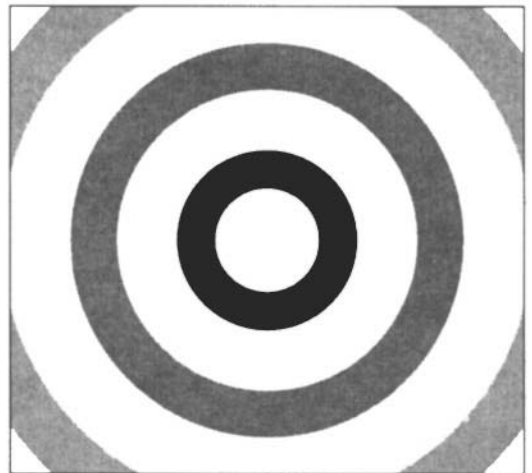


Fig. 5.5 Diffraction pattern from a pinhole

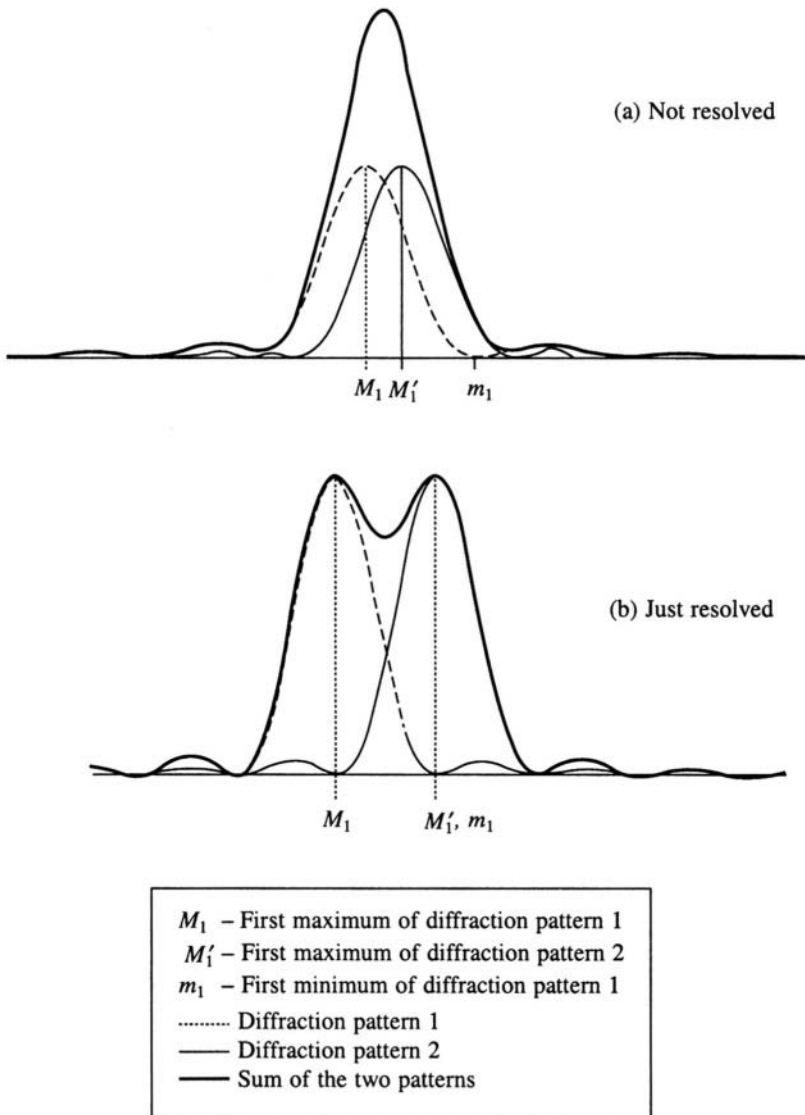


Figure 5.6 Rayleigh's criterion for the resolution of two objects

microscopes using ordinary light is about $\times 800$ ($\times 800$ or $800\times$ means 800 times). While the Rayleigh criterion is simple, there are certain disadvantages when applied to images formed by microscopes. Apart from being strictly applicable only to self-luminous objects, there is also the problem of analysing diffraction from a complex object. The approach of Abbe is probably more suited to microscopes. Abbe realized that the objective lens in a microscope acted essentially as a Fourier summation device. Fourier showed that any regular periodic function could be written as a sum of simultaneously varying waves of different wavelength and amplitude. When light falls on an object, the effect of diffraction is to split the wave into Fourier components. A summation of all these components will give back the original image and this task is performed by the objective lens.

However, owing to the large angles at which most of the components are scattered, the objective usually is able to capture only the low order components. This leads to a fuzzy representation of the objects. The angles at which the various components are scattered also depends on the spacing of the features in the object—the smaller the spacing, the larger the angle. Thus features which are very close together are not well resolved since fewer of the Fourier components of the diffraction pattern from these features are used by the object lens in the Fourier summation. Abbe suggested that a pair of features on the object be considered as well resolved if both the zeroth and the first orders of the Fourier components reach the objective. This criterion leads to essentially the same result as the Rayleigh criterion, for the resolving power of a microscope and the highest meaningful magnification. It has the additional advantage of providing a rigorous understanding of the similarities between the microscope lens system and the mathematical operations of Fourier transformation.

5.4 Different Types of Microscopy

5.4.1 *Bright field microscopy*

Bright field microscopy is obtained in the normal method of illuminating the object, when thin, more or less transparent samples (such as cells or bacteria) are being viewed. The illumination is usually as illustrated in Figure 5.4. The light is transmitted through the sample and the image is formed by the absorption of this light. Thus the image will appear darker than the background which is the bright field. In other words, the image is formed due to the contrast in the amplitude of the light waves coming from the object, and those that are transmitted unchanged through the object. This imaging technique may be called amplitude contrast.

5.4.2 *Dark field microscopy*

In the case of dark field microscopy the background illumination is cut off completely, leaving the field of the image dark. An extra condenser lens is added so that the rays of light fall obliquely on the object. The direct beam then passes straight through and is cut out by means of a diaphragm. Only the light scattered by the object is collected by the objective and sent on to the eyepiece. The light coming from the source itself has its central portion cut off and only an annular ring of light falls on the condenser lens. This technique requires a very powerful source of illumination since only the scattered light is used in the image formation. Nevertheless it is one of the oldest kinds of microscopy and still finds application when viewing very small objects. Bacterial flagella, for example, with diameters of only about 20nm can be followed in motion using a very strong source of light and dark field microscopy. This is also a form of amplitude contrast microscopy.

5.4.3 *Phase contrast microscopy*

Every wave has a characteristic phase. This is illustrated in Figure 5.7 where wave b has the same phase as wave a while wave c has a different phase. In a microscope if the phase of the rays scattered from the sample can be made to have a different phase from the unscattered rays, then the image of the specimen will be different in light intensity as compared to the background and will be more clearly visible. This is because when two waves (e.g. the scattered rays and unscattered rays) are in phase, they reinforce one another and the total amplitude is larger. Against this, when the two waves are out of phase, they combine together destructively and give a smaller amplitude. This effect is used in a phase contrast microscope to image thin transparent specimens. The basic principle of the method is that the unscattered beam in the direction of the incident light has its phase shifted (or changed) inside the microscope before being allowed to recombine with the scattered light. The phase

shifting is done as follows. The illumination of the specimen is made obliquely using a condenser lens as in the case of dark field microscopy. Instead of a diaphragm cutting off the direct beam, however, a phase plate is placed in the focal plane of the objective lens. The phase plate consists of a transparent disc in which there is an annulus of smaller optical path than the rest of the plate. Since the medium of the plate is dense, light passing through it will be retarded. The direct beam will be less retarded than the scattered light, thus introducing a phase difference. This enables the image to be formed clearly in a bright field. Two effects usually accompany phase contrast microscopy. First, some thick objects may show less contrast than thinner ones and may even reverse contrast so that they appear brighter than the background. Second, every object is surrounded by a diffuse halo of light, and does not represent the real structure. These effects are due to the fact that, apart from the phase plate, the objects themselves introduce a shift in phase. In the case of the thick objects the shift in phase may be such that when the rays recombine, the scattered and the unscattered light combine constructively, thus producing reverse contrast. The halo effects appear because some of the direct beam, instead of passing through the retarding portion of the phase plate, passes through the other portion. This creates the effect of a bright, blurred, low-resolution image of the object appearing superimposed on the sharper actual image. The phase contrast microscope is used only with bright field illumination. However by changing the shape of the phase plate such that, instead of a ring of reduced thickness for the unscattered beam, we have a ring of enhanced thickness, it is possible to have reverse contrast for thin objects instead of positive contrast.

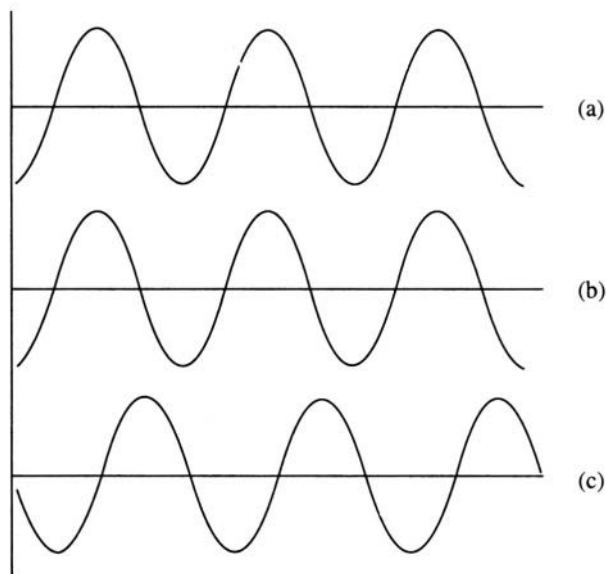


Fig. 5.7 Waves of different phases. (a) and (b) are in phase and (c) is 90° out of phase with the other two

5.4.4 Fluorescence microscopy

Fluorescence is the phenomenon by which certain substances absorb light of a particular wavelength. After a very short interval of time the light is re-emitted with its wavelength altered. If the delay between absorption and emission is less than 10 seconds, the effect is called fluorescence. Longer delays lead to the phenomenon of phosphorescence. Both go under the general name of luminescence. In general the wavelength of the emitted light is longer than the wavelength of the absorbed light.

Fluorescence microscopy uses this property to specifically image materials of interest. While performing the experiment, filters are used for both the incident light and the scattered light so that only a narrow band of wavelengths is passed in each case. The incident light filter allows only those wavelengths that will be absorbed to impinge on the specimen. The filter on the scattered light, also known as the barrier filter, allows only the emitted wavelengths to pass through. Since the emitted wavelengths are different from the absorbed wavelength, all the background is filtered out and only the image of the sample is allowed through. Many natural materials exhibit fluorescence. The amino acid tryptophan, for example, is an ubiquitous component of proteins and fluoresces in the 250 to 400 nm wavelength region. Such an auto-fluorescer, however, is of limited value.

Of much greater value is the use of fluorochromes. These are small fluorescent molecules, which may be linked chemically to materials of interest. Fluorescein isothiocyanate (FITC) absorbs blue light and emits green light. Rhodamine compounds absorb green light and emit red light. These compounds are especially useful in immuno-fluorescence microscopy. In this technique fluorescent dyes are linked to antibodies raised against the molecules of interest. The labeled antibodies are then cross-reacted with the sample containing the antigens. The fluorescent image of the sample will then clearly indicate the distribution, and even the size and shape of the antigen. In clinical medicine, fluorescence microscopy is a powerful technique to identify the presence of particular viruses in the sample. In biology, it enables the mapping of the distribution of even fairly small cell components. The antibodies are raised directly against the material extracted from similar cells and purified biochemically. Another method is to attach the fluorochrome to an anti-antibody, i.e. a general purpose antibody which will attach to the antibodies against the particular materials. By this method, the process of attaching the 'stain' to the antibody each time is avoided as the non-specific, labeled anti-antibody can be produced commercially and stored.

5.4.5 *Polarising microscopy*

As electromagnetic radiation, such as light, propagates, its electric field oscillates perpendicular to the direction of propagation. If the electric field is confined to one plane (say, the vertical 'up-down' plane) the light is said to be plane polarised. Light can be plane polarised by crystals or by special materials such as Polaroid. When a beam of unpolarised light falls on a sheet of Polaroid, only light polarised in one particular direction passes through. The rest is absorbed. Thus when a beam of polarised light falls on such a sheet, whether the light passes through or not depends on the relative polarisation directions of both the beam and the sheet. A device that polarises the light is called a polariser. A device used to analyse the direction of polarisation of the beam is called an analyser. Both the polariser and the analyser are normally the same material such as a Nicol prism or a sheet of Polaroid.

In a polarising microscope, light polarised by the polariser is allowed to fall on the object. Light scattered from the object is then viewed through an analyser. A polarising microscope helps to determine the structures of ordered domains within the sample. This is because such ordered domains possess the property of birefringence. In stretched biological fibres, for example, most of the chemical bonds will be oriented in a particular direction and the electric field of the incident light interacts very strongly if it is polarised in the same direction. In effect the fibres act as a kind of polariser. This property is called birefringence. If the direction of oscillation is along the fibre axis, the light travels slower than when the electric field oscillation is perpendicular to the fibre axis. This is called positive birefringence. For negative birefringence, the long fibre axis is the fast axis and the shorter axis is the slow axis. If therefore a beam of plane polarised light is incident on a birefringent object the effect is to turn the plane of polarisation by an angle. If now the object is viewed through the analyser placed

perpendicular to the polariser, all the direct light is cut off and only the light from the object will pass through (actually only a component of this light will pass through), and the birefringent object is clearly visible. By rotating the sample and observing the change in the intensity, it is possible to measure very precisely the direction of orientation of the birefringent portions of the sample. Typical applications of the polarising microscope in biology include the study of cell spindle formation and the behaviour of the components of a contracting muscle. Polarisation measurements can also be made in conjunction with fluorescence microscopy and information about the orientation of the marker dyes is obtained.

Electron Microscopy

6.1 Introduction

Electrons are negatively charged subatomic particles. Their use in imaging can be explained in terms of their quantum mechanical wave-like property, which allows a beam of electrons to be considered in exactly the same way as a beam of electromagnetic radiation such as light. Electrons possess two additional advantages for use in microscopy. The first of these is that according to quantum physics, the wavelength of the waves associated with an electron varies inversely as its energy. Thus by accelerating the electron between charged plates with a very high applied voltage, the wavelength of the ‘radiation’ used in the microscope could be made very small. As explained earlier, this implies a concomitant increase in the resolution and hence in the possible magnification. The second advantage, which actually allows the use of electrons for microscopy, is that, unlike electromagnetic radiation of short wavelengths such as X-rays, electrons can be focussed by means of magnetic lenses. This is dependent on the well-known principle in physics that a charged particle in a magnetic field is deflected in a direction perpendicular to its direction of motion. This principle has been used in the construction of electron microscopes.

In this chapter we will first study some elementary electron optics including the principle components of the electron microscope. Next we describe the different types of microscopic techniques, the preparation of the specimen, and some applications. We go on to an introduction to advanced image reconstruction techniques and electron diffraction. We end with a discussion of tunnelling electron microscopy and the atomic force microscope.

6.2 Electron Optics

The wavelength of the electron beam depends on its energy, which is in turn dependent on the voltage used to accelerate the electrons. Modern electron microscopes use accelerating voltages in the range 1000 volts to 1000 kilovolts. The most popular microscopes use about 100 kilovolts. According to the

de Broglie equation (see Chapter 1), this voltage will correspond to a wavelength of about $0.03 \text{ \AA}$. This should, in principle, allow the imaging of details on an atomic scale. However several factors come in the way of reaching this high resolution. Vibration, damage to the specimen and contamination are some of the problems encountered in microscope design. The most important factor limiting the resolution is the inability of the magnetic lenses to focus the beam accurately. This is called the aberration of the lens. Aberration is chiefly of two types. The first, called spherical aberration arises from the fact that all the electron beams scattered from the specimen which fall on the lens should be focussed on to the same spot but in practice are not. Figure 6.1 gives an illustration of how the images from a point will be spread out into a series of images. Thus the image of the point will be seen as a disk, called the disk of confusion. The spherical aberration can be reduced by cutting out some of the higher angle beams scattered from the point by means of a stop as illustrated in the figure. But obviously there is a limit to the narrowness of the aperture beyond which diffraction effects will come into play. The second important type of aberration, called chromatic aberration arises from the fact that the focal length of the lens is different for different wavelengths. Since the source cannot produce a uniformly energetic electron beam, therefore the image of the point again spreads out into a disk of confusion, this time caused by the chromatic aberration.

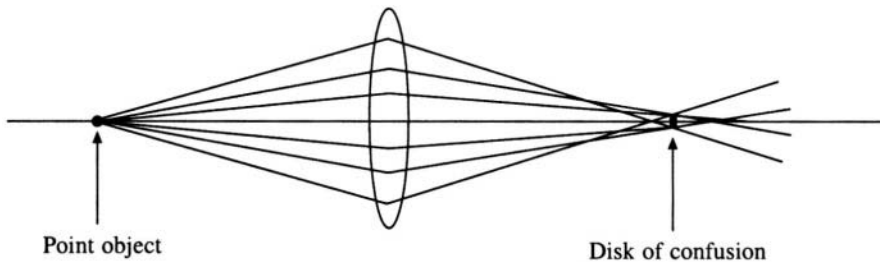


Figure 6.1 Spherical aberration in electron optics, caused by imperfections in the magnetic lens

The final quality of the image depends not only on the resolution but also on the contrast. Contrast is the difference in intensities between the object and the background as a fraction of the background intensity. For visibility, this ratio should be larger than 0.2. The contrast can be enhanced if the object is dense as compared to the background. Usually, the support for the specimen is a carbon film and biological materials, especially, have about the same density. The use of electron microscopy in biology is thus especially problem prone. As will be described later, one can use staining methods to increase the contrast. It is also possible to increase the contrast by increasing the intensity of the incident electron beam. This is because the background intensity varies as the square root of the incident intensity while the image intensity will vary directly as the incident intensity. However increasing the intensity of the incident beam leads to another severe problem in the study of biological materials, viz. radiation damage. This will destroy the sample leading to very poor quality images.

Thus because of the above problems and also because of others not fully discussed here, the actual practical resolution and magnification in biological electron microscopy is below what in principle is possible. Nevertheless, electron microscopy is far superior to the light microscope in its resolution and magnification. Objects such as cells, viruses, DNA strands, ribosomes, etc. can be imaged clearly at resolutions of a few tens of Angstroms, at magnification of 100000 X or more.

6.3 The Transmission Electron Microscope (TEM)

This is one of the most commonly used instruments. Its geometry is shown in Figure 6.2. The source

S of the electron beam may be a tungsten filament, but other materials like lanthanum hexaboride are also used. The beam path, the placement of the lenses, the specimen, the aperture etc, follow the plan of the light microscope as seen in the figure. The entire arrangement, including the specimen, has to be placed in high vacuum to avoid extraneous scattering and absorption of the electrons by air. The lenses are magnetic lenses in the electron microscope and unlike as in the light microscope, the image is not viewed directly through an eyepiece, but is projected on to a fluorescent screen on which the electrons form the image. An extra aperture called the objective aperture is placed in the electron microscope and not normally in the light microscope. This is because the image forming process in the TEM is different from that in light microscopes. In a light microscope, light is transmitted or absorbed by the specimen. This creates a contrast between the different parts of it and hence the image. In the TEMs the situation is quite different and most of the incident beam passes through the specimen unimpeded and very little is absorbed. Figure 6.2 shows that only a small portion is actually scattered in the forward direction. It is this forward-scattered fraction that is used to form the image. Contrast is created by differentiating between electrons scattered into wide angles from those scattered into small angles. The objective aperture cuts off the large angle electrons, and the amplitude of the electron beam from different portions of the specimen is different leading to amplitude contrast. Most biological materials however do not have enough contrast to be viewed unless they are stained. Dark field imaging is also possible and is achieved by cutting out the unscattered beam.

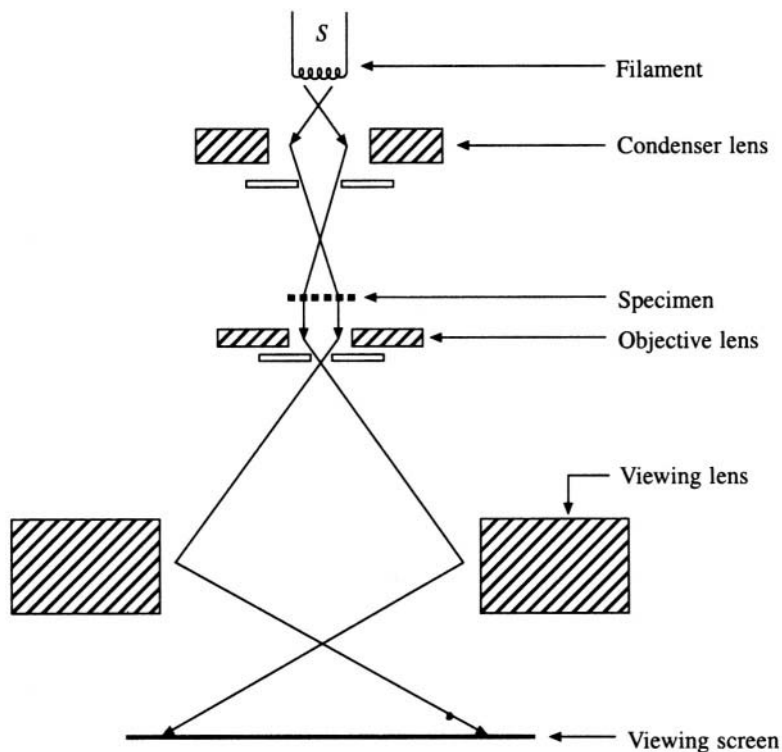


Figure 6.2 Schematic diagram of a transmission electron microscope

6.4 The Scanning Electron Microscope (SEM)

A SEM is very useful in obtaining images of the surface of thick specimens. In the scanning-

transmission mode (STEM), the SEM can also be used to study thin specimens and in this respect has some advantages over the conventional TEM. Figure 6.3 is a schematic illustration of the construction of the SEM. The most obvious difference as compared to a TEM is that the electron beam, after passing through the condenser lens is deflected in a raster pattern over the specimen stage, similar to the pattern in a television picture tube. The objective lens is split into two parts. One part is placed between the condenser lens and the specimen and can in fact be regarded as an additional condenser lens, which focuses the electron beam onto a small spot on the specimen. Image signals can be collected by the detector *D*, which is placed on the same side of the specimen as the source and the raster coils. This detector is used in the surface-scanning mode and collects the secondary electrons knocked out of the specimen by the primary beam, as well as the back scattered electrons reflected from the surface of the sample. Image signals can also be collected in the scanning transmission mode by the detector *D* after passing through the second half of the objective. These are the forward-scattered electrons.

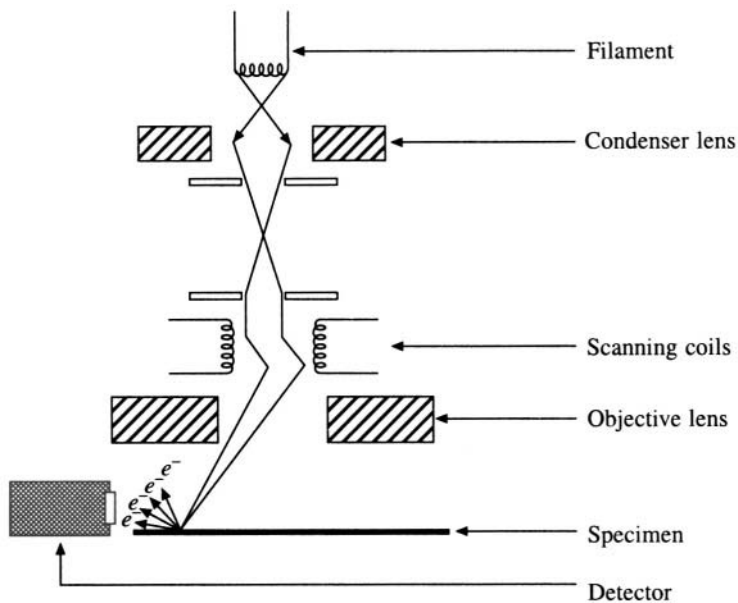


Figure 6.3 A schematic representation of a scanning electron microscope

Images obtained in the STEM mode are very similar to those obtained in the fixed beam TEM. The surface SEM image is quite different. In particular, the contrast in the surface imaging arises partly from the different scattering powers of the different atoms in the specimen, and partly from variations in the topography. In both the STEM mode as well as the surface SEM modes the images are generated in the following way: As the electron beam scans the specimen, the scattered electrons from each position is measured by the detector, one after the other in time. The measured intensities are displayed on the imaging screen one after the other in space, in the same type of raster scan as the electron beam. Thus if the scattering is high at particular point during the scan, then the corresponding point on the viewing screen is bright. If the scattering is low the corresponding point is dark. This creates an image of the object. The resolution of the image depends on the size of the electron spot used to scan the specimen and this may be as small as 50 Å. Both surface SEM and STEM have changed the use of electron microscopy quite dramatically and have made quantitative studies possible.

The scanning principle has also been used more recently in the construction of the atomic force microscope and the scanning tunnelling microscope. These will be discussed later.

6.5 Preparation of the Specimen for Electron Microscopy

Preparation of the specimen is one of the most important steps in the process of imaging by electron microscopy and often takes longer and requires greater skill than the actual observation. This is especially true in the case of biological samples. This is because biomolecules and assemblies are prone to disintegration in the high vacuum condition under which the imaging is done. Secondly without staining with heavy atoms, the lighter atoms such as carbon, hydrogen, nitrogen and oxygen, which are the principle components of biological samples, do not offer sufficient contrast against the support, and cannot therefore be observed. Finally, unless protected, the intense beam of electrons can easily destroy the sample.

The specimen preparation involves the following steps: (a) Preservation: The specimen has to be preserved in the relevant configuration. Care should be taken to avoid artefacts being introduced into the specimen during the preparation. (b) Staining: In order to increase the contrast between the sample and the background, the specimen has to be stained with a heavy, electron-rich metal. (c) The specimen has to be dried, i.e. water has to be removed in order that this does not interfere with the imaging. (d) The specimen has to be made into a form suitable for observation by replication and/or microtomy.

(a) *Preservation*: The two main methods of preserving the sample are by freezing and by chemical fixation. Freezing the sample has to be done with care to avoid formation of ice crystals that may radically alter the structures of interest. Rapid freezing is one way of reducing ice crystal formation and a cooling rate of about 5000 degrees per second is sufficient to freeze whatever water there may be in the sample into a glass like structure which does not affect the sample. Though the sample is usually cooled down to only liquid nitrogen temperatures, such cooling rates may be obtained if the whole process is accomplished in a few milliseconds. Once the specimen is frozen and is retained at the liquid nitrogen temperature or below, the structure, morphology and constitution do not alter even for months. The sample can also be preserved by chemical fixation. Chemicals, most commonly osmium tetroxide and glutaraldehyde are used to preserve the structure and shape of the specimen. Both chemicals work by creating covalent bonds between different molecules, especially proteins or lipids. Osmium tetroxide has the additional advantage of providing a heavy metal stain, doing away with the necessity of applying the stain separately. However, unlike glutaraldehyde, it does not act on a wide range of tissues. The great advantage of chemical fixation, as compared to freezing is that the procedure is simpler. Moreover, chemicals can be used to preserve much thicker samples since the penetration is greater. Handling and storage of chemically fixed samples is also easier.

(b) *Staining*: As mentioned earlier, both the biological samples studied and the carbon films on which they sit consist of light atoms and without contrast enhancement, the samples will be invisible with respect to the background. Thus staining the sample must be resorted to, in spite of the fact that this will result in a substantial loss of resolution. The technique of positive stain adds a specific heavy atom (uranium to DNA samples, or osmium to lipids) by chemically complexing them. Another technique of positive staining is called shadowing. Here heavy atoms, such as tungsten are evaporated from a filament onto the sample. If the sample is bombarded at an angle then walls of the metal atoms pile up on one side of vertical features in the sample, while no metal atoms are deposited in the shadow of the wall. This allows the easy visualisation of the contours of the raised features in the sample. In the technique of negative stain, the heavy metal stain fills the area where the molecules are not present. Thus the outline of the molecules or molecular complexes will be visible.

(c) *Dehydration*: For wet specimens the dehydration has to be carried out gradually in order that the sudden loss of water does not distort the structure. In the case of frozen specimens such sudden change in the water content is less likely, and dehydration is carried out by maintaining the sample in an atmosphere of very low pressure. A common method of dehydrating wet samples is to immerse it in alcohol solutions, slowly increasing the concentration of alcohol, such that the water is ultimately replaced completely by alcohol. In some cases the heavy metal stain can act also as the dehydrating agent. When a thin layer of the staining solution such as potassium phosphotungstate or ammonium molybdate or uranyl acetate is added to a thin specimen exposed to air, the solution takes on an amorphous, glassy shape and protects the morphology of the specimen during dehydration.

(d) *Replication, microtomy etc*: This step in the specimen preparation process renders it suitable for viewing in the microscope. Biological macromolecules such as DNA etc. can be deposited directly onto a very small (~1 mm diameter) fine grid made of copper which is then treated with the staining solution and inserted into the microscope for viewing. Replication is a method used to prepare samples when only the surface features are of interest. A thin layer of metals such as gold, tungsten, palladium or platinum, or a thin plastic film is cast over the sample and allowed to set. The specimen can then be discarded, as a replica of it is now preserved in the metal or plastic film. If the interior of the specimen is of interest, sectioning becomes necessary. In this case the specimen is embedded in a resin such as epoxy resins ('araldite') or polyester resins. It is then sectioned using a microtome. A microtome is a device that uses glass or diamond knives to produce very thin sections of the embedded samples. The thickness of the sections can be as small as 100 nanometers. These sections are then placed on the copper grid for viewing. The heavy metals that are used in the staining process also act to conduct away the excess negative charge that accumulates on a non-conducting sample. For very thin samples viewed on the copper grid, the grid will act to ground the excess charge. In cases where a thick non-conducting sample is used without staining, e.g. for surface SEM studies, it should be coated with a thin metal film solely to act as the conductor.

6.6 Image Reconstruction

Very often the electron micrographs of biological samples such as viruses or ribosomes or other large molecular assemblies consist of several repeated indistinct images of the object. The image can be made more distinct by reconstruction procedures. There are chiefly two such procedures. The first involves simply superposing all the images, such that the 'true' parts of the images, or the signal, gets reinforced, while the background 'noise' will be cancelled. The selection of the identical parts of the image for superposition can be accomplished visually. A more rigorous method is to calculate a superposition function in two dimensions. The superposition function or the cross correlation function will have a maximum value when one image is rotated and translated to superpose on another image optimally. Such optimisation can be accomplished on the computer if the images are digitised and fed to it. A more accurate way to accomplishing a clear image from several indistinct images of the same object is possible if the images are arranged in a regular repeating two-dimensional pattern. If the image is used as a kind of imperfect diffraction grating and is placed in the path of a source of light, a diffraction pattern will be formed. The diffraction pattern can then be filtered to allow through only those parts, which correspond to the periodic lattice. The filtered diffraction pattern is in turn used as the grating and placed in the path of a light beam. The image formed will correspond to a much clearer version of the original object. Mathematically, the procedure is equivalent to performing a Fourier transform, cleaning up the coefficients by filtering, and then performing a reverse Fourier transform. The method of image reconstruction has been widely used to study crystalline materials. Virus structures such as the stem of the *T* phage, membranes and two-dimensional arrays of ribosomes

have all been studied by this method. The method can be extended to three dimensions by taking several two-dimensional images obtained by tilting the specimen at different angles. In fact a single micrograph of a suspension of particles is bound to contain images of the particle in various three-dimensional orientations. Image reconstruction techniques can be fruitfully used to put together these images to form a 3-D picture of the particle.

6.7 Electron Diffraction

The electron microscope can be also be used with a few modifications, to obtain electron diffraction patterns from two-dimensional crystals. Instead of analysing these patterns through geometrical optics, it is possible to use Fourier theory to mathematically transform the diffraction pattern into a three-dimensional image of the particle. Since the electron wavelength is much smaller ($\sim 0.03 \text{ \AA}$) than that of X-rays, it may be thought that a much higher resolution will be obtained. However several obstacles come in the way of attaining such high resolution. Since electrons are easily absorbed, only very thin samples can be used. Electron diffraction is ideal for use in those cases where three-dimensional crystals do not grow but two-dimensional arrays form easily, such as to study membrane proteins. From such a thin 'crystal', almost the entire diffraction pattern will be in a single plane and most of the three-dimensional pattern required to perform a complete Fourier transform is not available. Another problem, also present in the case of X-ray diffraction, is the loss of phase information and the consequent inability to perform the Fourier analysis. One way in which this problem is got over is by using the electron micrograph to obtain some information about the approximate phases and using this information in combination with the measured electron diffraction intensities. In spite of these difficulties, electron diffraction is a popular technique to obtain low-resolution images of large particles that do not easily form crystals suitable for X-ray diffraction.

The experimental techniques of preparing the sample and obtaining the diffraction pattern are not very different from those for obtaining electron micrographs. Since the electron beam can be focussed on very small areas, the ordering in the sample needs to extend only up to a few microns (or less) to obtain the pattern. The intensities are then extracted from the electron diffraction photograph by densitometry. The intensities and the calculated or estimated phases are combined to perform a Fourier transform to obtain the electron density images of the particles. It is possible to obtain the diffraction patterns from tilted crystalline samples and calculate three-dimensional images. These will improve the quality of the final results.

6.8 The Tunnelling Electron Microscope

In the last decade, the tunnelling electron microscope has become an integral part of any laboratory interested in the study of surfaces. Its use in biological imaging is somewhat less widespread, though in recent years it has been used to produce images of DNA molecules that have enabled precise measurement of the pitch of the double helix, as well as of proteins and macromolecular assemblies.

The operation of the tunnelling electron microscope is based on the well-known quantum mechanical principle of electron tunnelling. The equations of quantum mechanics assign a finite non-zero probability for an electron to go from one region of low potential energy, to another such region, even if it does not possess sufficient energy to overcome the barrier between the two regions. This phenomenon may be equivalently visualised as a football going from one well into which it has been dropped into another one nearby, without coming to the surface or making a hole between the wells. In the tunnelling electron microscope, this phenomenon is used in the following way. A very, very sharp needle is brought close to the surface to be imaged. The distance is of the order of a few Angstroms. At such a distance electrons from the surface will tunnel across the gap and set up a measurable

tunnelling current in the needle. This current is a precise indicator of the distance between the tip and the surface. The needle can now be moved (in a raster pattern) across the surface, at each point its height being adjusted to maintain the tunnelling current at a constant value. Thus a record of the vertical portion of the needle at each position becomes equivalent to a record of the surface topography and can be converted into an easily observable picture.

Figure 6.4 is a schematic representation of a scanning tunnelling electron microscope. The tip of the needle can be made so fine that at the very end there is only one single atom. Also, modern electrical measurement technology allows even nano-amperes of tunnelling current to be measured. These technological developments have been instrumental in making possible the visualisation of individual atoms on the surface of materials.

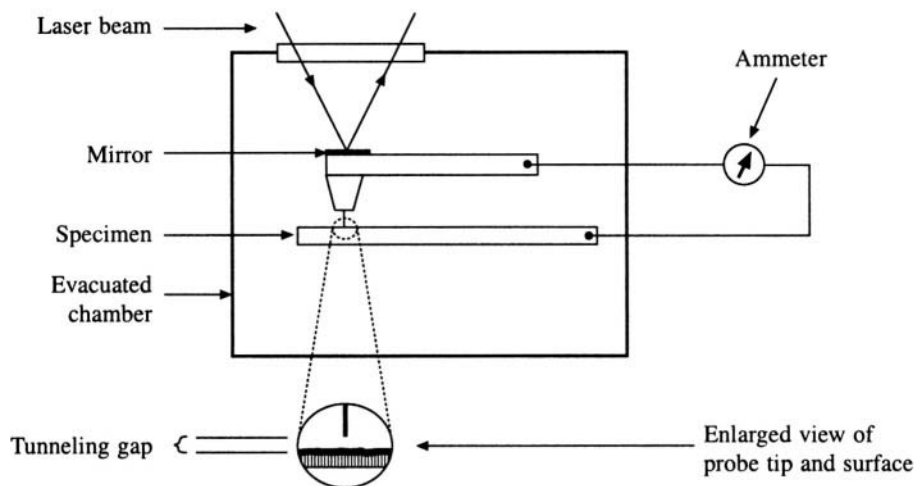


Figure 6.4 A schematic diagram of a scanning tunneling electron microscope

The use of this microscope in the study of biological samples is still not widespread for several reasons. The chief among these is that non-conducting or semi-conducting materials have to be stained with heavy metals before they can be imaged, leading to a loss of resolution. The preparation of the tip and its positioning above the surface also has to be more precise and controlled in the case of biological materials.

6.9 Atomic Force Microscope

The tunnelling microscope has been modified in the following way. Instead of keeping the needle tip at a small distance from the sample the needle is pushed right against the surface. The force present in the tip is kept constant and the tip is scanned across the surface like a phonograph needle running in the groove of a record. A record of the tip's vertical motion is made by reflecting a laser beam from a mirror fixed to the top of the needle and using an interferometer to accurately measure the distance travelled by the laser beam, and hence the position of the tip. This record can be converted and displayed as an image of the surface. Since the atomic force microscope does not depend on a current, it can be used to visualise the surfaces of conductors as well as non-conducting materials. However viewing biological samples such as cells or proteins is still a problem, as the tip tends to distort them.

Other microscopes have been built which are based roughly on the same principle of a tip scanning

the surface. In each different type of microscope a different force is measured. Thus we have friction force microscopes, magnetic force microscopes, electrostatic force microscopes, and scanning thermal microscopes. The last may be described as the world's smallest thermometer. A thermocouple attached to the needle tip measures the temperature at each position. Since heat is essentially motion at infrared wavelengths, it may soon be possible to measure the infrared spectrum of a single molecule!

X-ray Crystallography

7.1 Introduction

In June 1913 Laurence Bragg determined the structure of NaCl using X-ray analysis. Since then, in the intervening 90 years, continuous development of theory, experimental practice and technology has made possible the determination of the three dimensional structures of nearly 250,000 organic and biological molecules. This chapter discusses the powerful technique of X-ray crystallography in detail. Crystallography is the study of crystals. Since biological materials are only rarely crystalline in the natural state, one may wonder what biophysics has to do with crystallography. The answer lies in the fact that, except for NMR, X-ray crystallography is the only available technique of obtaining a detailed, accurate and unbiased picture of a molecule.

The method of X-ray crystallography (Figure 7.1) can be described very briefly as follows: Crystallise the molecule, subject it to irradiation by a beam of X-rays and make a record of the 3-D diffraction pattern. Analysis of this data through computer calculations results in a model of the molecule, which can then be refined against the observed X-ray data. Finally one obtains the three-dimensional coordinates, which describe the position in space of each atom in the molecule. Such a refined model can be displayed and manipulated either as plastic or wooden models, or on the computer graphics terminal. In the following portions of this chapter we will look into each of these aspects in detail.

7.2 Crystals and Symmetries

A crystal is a regular, repeating three-dimensional array of atoms or molecules or groups of molecules. If one imagines a point as representing each repeating unit, the array of points which represents the crystal is called a lattice (Figure 7.2). There are 14 different symmetrical patterns in which the points can be arranged in space and in addition be consistent with the requirement that the patterns be repeated in all three dimensions regularly to form the crystal. In other words, whatever other symmetry

the lattice may possess it has to be consistent with the translational symmetry that defines a crystal. These 14 patterns are called Bravais lattices.

The Bravais lattices exhibit mirror symmetry and rotational symmetries. A lattice possesses mirror symmetry, if there exists in the crystal a plane across which the lattice can be reflected without in any way changing the pattern of arrangement of the lattice points. Similarly, an n -fold axis of rotational symmetry exists if the lattice can be rotated about the axis by $360/n^\circ$ without any change in the pattern. Consistent with translational symmetry, only one-fold, two-fold, three-fold, four-fold and six-fold axes of rotational symmetry are possible.

Each Bravais lattice may be characterised by three axes, which are defined by joining a point to three neighbouring points that are not in the same plane. A unit cell is formed by completing the box and is described by the length of the three sides a , b and c and the angles between them, α , β and γ (Figure 7.3).

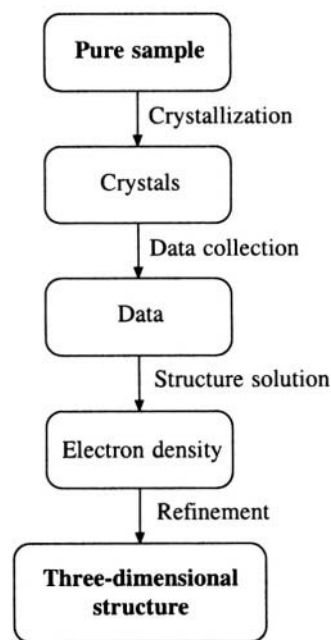


Fig. 7.1 Flowchart of the X-ray crystallographic method to obtain the three-dimensional structure of molecules

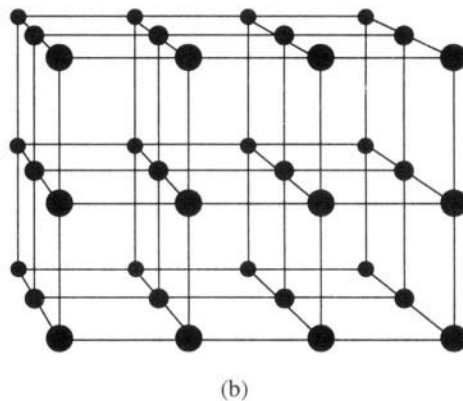
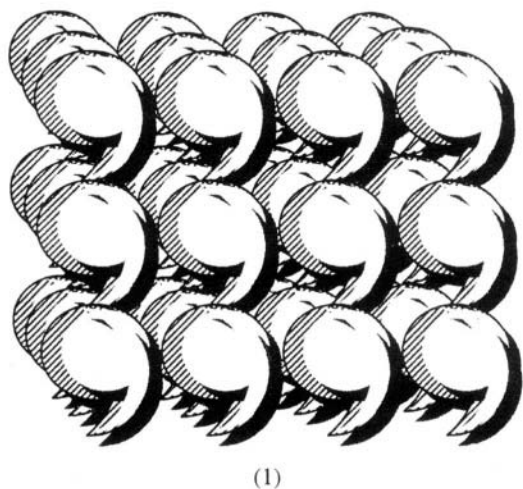


Fig. 7.2 (a) A regular periodic arrangement of three-dimensional objects. (b) The underlying lattice

7.3 Crystal Systems

A triclinic unit cell has three unequal axes and angles (Figure 7.4a).

A monoclinic unit cell (Figure 7.4b) has three unequal axes, two of the angles as right angles, and no limitation on the third one. This condition can be obeyed both in the case where the lattice points

are at the corners of the unit cell, as well as when there is an extra lattice point on one of the faces. This is conventionally taken to be the *ab* plane or the 'C' face. There are thus two possible Bravais lattices in this system.

An orthorhombic unit cell (Figure 7.4c) has three unequal axes, and all three angles are right angles. There are four possible Bravais lattices; with the lattice points at the corners; with an extra lattice points on the *ab* plane; with an extra lattice point at the body centre of the unit cell; and finally with extra lattice points on all six faces.

A rhombohedral system (Figure 7.4d) has three equal sides and three equal angles.

A tetragonal system (Figure 7.4e) has two equal sides and all angles as right angles. There are two possible Bravais lattices.

A hexagonal unit cell (Figure 7.4f) has two equal axes, and one angle as 120° , the other two as right angles.

A cubic system (Figure 7.4g) has equal sides and right angles. There are three Bravais lattices.

The Bravais lattices represent a complete set. No others are possible in a crystal. For example, a tetragonal lattice could have points at the centres of all the faces, like the face-centred cubic lattice in Figure 7.4g. However, this is not another tetragonal lattice, for a rotation of 45° around the *c* axis will give back the body-centred lattice of Figure 7.4e.

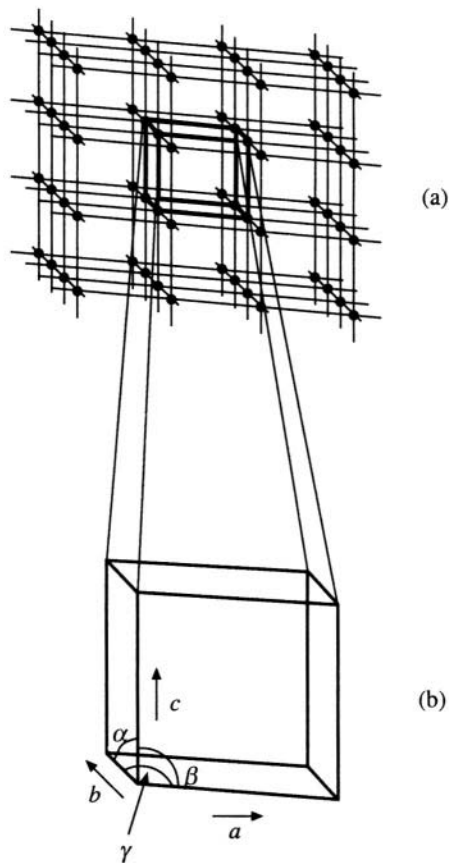


Fig. 7.3 (a) The lattice (b) The unit cell

7.4 Point Groups and Space Groups

Crystals can be classified in terms of the symmetry between their faces. In this case we consider the crystal as a whole and ignore for the present the lattice structure underlying it. Each crystal can be identified by the group of symmetry operations that pertain to it. The symmetry operations can be *n*-fold rotation, and mirror planes. In addition we may have centres of inversion—each point in space is matched by a similar one at an equal distance on the opposite side of the centre. Rotatory inversion is another symmetry operation that the crystal as a whole may possess—the operation consisting of rotation about an axis combined with inversion about a centre.

Each group of symmetry operations that identify a crystal is called the symmetry point group of that crystal. Since the point group of the crystal is also the point group of the underlying lattice, only symmetry operations consistent with lattice translations are allowed. With this restriction, a total of 32 point groups is possible for crystals and is divided among the seven crystal systems. For example, the monoclinic system has always one two-fold axis, and therefore the point groups 2, (i.e. one two-fold axis alone), *m* (i.e. one mirror plane alone) and 2/*m* (i.e. a two-fold axis and a mirror plane perpendicular to it) all fall in the monoclinic system. The point group 222 (three two fold axis

perpendicular to each other) requires three perpendicular axes and hence will fall in the orthorhombic system (Figure 7.5). Other point groups are similarly assigned to other systems.

In addition to the lattice, when we take into consideration the structural components of crystals, i.e. the atoms or molecules, additional symmetry elements, combining translation with rotation or reflection come into play. One of these is the screw axis which corresponds to a rotation and a translation, and is designated $\mathbf{n}_m$ which implies an n -fold rotation followed by a translation of m/n of the unit cell axis parallel to the rotation axis. Thus, for example, a 3_1 screw about c implies a 3-fold rotation about the c axis and a translation of $1/3$ times the length of c along the c axis (Figure 7.6). Similarly a glide plane arises when a mirror reflection is followed by a translation of $1/2$ along a unit cell axis (a , b , or c glides) or a unit cell diagonal (n glide).

The three dimensional assembly of planes, axes and centres of symmetry is called a space group.

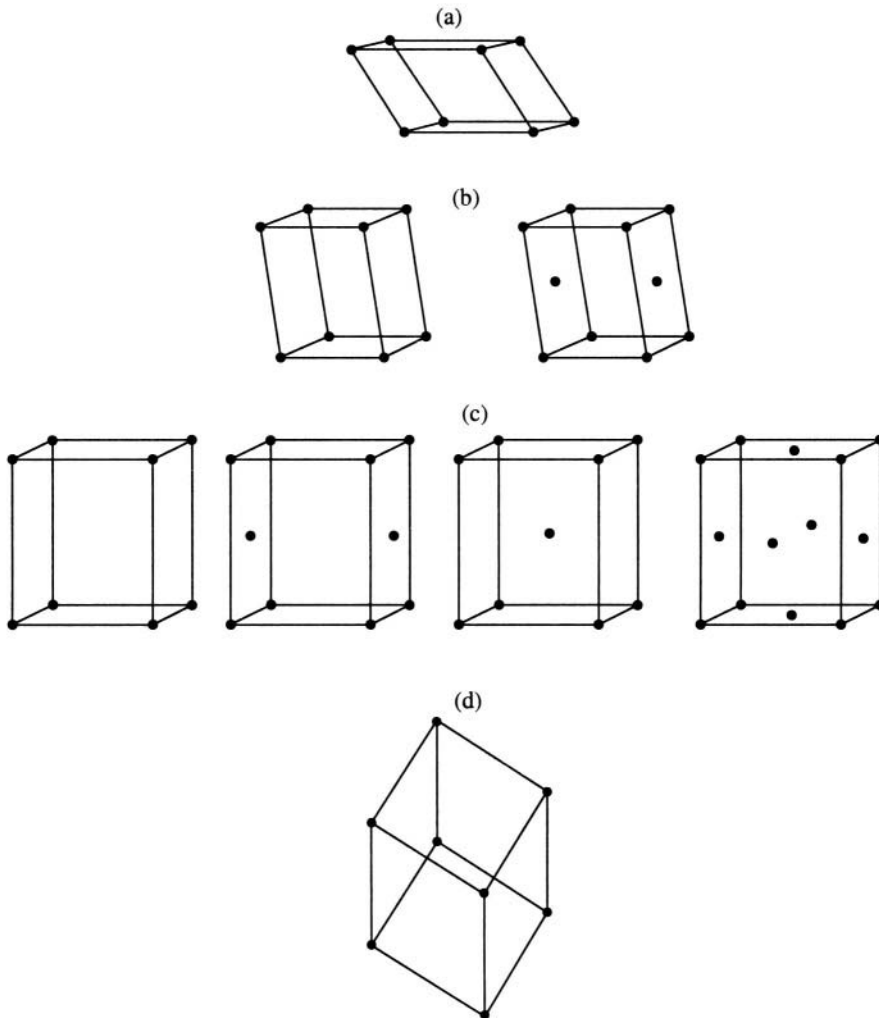


Fig. 7.4 (Contd)

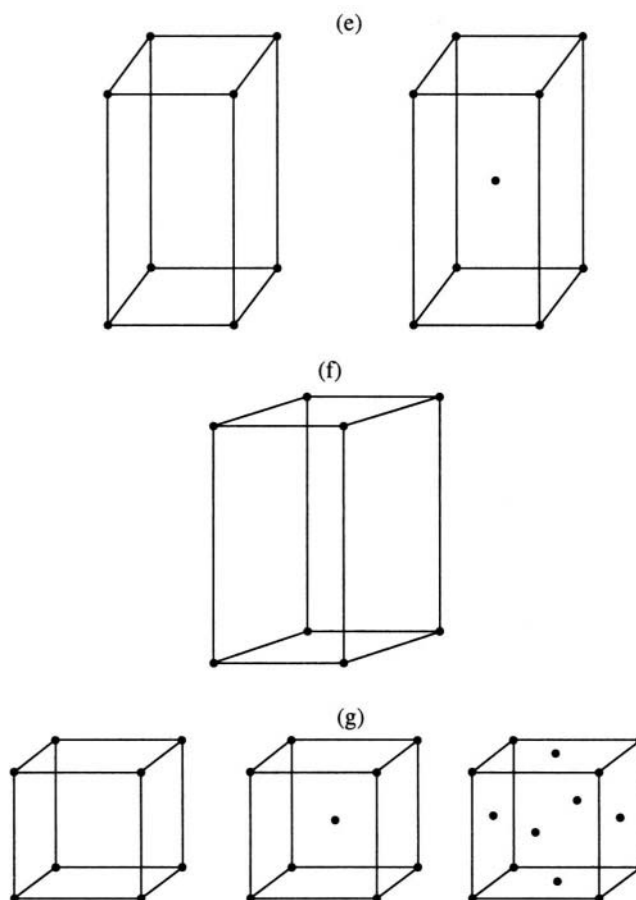


Fig. 7.4 The seven crystal systems and the fourteen Bravais lattices. (a) Triclinic. (b) Monoclinic. (c) Orthorhombic. (d) Rhombohedral. (e) Tetragonal. (f) Hexagonal. (g) Cubic

The symmetry operation of a space group, together with the unit cell translations, will transform a single object into a crystal. Just as there are only seven distinct crystal systems, fourteen Bravais lattices and 32 point groups, there are only 230 space groups, or 230 distinct ways in which an object, such as a molecule, can be arranged in three dimensional space to form a crystal. A description of the 230 space groups is beyond the scope of this book. The *International Tables for Crystallography*⁷ contain all the details. The symmetry of the crystal, i.e. its space group, can be deduced from its X-ray diffraction pattern and is an essential first step in the X-ray analysis. Before we can subject a crystal to X-ray analysis, however, we need to grow it.

7.5 Growth of Crystals of Biological Molecules

The chief difference between crystals of biological substances and those of other important materials such as silicon is that in the former the repeating units are invariably molecules or groups of molecules,

⁷*The International Tables for Crystallography, Vol A*, T. Hahn, (ed.), 1987. D. Reidel Publishing Company, Dordrecht, Holland.

while in the latter the repeating units are atoms or groups of atoms. This implies that the forces of bonding between one repeating unit and another are much weaker in the former, being van der Waals forces or hydrogen bonds. In the latter, on the other hand, the crystals are built up from strong covalent or ionic bonds. Crystals of biological materials are thus usually soft and brittle as compared to inorganic substances.

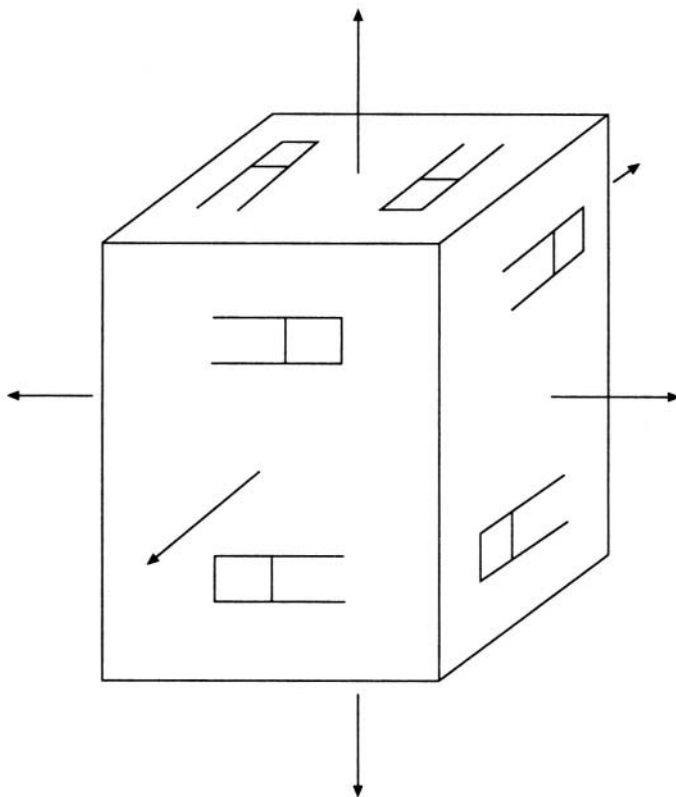


Fig. 7.5 An illustration of the 222 point group symmetry

Another difference is the size. Biological and organic molecules usually form very small crystals (approximately 1 mm^3) as compared to inorganic materials (about 1 cm^3). Bio-crystals are usually grown from solution and may have a large solvent content. This is especially true of crystals of macromolecules such as proteins and nucleic acids where the solvent can take up as much as 80% of the crystal. The growth of bio-crystals is a multi-parameter process. Table 7.1 gives a list of factors that could affect crystallisation.

To grow crystals of any compound from solution, the molecules have to be brought to a supersaturated state. When the solution returns to equilibrium, the substance will precipitate out, hopefully as crystals, and not as an amorphous precipitate. Supersaturation can be achieved by slow evaporation. Also by varying one of the parameters listed in Table 7.1 one may change the solubility and hence the saturation level of the molecule.

In order to obtain good, defect free crystals of a size suitable for X-ray analysis, the equilibration needs to be very carefully controlled, and techniques have been developed for this.

Slow evaporation (Figure 7.7) is conceptually the simplest of techniques. It is chiefly used to crystallise small organic molecules. The solution is placed in a small glass beaker and sealed with an

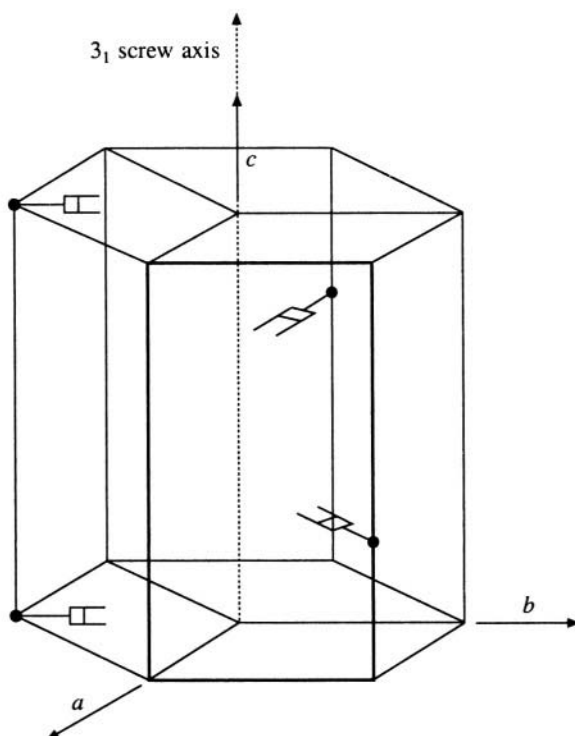


Fig. 7.6 The 3_1 screw symmetry axis in the trigonal system referred to hexagonal axes

airtight cover, except for one or two small apertures through which slow evaporation of the solvent takes place, bringing the solute to supersaturation and hence to crystallisation.

Table 7.1 Factors affecting crystallisation of organic molecules and biological macromolecules

— purity of the sample.
— concentration of the sample
— temperature, pH of the solution
— time, rate of equilibration
— ionic strength
— convection currents
— volume of the crystallisation solution
— pressure, vibrations
— additives, their character and concentration
— solvents (especially for small organic molecules)

For biological macromolecules, especially, dialysis can be used to obtain crystals. The solution of the molecule is separated from a large volume of the solvent by a semi-permeable membrane which gives small molecules (ions, additives, buffer, etc.) the freedom to circulate but prevents the movement of the macromolecule. The macromolecule solution is gradually built up to supersaturation by changing the concentrations of the small molecules in the large volume of the solvent. This has the effect of gradually changing the pH or the ionic strength or any similar parameter.

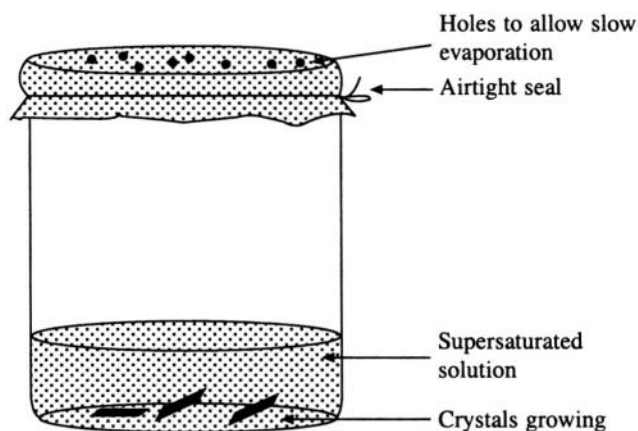


Fig. 7.7 Slow evaporation technique to grow crystals from a supersaturated solution

Vapour diffusion is another commonly used technique (Figure 7.8). A droplet of solution containing the molecule is equilibrated against a reservoir containing the precipitant, i.e. the crystallising agent, at a higher concentration than in the drop.

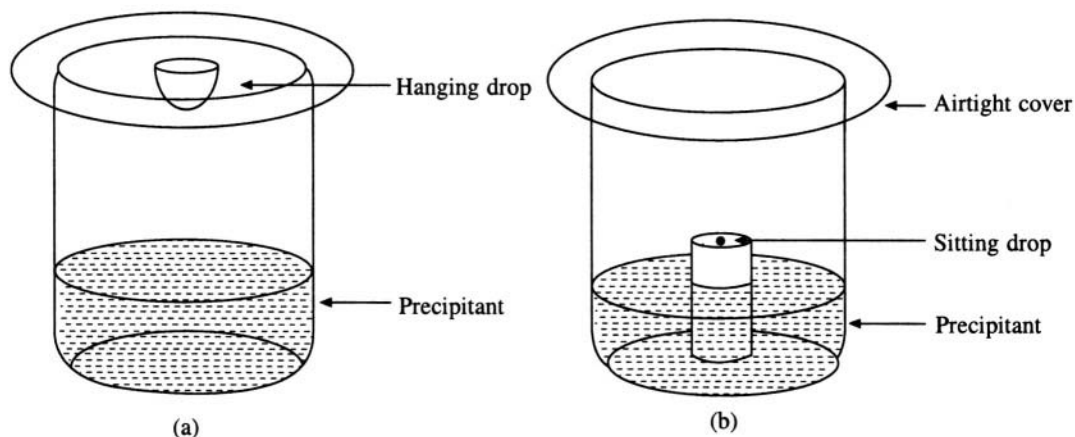


Fig. 7.8 Vapour diffusion method. (a) Hanging drop. (b) Sitting drop

In batch methods, the molecule to be crystallised is mixed with the crystallising agent at a high concentration so that supersaturation is immediately reached. Though this may lead to a large number of tiny crystals, it frequently gives rise to good crystals.

Interface diffusion (Figure 7.9) is used for small molecules as well as large biological macromolecules. The precipitant solution is gently layered over the solution of the molecule. Crystals usually grow at the interface.

Sometimes it is necessary to introduce seeds into the solution to induce the crystals to grow. These seeds are usually obtained from previous less successful crystallisation attempts, and may be extremely tiny crystallites. Table 7.2 gives some of the solvents, additives and precipitants used in growing crystals of organic and biological molecules.

7.6 X-ray Diffraction

Diffraction is a well-known phenomenon that occurs when two or more waves are combined. If the waves combine in such a way that the peaks of one wave fall on the peaks of the other, the total effect is to enhance the intensity. On the other hand if the combination is such that the peak of one wave falls on the trough of the other, the effect is one of cancelling each other out and the intensity of the combination is zero. If an incoming wave is split into two by slits and the two parts of the wave are allowed to interfere with each other, then a pattern is produced on a screen placed on the other side of the slits. Whether the two beams interfere constructively or destructively at a particular point on the screen, is decided by the difference in the length of the paths that each of the waves take from the slit to the point on the screen. If the difference corresponds to a whole number of wavelengths, the combination of the two waves is perfectly constructive and the intensity is greatly enhanced. A regularly spaced series of slits has the same type of effect and is called a diffraction grating (Figure 7.10). It is possible to have two and three-dimensional diffraction gratings. A crystal lattice is nothing but a three-dimensional grating (Figure 7.2(b)). However, it cannot diffract visible light. This is because for the diffraction effect to be noticeable, the spacing between the slits in the grating should be of the same order of magnitude as the wavelength of the incident radiation. In a crystal the distance between the lattice points corresponds to the dimensions of atoms and molecules i.e., a few Angstroms. X-rays are radiation with wavelengths of this size, and hence diffraction from crystals is observed with X-rays. The beginnings of a theory of X-ray diffraction from crystals was developed by Max von Laue. A few years later, W.L. Bragg came up with the idea of treating diffraction from a crystal as reflection from the lattice planes and put the theory on a firm footing. A crystal can be characterised as having lattice planes, each of which is identified by a set of integers h , k and l (Figure 7.11). These numbers are known as Miller indices. Miller indices help in identifying the plane from which reflection takes place. The interference of the beams of X-rays reflected from successive parallel planes corresponding to a particular triad of hkl

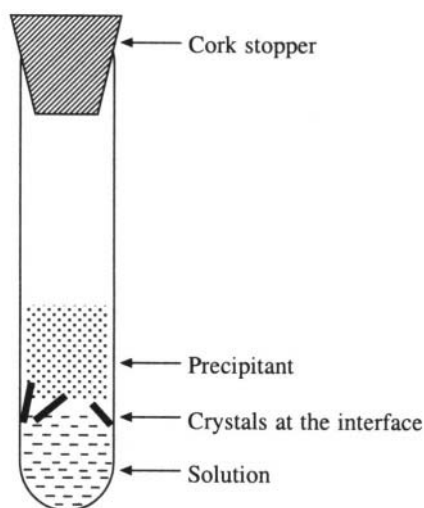


Fig. 7.9 The interface diffusion method of crystal growth

Table 7.2 Compounds that are required to crystallise biomolecules

Molecule	Solvents	Additives	Precipitants
Small organic Molecules	Organic solvents (e.g. CCl_4 , ethyl acetate etc.), alcohols, water	Metal ions, salts	Alcohols, organic solvents
Proteins	Aqueous solutions	Co-factors, sugars, metal ions, ion complexes, polyamines, salts	Ammonium sulphate, polyethylene glycol (PEG) Methylpentanediol (MPD)
Nucleic acids	Aqueous solution	Polyamines, metal ions, salts	MPD, PEG, Isopropanol, other alcohols

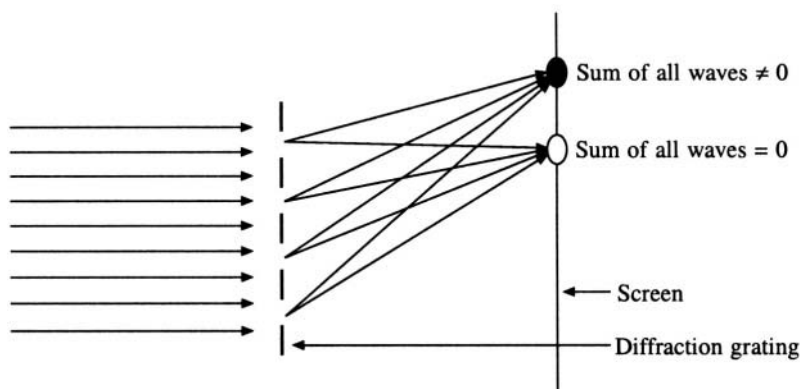


Fig. 7.10 A schematic sketch illustrating diffraction from a grating

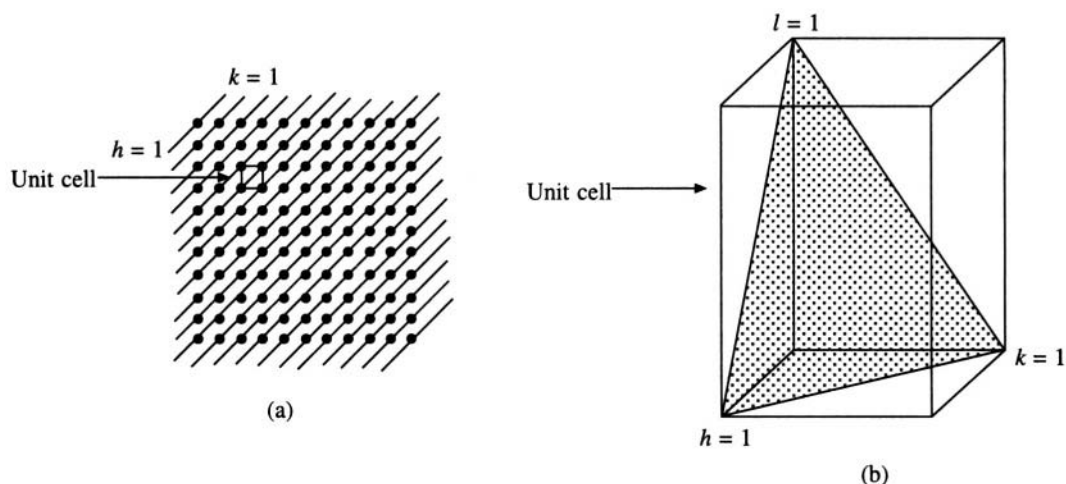


Fig. 7.11 Lattice planes and Miller indices. (a) A two-dimensional lattice showing the set of ‘planes’ (lines) with the Miller indices $h = 1$, $k = 1$ or $(1, 1)$. (b) A three-dimensional unit cell showing the plane $(1, 1, 1)$

values, leads to diffraction effects. For the interference to be constructive, as in the case of the one-dimensional diffraction grating, the difference in path lengths of the reflected beams should be equal to an integral number of wavelengths of the incident beam. A simple geometrical construction (Figure 7.12) leads to the famous Bragg reflection condition or Bragg’s law

$$2 d \sin \theta = n \lambda.$$

When a crystal is irradiated by an X-ray beam, Bragg’s law is used to predict the angles θ at which the diffracted intensities may be expected. This requires knowledge of the inter-planar spacing d and the wavelength λ of the X-rays. Conversely, if the angles of occurrence (and relative intensities) of the diffracted spots are measured in an experiment, the corresponding inter-planar spacing in the crystal may be calculated and the lattice structure can be determined. Therefore, the next step, after growing the crystal is to place it in a beam of X-rays and measure the angles and intensities of the diffracted spots.

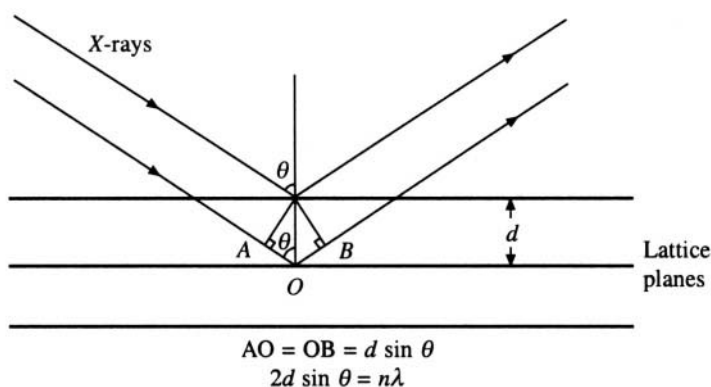


Fig. 7.12 Bragg's Law. For constructive interference to occur, the difference in path lengths, given by $AO + OB$ should be equal to an integral number of wavelength or $n\lambda$.

7.7 X-ray Data Collection

The process of collecting a complete set of X-ray diffraction data from a crystal consists of measuring the intensity of the diffracted beam from each set of planes. The experimental set up (Figure 7.13) consists of an X-ray source, a goniometer (Figure 7.14) on which to mount the crystal, and a detector to measure the intensity of the reflected X-ray beam.

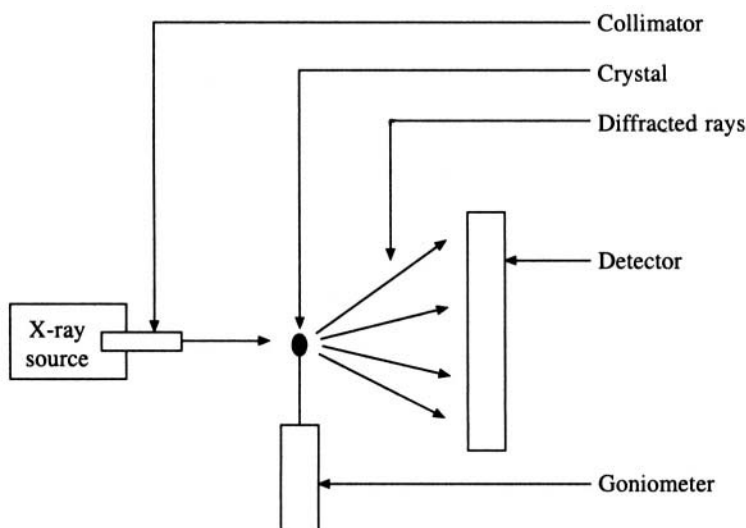


Fig. 7.13 A schematic sketch of the experimental setup for x-ray diffraction

The X-ray source is usually a sealed tube in which electrons are accelerated from one end and allowed to impinge at the other end on a metal target, usually copper or molybdenum for biologically relevant samples. This produces X-rays of wavelength $1.5418 \text{ \AA}$ (for Cu $K\alpha$) and $0.7107 \text{ \AA}$ (for Mo $K\alpha$). This process also produces a tremendous amount of heat and the tube has to be kept cool by circulating chilled water or by other means, such as rapidly rotating the metal target.

The collimated beam of X-rays then fall on the crystal mounted on a goniometer. This is a device

that allows one to position the crystal in different orientations in order to bring different sets of planes into the reflecting position.

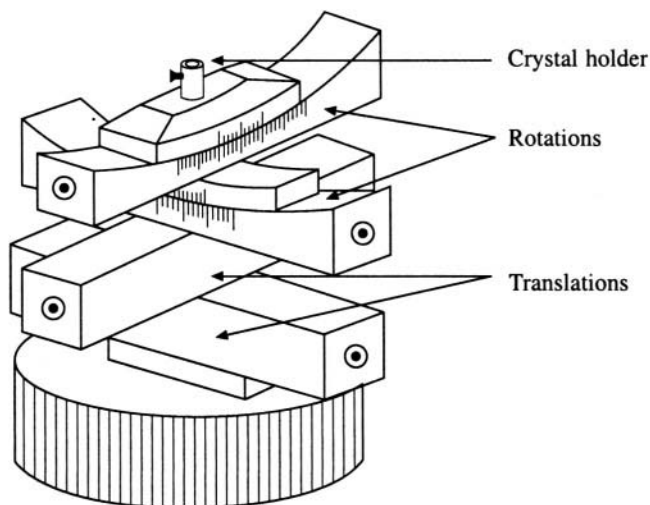


Fig. 7.14 A schematic diagram of a goniometer head, indicating the movements possible to centre the crystal in the X-ray beam

The detector of the diffracted X-rays may be a film, or a radiation counter. The first measurements of X-ray intensities used in the solution of the structure of NaCl, were made with an ionization chamber. This however allows the measurement of only one reflection at a time, since the crystal and the detector have to be set at different angles for each reflection. Films allow the recording of several reflections at a time, but suffer from the requirement of long exposure times, as well as imprecision in the conversion of the blackening of the film into a numerical value of intensity. Modern devices such as multi-wire proportional counters or imaging plates offer the advantages of both films and of counters and are now almost exclusively used in the collection of data from crystals of biological macromolecules.

The outcome of the data collection process is a list of angles, indicating the orientation of the crystal and the detector and the intensity of the reflection from the oriented lattice plane at this position. Instead of the angles, it is usual to identify the set of planes that give rise to each diffracted spot by the Miller indices hkl and associate with each value of hkl the appropriate value of the intensity. From the list of intensities, the amplitudes of each reflected wave is extracted by the simple process of taking the square root, since the measured intensity of a wave is the square of its amplitude. These amplitudes, represented as F_{hkl} , are now used in the analysis of the structure of the molecule that makes up the crystal.

7.8 Structure Solution

The structure of the molecule is obtained by a Fourier transform of the observed amplitudes F_{hkl} . This statement, which forms the core of the X-ray crystallographic method, is amplified and explained below. In order to first understand the mathematical process of Fourier transformation, consider the periodic function shown in Figure 7.15. Fourier theory shows that such a function can be represented as a sum of a series of simple waves. Note however that it is not only necessary to know the amplitude (and wavelength) of the simple waves but also to know the relative positions, or phases of the waves in order to reconstruct the original periodic function.

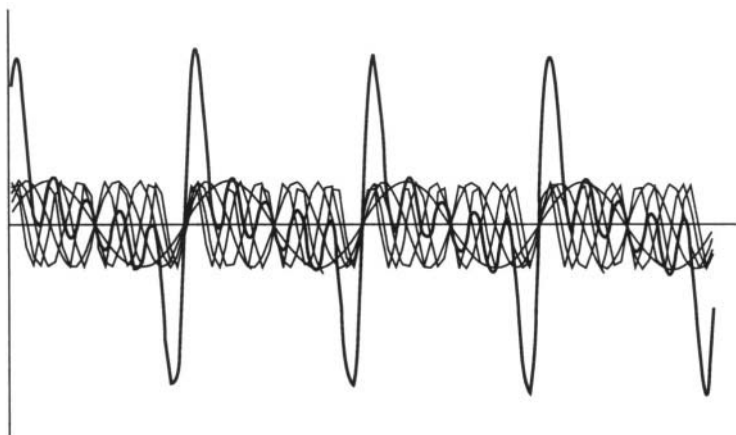


Fig. 7.15 The Fourier decomposition of a complex periodic function (thick line) into its sine and cosine components (thin lines). As more component waves at different frequencies (wavelengths) are added, the resulting wave approaches a square wave more and more closely. The Fourier decomposition of a square wave itself would result in components of all possible frequencies.

A crystal which has the unit cell, with its complex contents, repeated periodically, can be considered to be made up of simple planes of electron density arranged in three dimensions. This periodic function can be represented as series of simple waves. The amplitudes of these simple waves are measured, i.e. F_{hkl} . In order to reconstruct the crystal as a three-dimensional superposition of these simple waves of electron density, it is necessary to know the phases. Once the phases are known the mathematical operation of Fourier transformation, which corresponds to the physical superposition of the simple waves, will yield a picture of the distribution of electrons within the unit cell of the crystal. In other words, we obtain a picture of the structure of the molecules in the unit cell. The problem of X-ray crystallography now translates into one of obtaining the phase information. This is called the phase problem in crystallography since, in the X-ray diffraction experiments, the phases cannot be measured directly and the phases have to be determined by other indirect means. Some of these methods are described below.

7.8.1 Heavy atom or multiple isomorphous replacement (MIR) method

This method requires that the crystal of the sample be grown in at least two ways, once in the presence of a 'heavy', usually metal, atom and once in its absence. An additional, rather stringent, requirement is that in both cases the crystals are 'isomorphous', i.e. they have the same unit cell dimensions and crystal symmetry. In such a situation the only difference in the arrangement of the atoms in the crystal would be the presence or absence of the heavy atom. Whether a particular atomic species is called 'heavy' or not would depend on the contribution it makes to the total electron density and hence to the X-ray diffraction intensities. Thus for small organic compounds of molecular weight about 1000, cobalt, copper, iron and bromine would qualify as heavy atoms. For proteins, crystals may have to be grown in the presence and absence of platinum, samarium, mercury, etc. The heavy atom method depends on the assumption that, since a large portion of the scattering matter in the unit cell is concentrated into a single atom, a knowledge of the position of this atom and hence its contribution to the phases, would lead to a knowledge of the phases due to all the atoms in the unit cell, at least to a very good approximation. The position of the heavy atom itself is usually arrived at

by a special type of Fourier analysis called the ‘Patterson synthesis’. Once the position of the heavy atom is known, the phases can be calculated. On the basis of the above stated assumption, they are the phases due to all the atoms and a Fourier transform will give the approximate positions of the atoms. Usually, especially for macromolecules like proteins, there are ambiguities in assigning the phases, which can be overcome only by growing many crystals by substituting different heavy atoms and collecting and analysing the X-ray data for each crystal. This method is therefore usually called the ‘Multiple Isomorphous Replacement’ method.

7.8.2 *Direct methods*

H. Hauptman and J. Karle pioneered a statistical analysis of the X-ray data to obtain the phases. It was realised that the values of the phases have to be such that the electron density obtained has to correspond to physical reality, i.e. there can be no ‘negative’ electron density in the crystal (the presence of an electron is considered as ‘positive’ electron density) and the density should correspond to separate or bonded atoms. In addition the phases for some of the reflections called semi-invariants can be restricted to a few values depending on the space group. These facts can be used iteratively in a computer program to generate the phases for all the reflections. A Fourier transform then gives the structure. The theory is very well developed for crystals with less than about 50 atoms in the unit cell and the technique is routinely used to determine small molecule structures. For larger molecules such as proteins, the method is still not fully successful.

7.8.3 *Molecular replacement*

Pioneered by M.G. Rossmann, the molecular replacement technique is used when the unknown structure can be assumed to be similar to an already known structure. The problem then becomes one of first finding the orientation and position of the model molecule in the unknown unit cell, and next of finding the differences between the model and the actual structure. The first part of the problem is usually solved by means of a search procedure. The computer, in effect, generates the F_{hkl} for each position and orientation of the model in the unit cell and compares these calculated reflections with those actually observed in the experiment. The position and orientation corresponding to the best match is then taken up for the second part of the structure solution procedure, i.e. the refinement.

7.8.4 *The anomalous scattering technique*

In this method the wavelength of the incident X-ray beam is adjusted so that the scattered intensities would be different when measured from opposite sides of the crystals. Normally the two sets of intensities would be the same, but in particular instances when the wavelength is close to the so-called absorption edge of one of the atomic species in the crystal, the intensities are different. This difference is then exploited to obtain the phases. Since the wavelength has to be tuned to match the element, this method became widely used after the advent of powerful, tunable (and very expensive!) sources of X-rays called synchrotron radiation sources.

7.9 **Refinement of the Structure**

Once the approximate value of the phases are determined the next step is to refine them. This usually more conveniently done by performing the Fourier transform and refining the approximate positions of the atoms obtained by including other known data such as stereochemistry.

7.9.1 *Least squares method*

This is a well-known mathematical technique used to fit a model to the observed data. Small changes

are made in the model and with each set of changes, the expected data are calculated. The differences between the observed and calculated data are once again used to determine the changes in model in an iterative process that terminates when further changes do not lead to any further improvement in the calculated data. Since the differences can be either positive or negative, i.e. the calculated data points can either be more or less than the observed data points, the sum of the squares of the differences, which is always a positive quantity, is the term that is actually minimised in the least squares procedure. The whole procedure can be described in terms of a mathematical operation, with the changes to the model in each cycle being calculated automatically. The mathematical equations lend themselves to easy computerisation and the least squares method is widely used in many fields of science and engineering. As applied to crystallography, the function to be minimised is called the Residual or the R factor and is usually defined as

$$R = (\sum ||F_o - |F_c||) / (\sum |F_o|)$$

where F_c is the calculated structure amplitude for a particular reflection hkl , F is the observed structure amplitude for the same reflection and the summation is over all the reflections. Note that only absolute differences between the calculated and observed data are taken into consideration, in keeping with the idea that the actual sign of the displacement is not relevant to the least squares technique. The calculation of the F_c involves a Fourier transform of the model and it is performed in each cycle after shifting the atoms by the (usually small) amounts indicated in the previous cycle. Though the amount of computation required is quite large, modern technology allows the complete refinement of structure of molecular weight of about 1000 in less than one day. In order to work, the least squares technique requires that the initial model be close ($< 0.5 \text{ \AA}$) to the actual final model. Furthermore a very large number of observations are required for reliable results. As a rule of thumb, between seven and ten observed data points are required for every parameter refined. For example, a molecule containing 20 atoms will have three positional parameters (x, y, z) and six parameters corresponding to the thermal vibration, a total of nine parameters for each atom. To ensure accurate refinement therefore, more than 1500 reflections will have to be measured. In the case of biological macromolecules, this condition is seldom satisfied. Owing to various factors, including high solvent content of the crystals, the molecules are not very well ordered and reflection planes with spacing of less than about $2.0 \text{ \AA}$ are usually not sufficiently regularly spaced to give measurable diffracted intensities. In other words the limit of resolution of the data in such cases is usually not beyond $2.0 \text{ \AA}$. This means that the observation-to-parameter ratio is not good enough to allow refinement using the standard least squares techniques.

7.9.2 Constrained-restrained refinement

There are two ways in which the observation-to-parameter ratio can be improved without having to collect additional X-ray data. The first is to increase the number of observations by including known chemical information. For example, in the extreme case, the whole molecule can be treated as a single rigid unit. The only variables are then the orientation and the position of the molecule, a total of six parameters. This is called constrained refinement. In the intermediate cases various parts of the molecule can be considered rigid units, which are then linked to each other by flexible bonds. Each such unit will have six parameters, as compared to the three parameters for each atom of the unit. A typical example is the case of the DNA molecule (Figure 7.16) where each base can be considered to be a rigid planar unit. Likewise the sugar is a rigid unit, as also the phosphate group. Obviously the total number of parameters to be refined would reduce drastically, leading to a much better observation-to-parameter ratio. Another method of improving the ratio based on available stereochemical information

is called the restrained refinement technique, where the distances between the various atoms in the molecule are known. For example, the distance between two carbon atoms joined by a single bond cannot be very different from 1.54 Å. Similarly the angle at a sp^3 hybridised carbon atom has to be close to 109.5° . This chemical information is treated as additional observation points and for the same number of parameters, the ratio improves. The bond length and angle information, together with other stereochemical and physicochemical information such as van der Waals contacts, hydrogen bonds, steric hindrances and intra-molecular electrostatic forces can also be programmed as an energy function which is to be minimised. This is then added to the residual function mentioned earlier and the whole is minimised by least squares or some other optimisation technique.

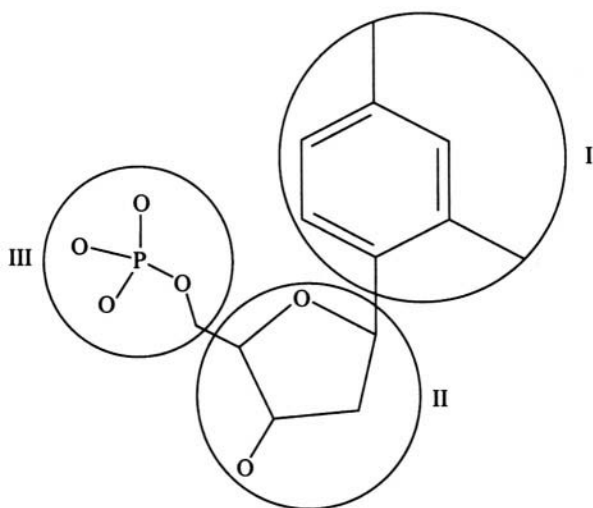


Fig. 7.16 The rigid groups in a nucleotide unit. I: atoms of the base are considered as one rigid unit. II: atoms of the ribose moiety. III: atoms of the phosphate group.

7.9.3 Computer graphics in refinement

The advent of high-resolution computer graphics has made possible the development of several new techniques of refinement and of utilising known chemical and biological information in the refinement process. Today much of the crystallographer's creativity is expressed at the computer graphics terminal. As mentioned earlier, once the approximate phases have been obtained, a Fourier transform using the amplitudes F_{hkl} as the Fourier coefficients produces a map of the electron density in the unit cell (Figure 7.17). The model is superposed on this map and wherever the fit of the model to the electron density is not good enough, adjustments are made to improve the fit. After a cycle of fitting, or changing various parts of the model, the map can be recalculated using the changed positions of the atoms in the model to obtain the new set of phases. This process iteratively brings the model closer and closer into a perfect fit with electron density. Simultaneously a constrained or restrained least squares refinement procedure is also carried out and the conventional Residual or R factor is also used as a check on the correctness of the structure.

At the end of the refinement process we obtain a set of coordinates for each of the atoms of the molecule and another number (or set of numbers) representing the thermal vibrations of the atom. These can be subjected to further geometrical analyses to understand the chemistry or biology of the molecule.

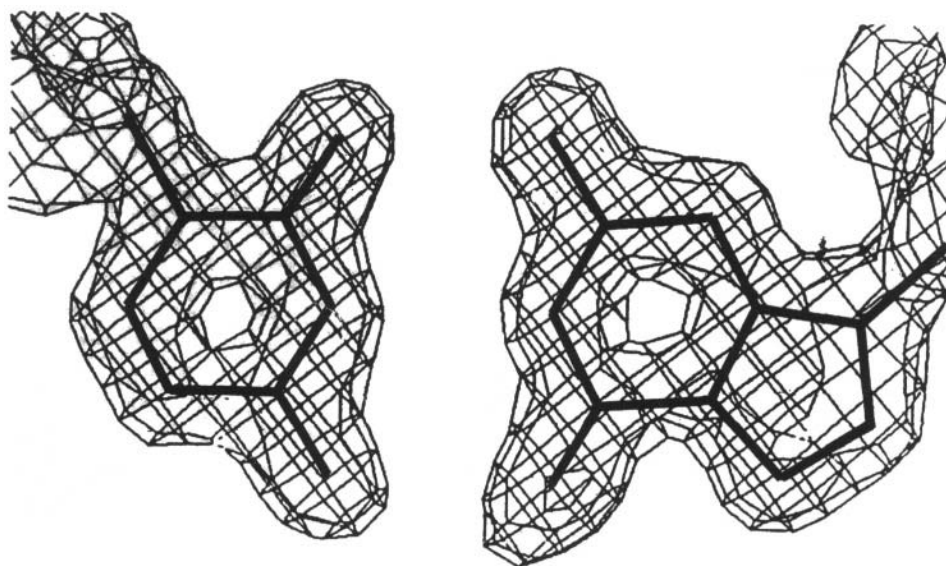


Fig. 7.17 Electron density net corresponding to a G.C. base pair in the crystal structure of a DNA fragment.

7.10 Note on the Resolution of an X-ray Structure

The number of X-ray reflections that may be obtained from a crystal depends on the degree of order present in it. A greater degree of order implies that lattice planes with smaller inter-planar spacing are regularly arranged in the crystal and that X-ray reflections from these planes can also be measured. This implies that in the final structure, features at smaller distances can be resolved (i.e. seen as separate entities). The terminology of the resolution can be somewhat confusing. Since a higher resolution means smaller separation of the observable features, a resolution of 6 Å, is lower than a resolution of, say, 2 Å. The larger the number the lower the resolution and vice versa. The resolution can also be expressed as the angular limit at which the intensity data can be observed. By Bragg's law

$$\sin \theta = n\lambda/2d$$

In other words, the lower the inter-planar spacing d (i.e. the greater the resolution) the higher the angle at which the diffracted beam corresponding to that set of planes is observed. The terminology in this case is more straightforward since $\sin\theta$ varies directly as the resolution. If we normalize with respect to the wavelength of the incident radiation, the resolution may be expressed as $\sin\theta/\lambda$ (or $1/2d$). Thus, the highest resolution achievable using radiation of a particular wavelength is $d = \lambda/2$ Å

NMR Spectroscopy

8.1 Introduction

Nuclear Magnetic Resonance Spectroscopy is one of the most important experimental techniques currently in use in biophysics. Until the early 1970's NMR was of interest mainly to physicists and chemists. Physicists use NMR Spectroscopy to study the nature and interactions of the atomic nucleus. Chemists use NMR Spectroscopy to determine chemical structure of small molecules. With the development of techniques such as Fourier transform NMR, and two-dimensional and three-dimensional NMR, biochemists and biophysicists use the method to determine the structures of large molecules such as proteins and also to study the dynamic behaviour of such molecules. Magnetic Resonance Imaging is becoming increasingly popular in medicine as a means to obtain accurate three-dimensional images of the interior organs of the human body in an essentially safe and non-invasive manner.

NMR is a rapidly growing field and already it is so vast that only a few of the specialised techniques can be explained in the present book. In this chapter we first explain the basic principles of the NMR technique. NMR theory is then dealt with in somewhat greater detail including a description of the parameters measured in the different types of experiments. Finally we discuss the applications of NMR in chemistry, biology and medicine.

8.2 Basic Principles of NMR

NMR is a spectroscopic technique. In common with other spectroscopic techniques, the NMR experiment consists of the measurement of the intensity of absorption (or emission) of electromagnetic radiation at each frequency over a specific range. In NMR this range is the radio-frequency range, i.e. from a few MHz to a few hundred MHz. At this frequency, the energy is absorbed or emitted when the nuclei of the atoms go from one quantized spin state to another. Spin is a property possessed by only certain atomic nuclei such as the hydrogen nucleus H or the phosphorous isotope ^{31}P (Figure 8.1). A crude classical analogy to this property of spin is the rotation of a body about an axis that passes through

its centre. Since the nucleus has a charge, the spinning motion will give rise to a magnetic field, and the spinning nucleus can be thought of as a small bar magnet. The axis of the bar magnet will be parallel to the axis of spin. When this nucleus is placed in a strong external magnetic field, then, by the laws of quantum mechanics and due to the interaction of the two magnetic fields, the axis of spin will orient itself in one of two possible orientations (Figure 8.2). Transitions between these two states can occur when another external oscillating magnetic field supplies sufficient energy for the spin system, i.e. the nucleus to go from the state or orientation of lower energy to the state of higher energy. The difference in energy of the two states corresponds to the energy of a radio-frequency field and hence a nuclear spin, placed in a static magnetic field, must be excited by radio-frequency electromagnetic radiation in order that the transition takes place. The energy absorbed from the radio-frequency field can be detected and this is the fundamental parameter measured in an NMR experiment. The frequency at which the transition occurs is called the resonance frequency. The resonance frequency of a particular type of nucleus, e.g. ^1H , will be different when it is placed in different chemical environments. For example the ^1H in a methylene group will have a distinctly different resonance frequency from one in a methyl group. By changing the frequency of the exciting radio-frequency field and measuring the intensity of the absorption at each frequency, it is possible to detect the nuclei and in each case identify their respective environments. It is this feature which is useful in identifying various chemical groups in molecules. If the magnetic field in which the spins are placed is very high, subtler effects become noticeable, i.e. the resolution of the spectrum increases. In this way it is possible to detect even conformational changes in the molecule.

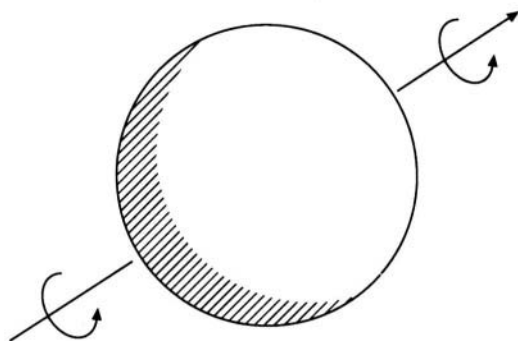


Fig. 8.1 The spin of a nucleus

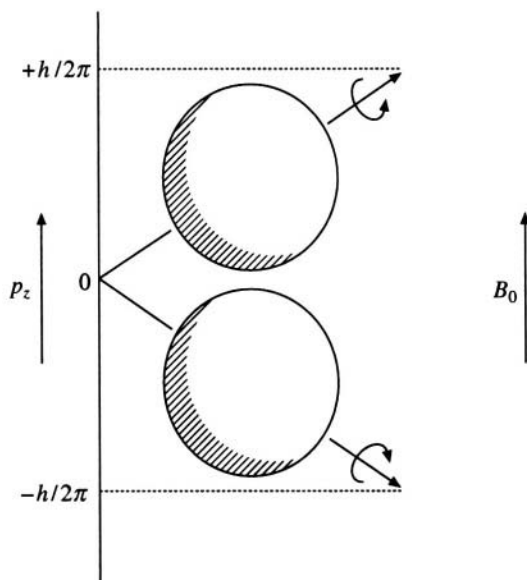


Fig. 8.2 The two possible orientations of nuclear spin in a magnetic field

Continuously changing the frequency of the radio field and measuring resonance is a tedious and time-consuming task. So much so, in the case of large molecules, where the changes in intensity are very small, formidably long data collection time would be required. Fourier transform NMR and multi-dimensional NMR have reduced the experimental time required immensely and have made it possible to use NMR to study large biological molecules and complexes.

Another aspect of the NMR experiment is used in medical imaging. After a nucleus makes a transition from a lower energy state to a higher energy state, it has a tendency to 'relax' back to the lower energy state by transferring the extra energy it has acquired either to another spin nearby, or to

the other atoms and molecules, i.e. the 'lattice'. The time required for this relaxation will depend upon the nature of the lattice. In a living body, different tissues have different relaxation times. A measurement of the relaxation time of the spin nuclei in different parts of the body can be made. These measurements can then be displayed as a picture of the interior of the body, enabling medical practitioners to detect malfunctions or tumours etc. in the various organs.

8.3 NMR Theory and Experiment

Rigorous NMR theory requires a knowledge of quantum mechanics and will not be attempted here. Some of the results obtained will be stated without proof. According to quantum mechanics the angular momentum of atomic nuclei or, in other words, the spin of an atomic nucleus can only have certain discrete values characterised by the spin quantum number I . The value of I for any nucleus can either be an integer or a half-integer ($1/2, 3/2, 5/2 \dots$) and depends on the number of protons (the atomic number and the number of neutrons present. (The sum of the neutrons and protons is the mass number. Table 8.1 gives the values of the spin for various possible combinations. For example the nuclei ^1H , ^{13}C , ^{31}P , and ^{19}F all have odd mass numbers and therefore a half-integral spin value. (In these particular cases the value is $1/2$). The nuclei ^{12}C and ^{16}O have even atomic numbers (6 and 8 respectively) and even mass numbers and the spin is zero. Nuclei of deuterium, i.e. ^2H , and ^{14}N have odd atomic numbers (1 and 7 respectively) and even mass numbers. The value of the spin is 1. In chemistry and biology, the most useful nuclei are those with spin $1/2$ and the following discussions will be restricted to such nuclei.

Table 8.1 Relation between the nucleon numbers and spin. Only nuclei with half-integral or integral spins are NMR-active

Atomic number	Mass number	Spin quantum number
Even or odd	Odd	Half integral
Even	Even	Zero
Odd	Even	Integral

When placed in a strong, uniform magnetic field, again according to quantum mechanics, the spin angular momentum vector can take up one of two positions (Figure 8.2) such that the component of the angular momentum along the direction of the applied magnetic field B_0 is

$$p_z = \pm 1/2(h/2\pi)$$

where h is Planck's constant. The z component of the magnetic moment of such a nucleus is also quantized and is given by

$$\mu_z = \gamma p_z = \pm 1/2(h/2\pi)\gamma$$

where γ is a parameter that depends on the particular nucleus and is called the magnetogyric (or gyromagnetic) ratio. The energy associated with these two positions due to the interaction of the magnetic moment with the applied magnetic field is

$$E = -\mu_z B_0 = \pm 1/2(h/2\pi)\gamma B_0$$

Thus the difference in energy between the two states is

$$\Delta E = \gamma(h/2\pi)B_0$$

Therefore, to induce the spin to make the transition from the lower energy state to the higher energy state, the applied oscillating magnetic field, i.e. the radio-frequency field should supply this energy. This implies that the frequency of oscillation should be ν where

$$\Delta E = h\nu_0$$

and

$$\nu_0 = \gamma B_0 / 2\pi$$

If we write $\omega_0 = 2\pi\nu_0$, where ω_0 is the angular frequency, then

$$\omega_0 = \gamma B_0$$

This is called the resonance condition and is the basic NMR equation.

Actual NMR experiments, of course, do not deal with single nuclei and single spins. The sample will contain a large number of nuclei, of the order of Avogadro's number ($\sim 10^{23}$). When the sample is placed in the static magnetic field, all the nuclei will not take up the lower energy state. Instead the ratio of the number of spins in the lower energy state to the number in the higher energy state is given by the Boltzmann formula

$$\exp(-\Delta E/kT)$$

where k is the Boltzmann constant and T the absolute temperature of the sample. Since

$$\Delta E = \gamma(h/2\pi)B_0$$

the ratio is

$$\exp(-\gamma(h/2\pi)B_0/kT).$$

This means that at room temperature and at medium magnetic fields, the energy difference between the two states is such that the population of spins in them differs by less than 1 part in ten thousand. This implies that the signal is very weak, since only a relatively few nuclei will absorb energy and make the transition. Note that if the applied static field is increased, the energy difference increases, leading to a larger population difference and hence a stronger signal.

8.4 Classical Description of NMR

The NMR condition

$$\omega_0 = \gamma B_0$$

may also be derived from considerations of classical mechanics, as follows. If we consider the nucleus to be analogous to a spinning bar magnet, classical mechanics tells us that such a system placed in a magnetic field will precess about the direction of the external field with the so called Larmor frequency given by

$$\omega_0 = \gamma B_0$$

The large numbers of nuclei in a typical experimental sample will all precess about $\mathbf{B}_0$ (Figure 8.3). This would mean that, while the net component of the magnetic moment in the xy plane is zero, there will be a net magnetisation along the z -axis (presuming $\mathbf{B}_0$ to be along z , as in Figure 8.3). In order to perform the NMR experiment, it becomes necessary to tilt the total magnetisation towards or onto the xy plane as illustrated in Figure 8.4. (This is equivalent to inducing a transition of many nuclei, from the lower energy quantum state to the higher energy state). To understand this better, we may

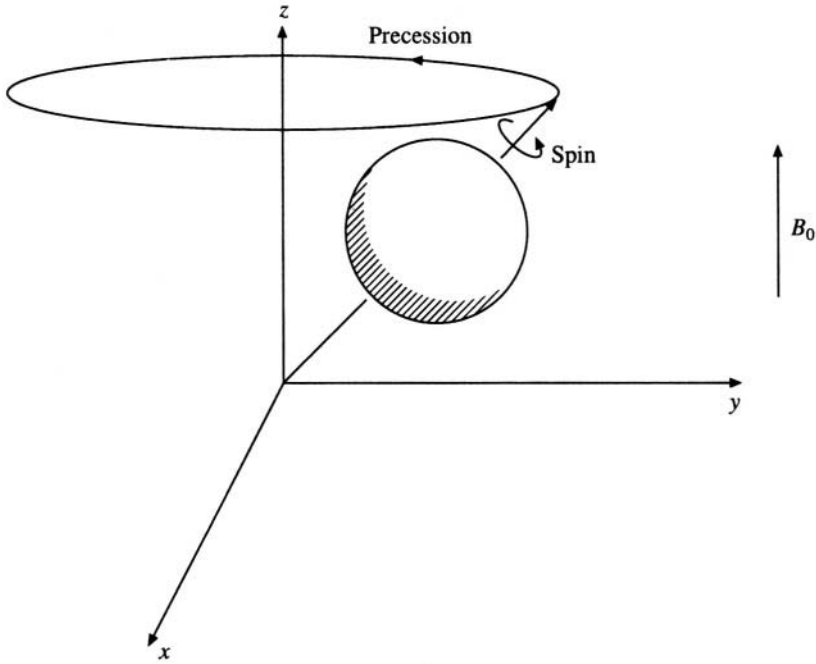


Fig. 8.3 Precession of a spinning nucleus. The precession axis is the z axis.

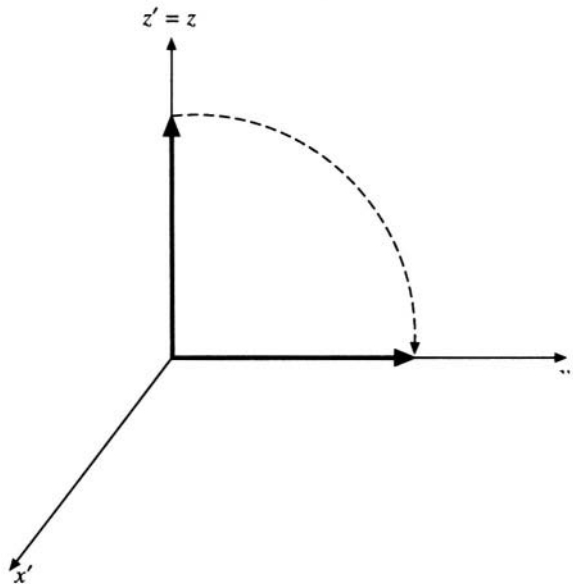


Fig. 8.4 Tilting of the total magnetisation on to the xy plane

change our reference axial system from the static xyz system in Figure 8.3 to the rotating frame of reference given by $x'y'z'$. This reference frame is assumed to rotate about the z -axis with the same angular frequency as the Larmor frequency ω_0 . Thus in this frame, the nuclear magnets do not precess

and the apparent static magnetic field is zero. Now if a static magnetic field $\mathbf{B}_1$ is applied along the $\mathbf{x}'$ direction in this field, then, just as the field $\mathbf{B}_0$ caused a precession in the stationary reference frame, so will the field $\mathbf{B}_1$ cause the nuclear magnets to precess about the $\mathbf{x}'$ axis, and the total bulk magnetisation will rotate about the $\mathbf{x}'$ axis. Of course the field $\mathbf{B}_1$ is not actually a static field, but only appears so in the rotating frame. In fact $\mathbf{B}_1$ is the radio-frequency field (of frequency $\omega_0 = \text{Larmor frequency}$) which is applied in addition to $\mathbf{B}_0$ in the NMR experiment. If the field $\mathbf{B}_1$ is applied for a time t , then the magnetisation will rotate through an angle given by $\theta = \gamma \mathbf{B}_1 t$. If the duration of the pulse of radio-frequency electromagnetic radiation is such that the magnetisation rotates by 90° onto the xy plane, then such a pulse is called a 90° pulse. In the same way a 180° pulse will reverse the direction of the magnetisation. The pulse produces a net component of the magnetisation in xy plane also. This component then produces an effect upon the detection coil in the NMR spectrometer, which is amplified and displayed as the NMR spectrum.

There are two main methods by which the NMR experiment is performed, viz. the continuous wave method and the Fourier transform method. In the continuous wave experiment the radio-frequency field is continuously applied to the sample, and the magnetic field is changed continuously in strength such that all the nuclei in the sample are able to come into resonance. Alternatively the magnetic field may be kept fixed and the oscillating radio frequency field may be varied. Either way, the continuous wave method, has been almost entirely replaced by the Fourier transform NMR (FTNMR).

In FTNMR the radio-frequency stimulus is applied as a pulse which may last typically for about $200\mu\text{s}$. The pulse is sufficiently strong and is constituted of a sufficiently large number of frequencies to excite all the resonances in the sample. As a matter of fact what is usually applied is a square pulse. As explained in a previous chapter (Chapter 7), according to Fourier transform theory, such a square pulse is constituted of all possible frequencies. The response to such an excitation would be an NMR signal from each of the spins in the sample. The composite would be another complicated wave which may be Fourier transformed to extract the NMR spectrum. In order to improve the detection of the signal, or, in other words, the signal to noise ratio, this process of giving the pulse and measuring the response is repeated several times, before the Fourier transform is taken. If the response is measured N times and added, the signal will increase N times while the background noise, which is random will increase only $\sqrt{N}$ times. The expression free induction decay or FID is used to describe the response on the application of the radio-frequency pulse – it describes the decay of the induced signal arising from the free precession of the nuclei in the field $\mathbf{B}_0$.

The nuclei in the sample will be present in several different environments, and the resulting FED will consist not only of several different frequencies, but also each different frequency will decay at a different rate. The Fourier transform of the FID therefore contains information not only about the resonance frequency of each nucleus but also about the decay of the resonance and hence the rate and mode of energy transfer from the nucleus to the environment.

8.5 NMR Parameters

In general, the following are the measurable NMR parameters: (1) chemical shift (2) intensity (3) line-width (4) spin-spin coupling constants.

8.5.1 Chemical shift

The effect of the external applied magnetic field on each of the protons in a molecule will not, in general, be the same. This is because, owing to the electron currents surrounding each nucleus, an additional small field is also created just around the nucleus. The total effective field at each nucleus

will thus depend on the neighbouring atoms and chemical groups. Since the magnetic field is different at each chemically distinct nucleus, the frequency at which the resonance occurs will also be different at each nucleus. Consider for example a molecule such as ethanol. A schematic sketch of its ^1H NMR spectrum at 100 MHz (i.e. B_0 , the applied field strength, corresponding to a resonance frequency of 100 MHz for a free H nucleus) is given in Figure 8.5. The frequency at which the hydroxyl (OH) proton resonates is different from the one at which the methylene (CH_2) protons resonate which is again different from the one at which the methyl (CH_3) protons resonate. Such a shift in the position of the resonance peak in the spectrum due to the chemical environment is called the chemical shift.

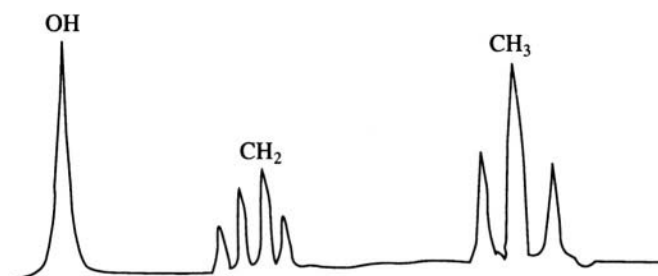


Fig. 8.5. The NMR spectrum of ethanol (schematic). The splitting of the peaks has been exaggerated.

The chemical shift is always measured with respect to some arbitrary chosen reference point in the spectrum. Usually an inert compound with a well-known resonance position, not much affected by the molecular environment, is used as the standard. The standard may be introduced either internally (i.e. mixed with the sample) or externally (in a small tube placed alongside the sample tube). Chemical shifts are expressed in the dimensionless units of parts per million (ppm) and are normalised to be independent of the applied magnetic field. The chemical shift

$$\delta = (\nu_s - \nu_R)/\nu_R \times 10^6$$

where ν_s is the resonance frequency of the sample, and ν_R is the resonance frequency of the reference. By convention, when displaying the NMR spectrum, the resonance frequency of the standard is placed at the right-most end of the scale at 0 ppm, and the scale, in ppm, increases to the left, in decreasing order of frequency. This corresponds to the fact that the chemical shift is actually produced by a shielding and therefore a reduction of the applied magnetic field by the electrons in the neighbourhood of the resonating nucleus. Table 8.2 indicates the typical chemical shifts for some important chemical groups.

8.5.2 Intensity

The signal strength or intensity of the resonance peak is a measure of the number of nuclei that contribute towards it. Signal intensity is obtained by measuring the area under the peak. To be mathematically rigorous, it is necessary to integrate each peak from $-\infty$ to $+\infty$. In practice this is not possible and one has to be content with measuring the area under those parts of the peak which are accessible. This usually accounts for only about 90% of the total area, and hence there is always an error of $\pm 10\%$ in the measurement of the signal strength.

Apart from the number of nuclei resonating at the given frequency, signal intensity is also influenced by the effects of saturation, the nuclear Overhauser effect, the effects of a finite pulse width and other experimental limitations. If these factors are taken into consideration and if appropriate corrections

Table 8.2. Typical ^1H chemical shifts (in ppm from tetramethylsilane or TMS. Where a range is given, the position of the peak is dependent on the attached groups as well as the solvent, pH, etc.)

Chemical group	Chemical shift δ
$-\text{CH}_3$	0.9
$-\text{CH}_2$	1.4
$-\text{CH}$	1.5
$-\text{C}_6\text{H}_6$	7.3
$-\text{CHO}$	9.7
$-\text{OH}$	0.5-4.0
$-\text{NH}_2$	2.0-5.0
$-\text{NH}$	2.0-5.0
$-\text{SH}$	1.0-2.0

are applied, then accurate measurements of relative concentrations of the different nuclei can be made. If absolute concentrations need to be measured, a known quantity of a standard sample can be used to calibrate the spectrum and measurements can then be made. This technique is especially useful in biomedical applications.

8.5.3 Line width

The width of each spectral line is related to the relaxation processes in the sample. There are primarily two ways in which the additional energy gained by the nucleus on transition can be released. The energy can either be transferred to a neighbouring spin or it can be transferred to the neighbouring lattice. A time constant is associated with each and referred to as T_2 and T_1 respectively. Line width is measured at half the signal height in Hz and is represented as $\Delta\nu_{1/2}$. The relationship to so-called spin-spin relaxation time T_2 is

$$1/T_2 = \Delta\nu_{1/2}$$

in the ideal situation where only the relaxation contributes to the line width.

8.5.4 Relaxation Parameters

As explained earlier, when a sample is placed in a magnetic field, it becomes magnetised. The direction of the magnetisation $\mathbf{M}_0$ is the same as that of the applied magnetic field, being conventionally taken as the z-axis. The components of magnetisation along the other two directions, i.e. $\mathbf{M}_{xy}$ is zero. This is the situation in the equilibrium condition. When a radio frequency pulse is applied, the magnetisation tilts over onto the xy plane and as the perturbing field is removed, the magnetisation returns to the equilibrium situation of $\mathbf{M}_z = \mathbf{M}_z$ and $\mathbf{M}_{xy} = \mathbf{0}$. This happens by the relaxation processes and lasts over a period of time ranging from milliseconds to a few seconds. As mentioned earlier, there are two main relaxation processes, spin-lattice relaxation and spin-spin relaxation.

Spin-lattice relaxation is responsible for the return of the z component $\mathbf{M}_z$ to its equilibrium value. It is also called longitudinal relaxation. The lattice referred to in the term is the environment and all the nuclear dipoles surrounding the particular spin, whether they belong to the same molecule or to other molecules. In solid state NMR, the word lattice may refer to the crystal lattice. In solution phase NMR it simply means the neighbourhood of each spin. The return of the nuclear spins to thermal